

DESIGN REPORT
30th ANNUAL
DBF COMPETITION
BY TEAM MIST DBF



P R E S E N T I N G

HEISENBIRD

**MILITARY INSTITUTE OF
SCIENCE AND TECHNOLOGY**

TABLE OF CONTENTS

ACRONYMS, ABBREVIATIONS AND SYMBOL	3
1 EXECUTIVE SUMMARY	4
2 MANAGEMENT SUMMARY	5
2.1 Team Organization	5
2.2 Milestones and Schedule	6
3 CONCEPTUAL DESIGN.....	7
3.1 Mission Requirements	7
3.2 General Requirements and Constraints	9
3.3 Deriving Design Requirements.....	10
3.4 Scoring Sensitivity Analysis	10
3.5 Configuration Selection	12
4 PRELIMINARY DESIGN.....	17
4.1 Design Methodology	17
4.2 Mission Model	17
4.3 Weight Estimation	18
4.4 Wing Sizing	18
4.5 Aerodynamic Analysis.....	19
4.6 Stability and Control	22
4.7 Propulsion Package Selection	24
4.8 Operational Wind Limits and Gust Disturbance Analysis	27
4.9 Mission Uncertainties	28
5 DETAILED DESIGN	28
5.1 Dimensional Parameters	28
5.2 Structural Characteristics and Capabilities	29
5.3 Subsystem Integration	34
5.4 Weight and Mass Balance Analysis	37
5.5 Flight and Mission Specific Performance	38
5.6 Drawing Package.....	38
6 MANUFACTURING PLAN	43
6.1 Manufacturing Processes Investigated	43
6.2 Selected Manufacturing Processes	43
6.3 Manufacturing Timeline and Milestones	44
7 TESTING PLAN.....	45
7.1 Test Objectives.....	45
7.2 Testing Schedule	46
7.3 Ground Tests.....	46
7.4 Aerodynamics Tests	48
7.5 Propulsion and Electronics Tests	50
7.6 Mock Tests.....	51
7.7 Flight Checklists	52
7.8 Flight Tests.....	53
8 PERFORMANCE RESULTS	54
8.1 Demonstrated Performance of Key Subsystems.....	54
8.2 Demonstrated Flight and Mission Performance.....	56
8.3 Design Changes and Lessons Learned.....	57
9 BIBLIOGRAPHY	58

ACRONYMS, ABBREVIATIONS AND SYMBOL

AIAA	American Institute of Aeronautics and Astronautics
AR_b	Banner aspect ratio
AR	Aspect Ratio
AVL	Athena Vortex Lattice
BMD	Bending Moment Diagram
BWB	Blended Wing Body
C_{Lδ_e}	Lift Change due to Elevator Deflection
C_{Lq}	Rolling moment due to pitch rate
C_{Lα}	Lift curve slope
C_{y_p}	Roll-induced side force.
C_{y_r}	Yaw-induced side force.
C_{yβ}	Side Force due to Sideslip
C_{lp}	Rolling moment due to roll rate
C_{lr}	Rolling moment generated by yaw rate
C_{lβ}	Rolling moment due to sideslip
C_{mδ_e}	Pitching moment change from elevator deflection
C_{m_q}	Pitching moment due to pitch rate
C_{mα}	pitching moment derivative
C_{n_p}	Yaw due to Roll
C_{n_r}	Yaw Damping due to Yaw
C_{nβ}	Directional Static Stability
C_D	Coefficient of Drag
C_L	Coefficient of Lift
CAD	Computer-Aided Design
CFD	Computational Fluid Dynamics
CF-PLA	Carbon Fibre PLA
CFRP	Carbon-fiber-reinforced polymer
CG	Center of Gravity
COTS	Commercial Off-the-shelf
DBF	Design/Build/Fly
Dept.	Department
EF	Efficiency Factor
ESC	Electronic Speed Controller
FAA	Federal Aviation Administration
FEA	Finite Element Analysis
FoM	Figures of Merit
FoS	Factor of Safety
GM	Ground Mission
L/D	Lift-to-Drag Ratio
LiPo	Lithium Polymer
M1, M2, M3	Mission 1, 2, 3
MAC	Mean Aerodynamic Chord
MGTOW	Maximum Gross Take-Off Weight
MLG	Main Landing Gear
MTOW	Maximum Take-Off Weight
Ni-CAD	Nickel - Cadmium
PLA	Polylactic Acid
RAC	Rated Aircraft Cost
Re	Reynolds Number
ROC	Rate of Climb
SFD	Shear Force Diagram
TOFL	Take Off Field Length
TOGW	Take-Off Gross Weight
TX-RX	Transmitter-Receiver
W-h	Watt-hour
WS	Wing Span

1 EXECUTIVE SUMMARY

This report documents the design, manufacturing and testing of “HeisenBird”, developed by MIST DBF team for the 2026 AIAA DBF competition. It is optimized to complete three flight missions and one ground mission under competition constraints: wingspan ≤ 60 in, total propulsion energy ≤ 100 W-hr and a deployable banner of length ≥ 10 in with aspect ratio ≤ 5 . M1 demonstrates basic flight and operational capability with no payload by completing 3 laps within a 5-minute window. M2 is a revenue-generating transport mission, scored by carrying passengers and cargo for the maximum possible laps within 5 minutes. M3 is a banner-towing mission, scored on lap count and banner length within 5 minutes. GM evaluates rapid payload handling and configuration changes between these missions [1].

A MATLAB-based parametric scoring model was developed to determine the influence of design variables on total score. Sensitivity analysis demonstrated that lap time in M2 and M3 is the dominant scoring driver, followed by GM duration and payload/banner parameters. While increased payload and banner length increase revenue, excessive weight and drag degrade lap performance, creating a diminishing-return trade-off. The optimal balanced strategy targets 24 passengers and 8 cargo in M2, and a 60-inch banner (5ft x 1 ft) in M3, while fully utilizing the 100 W-hr energy allowance ($EF \approx 1$) and maximum span ($RAC \approx 1$). GM time was minimized through modular payload bays and rapid reconfiguration mechanisms.

The finalized configuration is a 60-inch low-wing tractor monoplane with a conventional tail and tricycle landing gear, selected for aerodynamic efficiency, CG stability, and ground handling. Clark Y wing and NACA 0009 tail airfoils were selected for their reliable performance. AVL analysis confirmed adequate static stability across all configurations, including banner-tow, while coordinated-turn and constraint analyses ensured stall margin during mission flight. FEA verification validated wing, fuselage, and motor mount structures under representative 2.5g-4g loads with acceptable safety margins. Winglets were added to reduce induced drag within the span limit.

Table 1.1 Predicted mission-level performance targets for HeisenBird in M1-M3 and GM

Parameter	M1	M2	M3	GM
MTOW (lbs.)	8.82	12.87	9.2	N/A
Payload Weight (lbs.)	N/A	4.06	0.41	
Passengers Carried	0	24	0	
Cargo Units Carried	0	8	0	
Banner Length (in)	N/A	N/A	60	
Predicted Lap Time (s)	35	30	37	
Predicted No. of Laps	3	10	8	
Mission Time (s)	95	N/A	N/A	
Efficiency Factor, EF	0.99	0.99	0.99	
Rated Airplane Cost, RAC	1	1	1	
Cruise Airspeed (ft/s)	89	108	89	
Predicted Net Income	N/A	964	N/A	
Predicted Mission Score	1	1.643	2.144	0.25
Total Predicted Mission Score	5.03			

Propulsion is based on a Scorpion A-4220 V2 (540 kV) motor with Falcon propellers (15x8, 16x7) powered by a 6S 4500 mAh battery. Static thrust and battery discharge tests confirmed takeoff/climb margin and mission-energy compliance, while avionics validation included full control-surface travel checks, receiver range testing, and failsafe verification. The M3 banner system is a removable external module, incorporating a recessed stow cavity and a servo-actuated claw to control banner phases. Banner fabric trade testing across wind-tunnel/ground/flight evaluations selected Taffeta 240T as the lowest-drag option. The final 5 ft $\times$ 1 ft banner integrates a 5-mm carbon leading-edge rod sewn into the fabric, reinforced attachment points with bridle-to-single-line towing, adjustable ballast for vertical alignment, and sewn contrasting “MIST” lettering on both sides.

A hybrid manufacturing approach combining laser-cut balsa structures, plywood reinforcement, carbon components, and additive manufacturing ensured structural reliability and modular transportability. Ground tests, mock missions, and phased flight trials demonstrated stability, subsystem reliability, and close agreement between analytical predictions and measured lap performance.

HeisenBird delivers a scoring-optimized, aerodynamically efficient, and structurally robust solution capable of competitive performance across all DBF 2026 missions. Table 1.1 summarizes the predicted mission-level performance targets.

2 MANAGEMENT SUMMARY

The 2026 MIST DBF Team consisted of 41 undergraduate members, including 22 seniors and 19 underclassmen among which 14 are juniors, 4 are sophomores and 1 is freshmen. The 41-member team was organized into two departments based on functionality: Technical and Administrative. The team was fully student-led and advised by 3 faculty members who provided guidance and shared technical expertise at each weekly meetings and scheduled design review sessions. The technical and operational complexity of this year's project necessitated a larger team to enable efficient workload distribution and specialized task assignment, improving technical depth as well as reducing any schedule and execution risk.

2.1 Team Organization

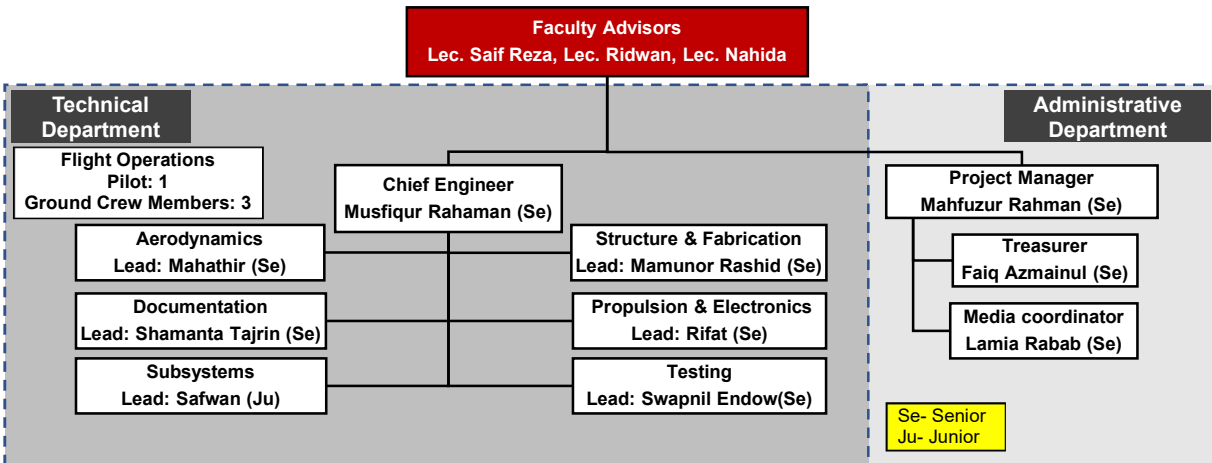


Figure 2.1 Team Organization Chart

The team was organized using a matrix leadership framework reflective of typical aerospace firm management structures. The 2025-26 leadership organization is shown in Figure 2.1.

The team was led by the Project Manager and the Chief Engineer, overseeing administrative and technical departments, respectively. The Technical Dept. was responsible for translating competition requirements into verified aircraft design, fabrication, testing, and flight performance, while the Administrative Dept. governed resources, schedule control and external coordination to ensure execution remained compliant and timely. The Technical Dept. consisted of six technical sub-teams, each with 4-8 members led by a sub-team lead. The administrative department consisted of a treasurer and a media coordinator. The technical sub-team leads delegated tasks to members of their respective sub-teams, monitored progress and arranged sub-team meetings. The Flight Operations sub-team under the Technical Dept., comprised the pilot and ground crew, jointly supervised by the Pilot and the Chief Engineer, was responsible for safe, efficient mission execution, ground operations, and compliance with fly-off competition procedures. The responsibilities and skills of each technical sub-team and all personnel are listed in Table 2.1.

Regular communication and defined responsibilities are central to the team's management strategy. The entire team engaged in weekly meetings to coordinate major design, manufacturing and testing decisions, and to review progress and address interdisciplinary issues. In addition, each sub-team met twice weekly to plan and track their specific tasks and report updates to the leads. This meeting cadence promoted accountability and rapid problem-solving. The overall team structure established a clear chain of command and efficient workflow while paving the path for sharing knowledge and hands-on learning for underclassmen, collectively improving project coordination and execution reliability.

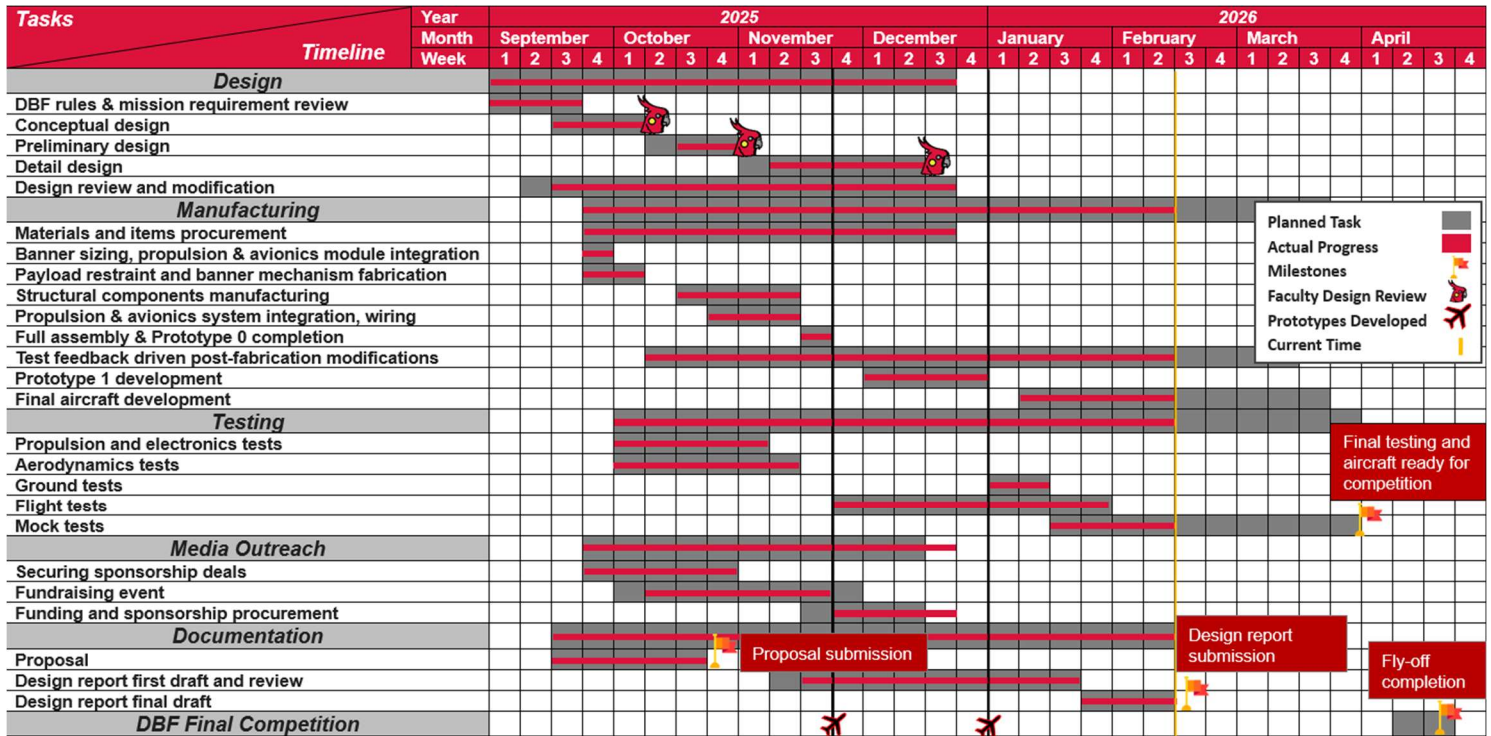
Table 2.1 Sub-team & Personnel Responsibilities and Skills

Sub-team/ Personnel		Responsibilities	Skill Requirements
Technical sub-teams	Aero- dynamics	Perform aerodynamic analysis for sizing, mission performance, stability and control. Conduct trade studies & sensitivity analysis. Shape, size, optimize aircraft components & control surfaces. Develop CAD model and run aerodynamic simulations.	Knowledge of aerodynamic principles, stability and control concepts. Proficiency in SolidWorks, MATLAB, XFLR5, ANSYS and CFD tools.
	Propulsion & electronics	Determine most suitable motor, battery, propeller and ESC through performance analysis. Conduct wiring design & integration of propulsion and electronic systems into aircraft ensuring connectivity of onboard controls.	Knowledge of propulsion, power, electrical systems, signals and circuitry. Proficiency in MATLAB, eCalc, MotoCalc and Proteus.
	Structure & fabrication	Design and optimize the aircraft structural framework. Conduct structural analysis, select suitable materials for the aircraft and run structural simulations. Manufacture all the components and sub-systems, conduct full assembly of prototypes ensuring safety compliance.	Knowledge of material properties, aircraft structures, different manufacturing processes. Proficiency in SolidWorks, MATLAB, handling machine tools.
	Sub- systems	Design and optimize the payload restraint and banner stow/deploy/release mechanism. Ensure all aircraft sub-systems meet the requirements and follow DBF rules.	Knowledge of servo-based mechanisms and mechanical design. Proficiency in SolidWorks, ANSYS and MATLAB.
	Docu- mentation	Maintain records of important design data and modifications. Prepare and format proposals, design reports in compliance with official DBF documentation requirements.	Technical writing, formatting and organizational skills. Proficiency in LaTeX, MS Word & Excel.
	Testing	Test fabricated individual components, sub-system mechanisms, propulsion & avionics systems. Record performance and aerodynamics data for further improvement.	Test instrument setup, data logging and analysis skill. Proficiency in MATLAB, Python, MS Excel.
Technical personnel	Pilot	Execute all flight missions with precise airspeed, trajectory, and energy management. Conduct flight checks and post-flight feedback to support refinement and reliability. Coordinate with chief engineer to guide ground crew and flight operations.	Strong situational awareness and decision-making ability. Proficiency in fixed-wing RC aircraft operation under varying payload and drag conditions.
	Ground Crew	Perform rapid aircraft assembly, payload loading/unloading and mission reconfiguration in accordance with GM process. Execute inspections, safety checks & ensure rule compliance.	Clear and in-depth understanding of aircraft structure, subsystems and DBF mission operations.
	Chief Engineer	Oversee and guide all engineering design, analysis, fabrication & testing tasks. Ensure coordination between all technical sub-teams and faculty advisor. Coordinate with project manager to ensure engineering decisions meet project goals and with pilot to guide ground crew and flight operations. Resolve conflicts between technical sub-teams and personnel.	Planning, analytic and leadership skills. Broad technical knowledge in aerodynamics, stability and control, structure, propulsion and electronics. Proficiency in FEA, CAD and programming tools.
Administrative personnel	Project manager	Oversee work-quality, take lead in budgeting, resource allocation, outreach, schedules, deadlines and submissions. Arrange timely team meetings and coordinate between media, treasury and faculty advisors. Coordinate with chief engineer to align timeline and resource constraints with technical decisions. Resolve conflicts within administrative personnel.	Strong planning, organizational and leadership skills. Experience in quality control and management. Proficiency in MS Word and Excel
	Treasurer	Manage funds, plan budgets and distribute resources to sub-teams and personnel while maintaining financial compliance. Track all expenses and keeping record.	Financial management and organizational skills Proficiency in MS Excel
	Media co- ordinator	Approach companies or institutes for sponsorships and manage funding activities. Coordinate public engagement and publications.	Communication, networking and marketing skills. Proficiency in image and video editing tools

2.2 Milestones and Schedule

The Gantt Chart in Table 2.2 represents the team's schedule. The project manager maintained the schedule to structure workflows and track tasks to completion and referenced the chart in each full team meeting to ensure timely completion of all the scheduled activities and milestones. The schedule set at the beginning of the year was updated accordingly as shown by 'Actual Progress'. The parallel workflow employed for the project enabled simultaneous task execution, significantly improving resource utilization and accelerating project progress. The schedule was designed for the construction of 3 iterative prototypes with sufficient window for testing and subsequent design improvements between each, eventually leading to the final competition aircraft.

Table 2.2 Gantt Chart



3 CONCEPTUAL DESIGN

This phase systematically translated the 2026 AIAA DBF mission requirements into a scoring-driven aircraft configuration. Beginning with an analysis of competition rules and scoring equations, key performance drivers were identified and quantified through parametric sensitivity studies. These insights informed a structured selection methodology using weighted Figures of Merit (FoMs), enabling objective trade-offs across all major subsystems. The process ensured that design decisions were rigorously aligned with maximizing total competition score while adhering to all constraints.

3.1 Mission Requirements

3.1.1 Problem Statement

The 2026 AIAA DBF competition requires the design of a banner-towing bush plane that earns revenue through charter flights and aerial advertising. The aircraft must complete three flight missions and one ground mission under strict constraints: a maximum 5-ft wingspan (corresponds to $RAC=1.0$), 100 W-hr total propulsion energy (corresponds to $EF=1$) and a banner of at least 10-in length (aspect ratio ≤ 5). Performance is scored through mission-specific equations that create direct trade-offs between aircraft size, weight, efficiency and operational speed.

3.1.2 Mission Scoring

The overall score for the 2026 AIAA Design/Build/Fly competition is determined by a combination of flight mission performance and written report evaluation. The competition structure emphasizes balanced aircraft design, operational efficiency, and technical justification.

$$\text{Total Mission Score} = GM + M1 + M2 + M3$$

$$\text{Report Score} = 0.85 \times \text{Design Report Score} + 0.15 \times \text{Proposal Score}$$

$$\text{Overall Competition Score} = \text{Total Mission Score} \times \text{Report Score} + \text{Participation Score}$$

Participation points are awarded as: +1 for attending the fly-off, +2 for passing technical inspection and +3 for attempting M1.

Before any mission attempt, the aircraft must successfully pass Technical Inspection to verify compliance, airworthiness, mechanical fastening requirements, and payload configuration rules.

Mission 1 (Test Flight)

It is a binary demonstration of basic airworthiness and handling. The airplane flies three laps of course within a 5-minute window with no payload and must land successfully on the paved runway to receive a score of $M1 = 1.0$.

Mission 2 (Charter Flight)

It represents revenue flights with passengers and cargo. The team selects the mix of 2.3-in rubber-duck passengers and hockey-puck cargo units up to the maximum declared at Tech Inspection, then flies laps for 5 minutes. Net income is computed from per-passenger and per-cargo fixed and per-lap income terms minus an operating cost that scales with laps, payload count and the Efficiency Factor, $EF = R_{batt}/100$. M2 score is normalized as:

$$M2 = 1 + \frac{(\text{Net Income})_{\text{Team}}}{(\text{Net Income})_{\text{Max}}}$$

Mission 3 (Banner Flight)

It evaluates aerial endurance. The aircraft must tow a team-supplied banner, deploying it after the first turn and releasing it after the final lap. Score is a function of laps completed, banner length (L_{banner}) and RAC, where $RAC = 0.05 \times \text{Wingspan (in)}/12 + 0.75$, normalized as:

$$M3 = 2 + \frac{(\text{Laps} \times L_{\text{banner}} / \text{RAC})_{\text{Team}}}{(\text{Laps} \times L_{\text{banner}} / \text{RAC})_{\text{Max}}}$$

Ground Mission

It is a timed demonstration of loading and unloading the maximum declared passengers and cargo, installing the banner in the stowed configuration, and demonstrating deployment and release on the ground, with the score:

$$GM = \frac{\text{Fastest Time}}{\text{Team Time}}$$

3.1.3 Staging and Flight Profile

Prior to each flight mission, the aircraft is staged in a non-flight configuration within a 2.5-ft (0.76-m) wide parking space, with propulsion battery and payloads removed. The team is allotted five minutes of staging time to transition to a flight-ready state: installing the battery, loading and securing payloads and verifying restraint systems. Upon staging completion, the five-minute mission flight window begins when throttle is first advanced for takeoff.

All flight missions follow the standardized competition flight course illustrated in Figure 3.1.

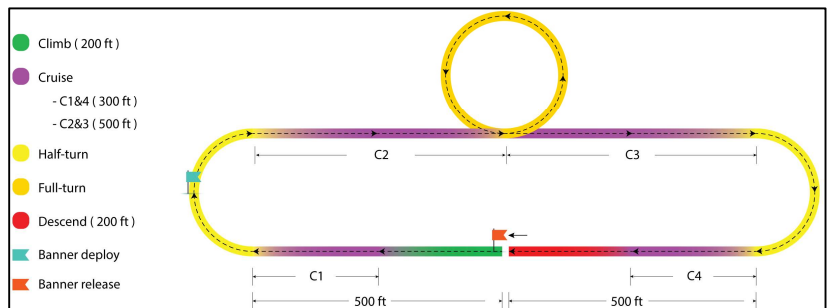


Figure 3.1 Flight Mission Profile

Each lap consists of two 1,000-ft straightaways, two 180° turns and one 360° turn (opposite to the direction of 180° turns), completed by crossing the finish line. A successful landing is required for any flight mission to receive a valid score.

3.2 General Requirements and Constraints

Table 3.1 Compliance and Allowance Matrix

Category	Requirement	Moc	Soc	Allowance
General	Airplane must be fixed-wing, unmanned	R	C	Any fixed-wing configuration is permitted within all other constraints, except rotary wing or lighter-than air.
	Max. takeoff gross weight (TOGW) with payload must be ≤ 55 lb.	R	C	
	All components must remain attached with airplane during flight except where banner deploys and releases in M3	R	C	
	Must be flown with human pilot, no autopilot	R	C	
Sizing	Maximum wingspan (WS) must be ≤ 5.0 ft	R	C	Design may trade wingspan and therefore RAC against mission performance.
	Minimum allowable RAC is 0.90 (≈ 3 ft wing span)	R	C	
	Assembled aircraft must fit inside an approximately 10 ft x 10 ft box	R	C	
Structure	Aircraft must pass wingtip load test in flight configuration	T	C	Safety wire, thread-locker or mechanical interference-fit is allowed as secondary fastener.
	All permanent aircraft components must be secured using locking devices	R	C	
Propulsion & Avionics	Airplane must be electric powered & propeller driven	R	C	Multiple motors/ propellers/ propulsion systems are allowed. Commercially procured flight stabilization systems/gyros are allowed; secondary transmitter is allowed. Any commercial brushless or brush electric motor is allowed. Motors maybe direct drive or with gear or belt reduction. Arming system may be a blade-style fuse, arming plug, or equivalent. Avionics battery chemistry may differ from propulsion battery.
	Airplane must be radio-controlled; a receiver power switch must be externally accessible	R	C	
	Propulsion arming must use an externally accessible arming fuse/plug hard-mounted on the outside of airplane	R	C	
	Propulsion battery must be separate from avionics battery	R	C	
	Total propulsion energy stored on the airplane (sum of all propulsion batteries) must be ≤ 100 W-h	R	C	
	All propulsion batteries must consist of a single type			
	Propulsion batteries must be COTS and unaltered	R	C	
	Each individual battery pack must be ≤ 100 W-h (FAA carry-on limit defined as rated capacity × rated voltage)	R	C	
	Batteries must be either NiCad/ NiMH or LiPo			
Payload Compartment & Restraint System	Arming fuse maximum continuous discharge current rating must be below 100 A	R	C	Doors/hatches count is flexible within access rules. Number of passengers and cargo is selectable by the team for M2 within limits and declared during Tech Inspection, prior to staging assembly. Cargo bay location (forward/aft/below passengers) and number of cargo bays are selectable by the team. Small openings in bulkheads or floor are allowed for the pass through of wiring. Multiple cargo bays may be used, each with its own single access door/hatch. More than one passenger door is permitted.
	Passengers are standard ~2.3 in. rubber ducks provided at contest; airplane must carry at least 3 passengers	R	C	
	Passengers must be carried in a single passenger compartment on a single plane parallel to the horizontal plane while in flight	R	C	
	Passenger accommodation system should accommodate small duck size variations	R	C	
	Passengers must board individually (no speed loaders) and be restrained in a vertical, upright orientation	T	C	
	Passenger compartment must be accessed via door(s)/ hatch(es) providing access only to passenger compartment	R	C	
	Cargo is standard hockey pucks; Airplane must carry at least 1 cargo item; may use multiple cargo bays separated from passenger compartment and from each other by solid bulkheads/ floors	R	C	
	Passenger and cargo restraints must prevent movement during all phases of flight, including landing	T	C	
	Each cargo bay may have only one door/hatch; each cargo door/hatch may provide access only to cargo bay(s)	R	C	
Fuse-lage	Cargo orientation and whether cargo touches other cargo or parts is unrestricted	R	C	Door actuation method (hinge / pivot / slide, etc.) is at the team's discretion.
	Primary retention method must be mechanical, not frictional	R	C	
Banner Mechanism	All passenger and cargo access doors/ hatches must remain rigidly attached to airplane when opening and in the open configuration; strings/ lanyards are not acceptable attachment methods	R	C	The banner components or mechanisms may be either internal or external to airplane or a combination of both. A second transmitter may be used for banner deployment/ release and may only be operated by the Observer. Release mechanism design is flexible provided it meets banner deploy and release requirement.
	Banner must have minimum length 10 in; maximum aspect ratio 5	R	C	
	Banner must display the team's university name/ logo on both sides	R	C	
	Banner must be compactly stowed external to the airplane for takeoff; stowage may not interfere with flight or control functions	R	C	
	Banner must be remotely deployed in flight and remain in the vertical orientation during flight	T	C	
Banner Mechanism	Banner must be remotely released in flight; the banner must not sustain any damage during flight or upon ground impact after release (excluding the tow line or replaceable tow structure)	T	C	

Following the mission analysis, all general and subsystem-specific requirements from the 2026 DBF rules were compiled into a compliance and allowance matrix (Table 3.1) [1]. This matrix categorizes constraints by structure, propulsion &

avionics, fuselage, payload and banner systems. For each requirement, an appropriate Means of Compliance (MoC) was identified, where T denotes verification by test and R denotes verification by design review. As the design was developed with strict adherence to official rules, the State of Compliance (SoC) for all listed requirements is compliant (C). The allowance outlines all permitted components, configurations and modifications and thus reflects the design freedom. Adherence to these constraints ensured a rule-compliant baseline before conducting scoring sensitivity studies and configuration trade-offs.

3.3 Deriving Design Requirements

To maximize the scores, various coupled requirements arise which define HeisenBird's core design drivers:

Table 3.2 Design Goals Derivation from Constraints

Mission	Trade-offs	Design Goals
M1	Lighter weight improves climb rate and lap time but reduces structural margin for later payload missions. Higher power allows faster lap times but increases energy consumption, risking the battery limit for subsequent flights. Aerodynamic refinement reduces drag and increases laps but may complicate construction and increase build time.	Achieve predictable, consistent lap times through stable aerodynamics and precise control. Maintain energy margin to avoid battery exhaustion before completing all missions. Design for repeatable takeoff and landing performance across varying payload states.
M2	Carrying more passengers and cargo increases income per lap but reduces lap count due to added weight and drag. A larger battery supports more laps but raises the EF, increasing the operating cost per lap. Flying more laps multiplies both income and cost, requiring an optimal balance between payload quantity and lap count.	Optimize passenger-cargo mix to balance high per-unit revenue with low operating cost. Maximize payload without compromising stability and speed. Design and develop robust and secure passenger and cargo restraint mechanism with rigid locks and hatches. Minimize battery capacity while achieving target lap count to reduce EF penalty.
M3	A longer banner adds to the score but creates more drag, reducing lap count and overall score. A smaller wingspan lowers the RAC and improves the score but reduces aerodynamic efficiency and potential lap speed. Minimizing structural weight helps lap count but may compromise banner-tow strength and deployment reliability.	Design and develop robust, servo-driven deployment and release system. Reduce drag by selecting proper banner material, aerodynamics stowage and streamlined towing. Optimize banner length for maximum score without excessive drag. Ensure propulsion system produces high thrust to overcome banner induced drag. Minimize RAC by using full 5-ft WS efficiently, not necessarily minimizing it.
GM	Quick access payload bays speed loading time but may increase airframe complexity and empty weight-access payload bays speed loading time but may increase airframe complexity and empty weight. Lightweight restraint systems reduce handling time but must securely hold payloads during all flight maneuvers. A simple, reliable banner mechanism minimizes deployment time but must also be compact and low drag in flight-drag in flight.	Design rapid-access payload bays with simple, secure restraint systems. Implement quick-connect banner mechanism for fast stow/deploy cycles.

3.4 Scoring Sensitivity Analysis

3.4.1 Parametric Sensitivity Analysis

A high-fidelity parametric scoring model was developed in MATLAB to systematically explore the design space and identify an optimal configuration within the 100 W-hr energy constraints. The model computed total score as a function of key design and operational variables including: airframe geometry (wingspan, wing loading, tail configuration), propulsion system characteristics, (motor efficiency, propeller performance, total battery capacity) and mission parameters (passenger count, cargo count, banner length and drag, target cruise speed, achievable lap time, and ground mission

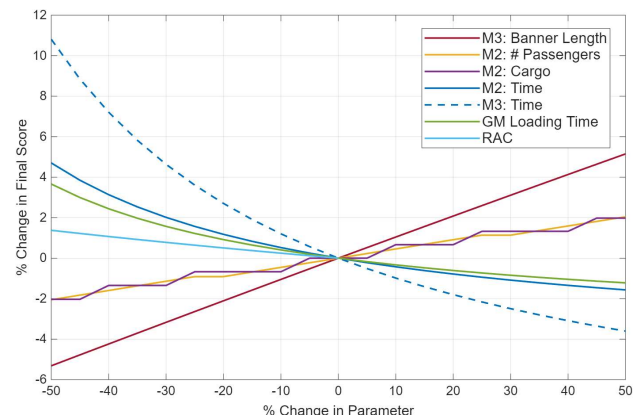


Figure 3.2 Overall mission sensitivity results

duration). Fixed scoring constants from the 2026 rules were embedded to ensure accurate mission revenue and cost calculations.

The analysis confirmed that, lap time is the most sensitive parameter for both M2 and M3, with minor reductions yielding disproportionately large score gains (Figure 3.2). This is because lap time strongly influences both the number of laps completed in M2 (via revenue scaling) and M3 laps-completed term, creating a nonlinear relationship between performance and points. GM time was identified as the second-most influential parameter, as it scales the GM score directly through a simple inverse relationship. While increasing payload mass (additional passengers or cargo) and banner length raise the potential revenue and M3 scoring numerator respectively, both exhibit strong diminishing returns. Beyond carefully identified optimal points, the added drag and weight increase lap time sufficiently to degrade the net score, creating a classic trade-off between capacity and performance.

3.4.2 Energy-Impact Analysis

To resolve these fundamental trade-offs and determine the most efficient allocation of the 100 W-hr energy budgets, a comparative energy-impact analysis was conducted. This analysis evaluated the marginal propulsion energy required for incremental score improvements across all parameters, leading to a critical strategic insight expressed as:

$$\frac{\partial E}{\partial T_{M2}} \gg \frac{\partial E}{\partial L_b} > \frac{\partial E}{\partial N_{pax}} > \frac{\partial E}{\partial T_{GM}}$$

where E is total propulsion energy consumption, T_{M2} is M2 lap time, L_b is banner length, and N_{pax} is passenger count.

The results showed that improving speed (reducing T_{M2}) is energetically very expensive because higher speeds require operating at significantly increased drag and electrical power. In contrast, adding banner length or passengers represents a more energy-efficient way to gain points, as their primary impact comes through linear increases in weight and parasite drag rather than the velocity-cubed penalty. GM time reduction, while highly influential to the overall score, carries a negligible in-flight energy cost. Its optimization is therefore a function of mechanical design and crew procedure rather than propulsion energy allocation.

3.4.3 Integrated Design Space Exploration

An energy-constrained, brute-force design space exploration was implemented to validate these trends and quantify optimal configurations. The model iterated over combinations of passenger count, M2 flight time, banner length, M3 lap count, and GM time, enforcing the 100 W-hr limits. Each point was converted to a representative aircraft using conservative efficiency assumptions and scored directly. The resulting design space (Figure 3.3) shows total score as a function of payload and performance. Three representative configurations are summarized in Table 3.3.

Balanced configuration achieves the highest overall score by optimally trading moderate passenger count, competitive lap time, long banner length and efficient ground operations. The M2-Focused point prioritizes passenger revenue but sacrifices banner length and ground speed, falling short in total score. The M3-Focused point maximizes banner exposure but carries fewer passengers, also yielding a lower aggregate result.

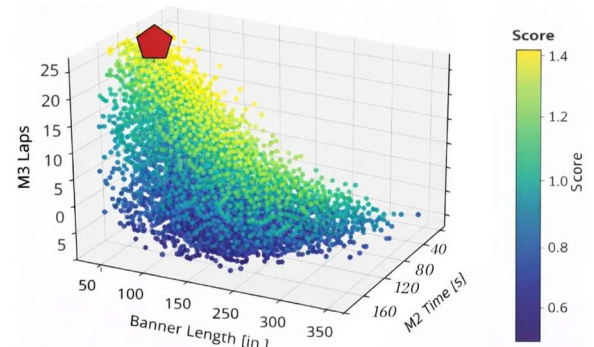


Figure 3.3 Energy-constrained design space results

Table 3.3 Representative Energy-Constrained Configurations

Focus	N_{pax}	$T_{M2}(s)$	$L_b(in)$	N_{laps}	$T_{GM}(s)$	Mission Score
Balanced	24	67.5	80	24	58	1.45
M2-Focused	36	61.0	60	18	75	1.32
M3-Focused	18	72.0	120	26	52	1.41

This confirms that a balanced design, neither overly specialized for a single mission nor too conservative in any parameter, maximizes the total score within the fixed energy budget.

3.4.4 Derived Design Strategy

Derived from the parametric sensitivity, energy-impact, and integrated design space analyses, a clear design-strategy hierarchy emerges for allocating the 100 W-hr energy budget and maximizing total competition score.

Lap time is the most sensitive parameter for scoring, but energy-impact analysis reveals that, it is also the most energy-intensive to improve. Therefore, the design focuses on maximizing aerodynamic and propulsive efficiency to achieve a competitive lap time without requiring prohibitive power increases. These can be achieved through careful motor-propeller matching and minimizing airframe drag via streamlined shaping and smooth surface finishes. The full 100 W-hr battery capacity should be utilized ($EF \approx 1.0$), accepting the associated per-lap operating-cost penalty in exchange for the substantially greater revenue gained from additional laps. Wingspan should be maximized to the full 5-ft limit. A larger span reduces induced drag, which dominates during climb and turning flight, more than the resulting increase in RAC penalizes the M3 score. This yields a net performance benefit across all flight missions. Finally, GM time must be minimized through rapid-access payload bays and a quick-deploy banner mechanism. Since GM time scales the GM score directly and imposes negligible in-flight energy cost, its reduction is a pure benefit, demanding careful attention to ergonomic and mechanical design.

Thus, 24 passengers and 8 Cargo units were selected as it provides high revenue potential without pushing the aircraft into a regime where added weight severely degrades lap time. The sensitivity analysis confirmed this as the point where the marginal energy cost of adding more payload begins to outweigh the scoring benefit. In the analysis, 80-inch length was seemed optimum. However, 60-inch length was selected, as stowage becomes challenging with longer banners, in addition to the added drag and wind sensitivity, as the banner must be stored at aircraft exterior.

This baseline, 24 passengers + 8 cargo units for M2 and 60-inch banner for M3, strategically balances the highest-scoring region from the design space exploration with practical considerations for contest-day weather reliability, drag margin and operational simplicity.

3.5 Configuration Selection

Following the scoring sensitivity analysis, a structured down-select process was implemented to determine the primary aircraft configuration. This process translated mission requirements into weighted Figures of Merit (FoMs), prioritized according to their influence on total competition score.

3.5.1 Methodology and Key FoM Parameters

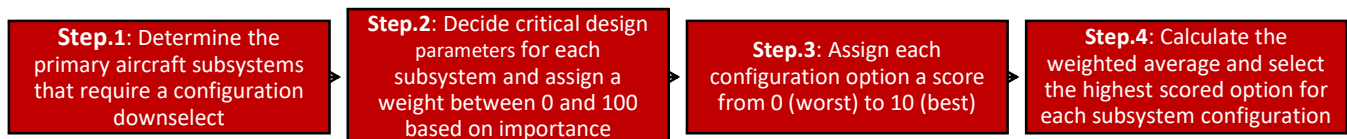


Figure 3.4 Configuration Downselect Process

Candidate configurations were evaluated objectively against these FoMs, ensuring the selection of an optimal design aligned with scoring strategy. Prior to conducting the FoM analysis, a systematic configuration-selection workflow was defined (Figure 3.4) and key performance parameters and their rationales were established (Table 3.4).

Table 3.4 FoM parameters and reasoning

FoM Category (Explanation)	FoM Parameters	Primary Score Driver	Mission Impact
Aerodynamic Performance (Parameters governing time-per-lap and energy efficiency during cruise, climb, and turning flight.)	Aerodynamic Drag / Efficiency	Lap Time, EF	M2, M3
	Lap Time Efficiency (Thrust–Drag)	Lap Count	M2, M3
	Flight Speed	Lap Count	M2, M3
	High-Lift / Payload Capability	Payload Revenue, Takeoff Margin	M2
	Maneuverability (Turn Efficiency)	Lap Time	M2, M3
	Wake Interaction	Stability, Control Authority	All
Stability & CG Management (Parameters ensuring predictable handling and minimal trim penalties across payload states.)	Banner-Mount Interference	Mission Reliability	M3
	Aerodynamic Stability / Controllability	Lap Consistency	All
	Longitudinal CG Shift	Trim Drag, Stability	M2
	Lateral CG Shift	Roll Stability	M2
	Stability on Ground / Handling	Mission Completion	All
Weight & Structural Efficiency (Parameters controlling mass growth, structural robustness, and load transfer efficiency.)	Weight / Structural Weight Efficiency	Payload Revenue, EF	M2, M3
	Structural Integrity / Restraint Efficiency	Mission Validity	All
	Structural Simplicity	Reliability, Turnaround	All
Payload & Volume Optimization (Parameters governing internal packaging efficiency and achievable revenue configuration.)	Payload Capability	Revenue Score	M2
	Internal Volume Utilization	Payload Packing Efficiency	M2
	Integration w/ Wing & Payload	GM Efficiency	GM, M2
Operational & Ground Mission Efficiency (Parameters directly affecting staging time, turnaround speed, and contest-day workflow.)	Payload Accessibility	GM Score	GM
	Ease of Loading/Unloading	GM Score	GM
	Assembly & Maintenance	Turnaround Efficiency	All
	Ground Clearance	Mission Reliability	All
Mechanism & Deployment Performance (Parameters ensuring reliable and repeatable banner deployment and release operations.)	Reliability (No Premature Release)	Mission Validity	M3
	Operational Repeatability	Score Protection	M3
	Actuation Speed	Time Efficiency	M3
Manufacturing & Build Practicality (Parameters ensuring repeatable fabrication quality and dimensional control)	Manufacturing Simplicity / Manufacturability	Performance Consistency	All

3.5.2 Wing Configuration

Table 3.5 Wing Configuration Trade-off

				
Parameters	Weight (%)	Monoplane	Bi-plane	BWB
Aerodynamic Efficiency	30	9	6	8
Payload Accessibility	25	8	5	4
High-Lift / Payload Capability	20	7	8	9
Stability	15	9	4	5
Manufacture and Assembly	10	9	7	2
Total Weighted Score	100	8.35	5.95	6.15

The monoplane has superior aerodynamic efficiency and is simple to construct. While alternative configurations offer some advantages in payload capability, they are far more troublesome to construct, leading to the selection of monoplane.

Table 3.6 Wing Vertical Position Trade-Off

				
Parameters	Weight (%)	High-Wing	Mid-Wing	Low-Wing
Lap Time	30	6	7	9
Payload Accessibility	25	3	5	10
High-Lift / Payload Capability	20	8	7	6
Stability	15	9	5	4
Manufacture and Assembly	10	8	5	7
Total Weighted Score	100	6.30	6.10	7.7

The wing configuration presented a classic engineering trade-off. The high-wing offered superior inherent stability but obstructed critical top-access to the payload bay. Conversely, the low-wing, while sacrificing some passive stability, provided the unimpeded top payload access essential for rapid loading and unloading during GM. The mid-wing option, compromising both stability and accessibility, was rejected due to its complex structural integration and inefficient use of internal volume. Therefore, the low-wing configuration was selected, with its stability deficit to be addressed through deliberate aerodynamic design [2].

3.5.3 Banner Subsystems

Table 3.7 Banner Subsystem Trade-off

				
Parameters	Weight (%)	Claw Mechanism	Rubber Band Release	Pivot-Rod Trigger Mechanism
Reliability (no premature release)	35	9	5	8
Actuation Speed	20	8	9	9
Structural Weight	15	7	9	6
Operational Repeatability	20	9	6	8
Manufacturability	10	8	9	8
Total Weighted Score	100	8.4	7.0	7.9

The claw mechanism ensures the banner remains securely stowed until intentional deployment, a critical mission-success factor. While alternative designs scored well in actuation speed or weight, the claw's robust performance in reliability and repeatability justified its selection to ensure consistent, controlled deployment and release during M3.

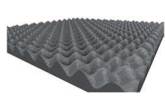
Table 3.8 Banner Installation Location

				
Parameters	Weight (%)	Above Fuselage storage	Underwing Mount	Tail-Mounted Module
Aerodynamic Drag	30	8	7	9
Deployment Reliability	25	3	8	8
Ground Clearance	20	9	7	8
Structural Simplicity	15	7	6	9
Manufacturability	10	8	7	8
Total Weighted Score	100	6.8	7.1	8.4

The tail-mounted module sits within the fuselage wake region, minimizing additional frontal area and interference drag compared to an above-fuselage or underwing installation. Its deployment has reduced exposure to prop-wash and ground disturbances. Direct attachment to the aft fuselage enables efficient load transfer with minimal structural reinforcement, resulting in a mechanically simple and manufacturable solution compared to the alternative configurations.

3.5.4 Passenger Restraint Systems

Table 3.9 Passenger Subsystem Trade-off

				
Parameters	Weight (%)	Foamed Cutouts	Velcro Straps	Top Compression Plate
Restraint Efficiency	35	8	6	9
Ease of Loading/Unloading	20	6	7	9
Weight	15	6	9	7
Structural Integrity	20	8	6	9
Manufacturability	10	5	9	8
Total Weighted Score	100	7	6.95	8.6

The top compression plate provides the most secure in-flight payload retention and the fastest ground operation, directly optimizing both flight safety and GM score. While alternative systems scored better in weight or manufacturability, the compression plate's superior performance in the mission-critical areas of security and speed justified its selection as the optimal solution.

3.5.5 Propulsion System Configuration

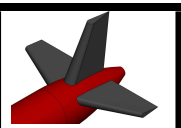
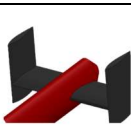
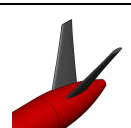
Table 3.10 Propulsion Layout Trade-off

				
Parameters	Weight (%)	Tractor	Pusher	Double Tractor
Lap Time Efficiency (Thrust and Drag)	35	9	5	7
System Weight	25	8	8	6
Payload and Banner Integration	20	9	6	5
Accessibility	15	9	4	7
Handling	5	8	8	6
Total Weighted Score	100	8.7	5.95	6.30

Tractor configuration allows clean propeller inflow and minimal fuselage wake interaction. It also offers unobstructed rear fuselage access for banner integration, critical for M3 and GM. Pusher systems suffer from propeller wake ingestion and reduced static thrust, while double-tractor designs add complexity, weight, and accessibility challenges without proportional gains in lap-time performance. Therefore, the conventional tractor configuration was selected for its balance of aerodynamic efficiency, operational simplicity and mission-specific integration advantages.

3.5.6 Tail Configuration

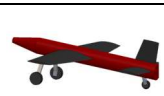
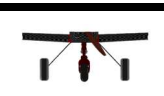
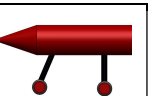
Table 3.11 Tail Configuration Trade-off

					
Parameters	Weight (%)	Conventional	T-Tail	H-Tail	V-Tail
Aerodynamic Stability	30	8	9	7	6
Drag	15	8	7	7	6
Wake Interaction	15	9	8	7	7
Assembly and Maintenance	20	9	6	5	4
Banner-Mount Interference	20	8	9	8	6
Total Weighted Score	100	8.35	7.95	6.8	5.75

The trade-off largely centered on T-tail versus conventional tail configurations, as alternative tail designs presented significant aerodynamic and construction complexities. T-tail offers superior aerodynamic stability and minimal interference with banner-tow operations in M3. However, it introduces a tail-heavy weight distribution, challenging longitudinal stability and requires additional structural reinforcement, increasing overall aircraft mass. In contrast, the conventional tail provides proven aerodynamic stability, simpler construction and lower weight, leading to its selection for this design.

3.5.7 Landing Gear Configuration

Table 3.12 Landing Gear Trade-off

				
Parameters	Weight (%)	Tail-Dragger	Tri-cycle	Bi-cycle
Ground Clearance	35	9	8	7
Stability on Ground	25	6	9	5
Payload Accessibility	25	7	9	6
Banner-Mount Interference	10	7	9	8
Assembly and Maintenance	15	7	8	6
Total Weighted Score	100	8.15	9.4	6.9

The tricycle arrangement provides inherent ground stability during taxi, takeoff, and landing. It also keeps fuselage level during GM. A tail-dragger offers slightly better ground clearance, but it has inherent directional instability and loading

passenger/cargo is difficult. The tricycle arrangement provides a stable, level platform for efficient payload handling and minimizes interference with the aft-mounted banner system, directly supporting the strategic priorities of operational speed and mission reliability.

3.5.8 Cargo Configuration

Table 3.13 Cargo Arrangement Trade-off

					
Parameters	Weight (%)	In-Line	Stacked (Vertical)	Side-by-Side (2x2)	Array (2x2)
Longitudinal CG Shift	30	9	5	7	3
Lateral CG Shift	25	9	8	4	2
Loading/Unload Speed	30	8	5	6	4
Fuselage Volume Efficiency	15	7	9	8	9
Total Weighted Score	100	8.40	6.35	6.10	3.95

The cargo along the fuselage centerline minimizes longitudinal and lateral CG shifts, keeping the center of gravity near the low wing's aerodynamic center for consistent flight stability. This layout also allows for rapid, linear loading/unloading, critical for achieving a fast GM time.

3.5.9 Fuselage Configuration

Table 3.14 Fuselage Shape Trade-off

				
Parameters	Weight (%)	Square	Circular	Elliptical
Manufacturing Simplicity	25	9	6	6
Aerodynamic Drag	20	6	9	8
Internal Volume Utilization	30	8	7	5
Structural Weight Efficiency	15	7	5	7
Stability	10	6	6	5
Integration w/ Wing & Payload	15	9	5	6
Total Weighted Score	100	8.85	7.50	7.05

The square fuselage allows maximum utilization of internal volume for passenger and cargo, direct integration with wing and payload systems and is simple to manufacture. Although circular and elliptical sections offer lower aerodynamic drag, they have complicated internal layout and structural assembly, leading to the selection of square-shaped fuselage.

3.5.10 Configuration Summary

The monoplane with low-wing layout balances aerodynamic efficiency for a competitive lap time with direct top bay accessibility. A conventional tail and tricycle landing gear ensure stable flight and ground handling, simplifying pilot operations and accelerating mission turnover.

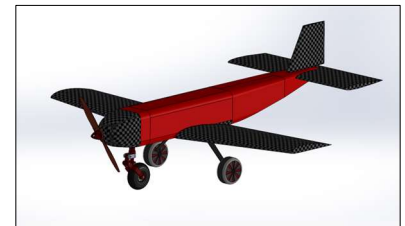


Figure 3.5 Conceptual Visualization

For payload integration, the design accommodates 24 passengers and 8 cargo pucks in dedicated compartments. A square fuselage cross-section maximizes internal volume and manufacturability, while an in-line cargo layout aids in maintaining a controllable center of gravity. Passengers are secured with a quick-acting top compression plate for efficient GM execution.

Propulsion system is a tractor configuration for optimal thrust efficiency, directly supporting the high-lap-count objective. The 60-inch banner is stowed externally in a tail-mounted module to minimize drag in flight. Deployment and release are handled by a robust, mechanically simple claw mechanism, ensuring reliable performance.

4 PRELIMINARY DESIGN

The preliminary design phase transforms the scoring-optimized strategy into a quantitatively sized and performance-validated aircraft. This section formalizes the analytical methodology, mission modeling framework, and subsystem sizing required to meet the mission requirements. Iterative trade studies were conducted to converge on an aerodynamically efficient, structurally feasible, and operationally reliable configuration that maximizes total competition score. The resulting design establishes a validated performance baseline prior to detailed design, manufacturing, and flight test implementation.

4.1 Design Methodology

The aircraft development followed a structured, iterative systems-engineering approach (Figure 4.1) starting from conceptual definition to validated prototype. The conceptual phase translated mission rules and scoring sensitivity into measurable design targets, which were then refined during preliminary sizing through constraint analysis, propulsion-aerodynamic matching, stability evaluation, and performance modeling. Each subsystem was analyzed independently before integration to ensure compliance with mission and energy limits. Following detailed CAD layout and structural verification, manufacturing processes were selected and prototype modules were fabricated and bench-tested (aerodynamic, propulsion, structural, and avionics validation). At each stage, performance checkpoints determined whether the configuration met requirements; deficiencies triggered targeted modifications before proceeding. This closed-loop validation process ensured that the finalized configuration entering full assembly was analytically justified, experimentally verified, and competition-ready.

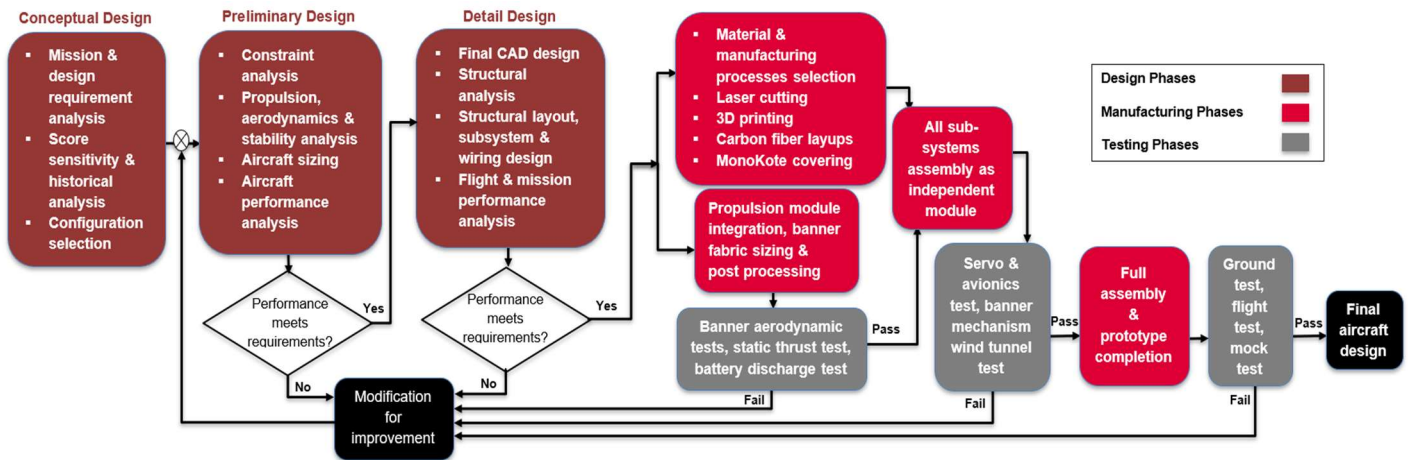


Figure 4.1 Design Methodology Flowchart

4.2 Mission Model

A MATLAB-based framework was developed to simulate the full mission profile, which integrates aerodynamic, propulsion, structural, and battery models to predict aircraft sizing, performance and evaluate design feasibility. It models each flight phase: takeoff, climb, cruise, and coordinated turns using:

- Aerodynamic data from XFLR5 and AVL for lift, drag, and stability.
- Propulsion inputs from eCalc/MotoCalc for motor efficiency, propeller thrust, and power consumption.
- Battery energy usage based on current draw and discharge limits.

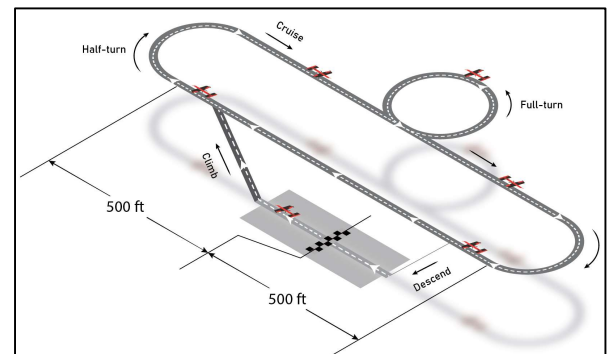


Figure 4.2 Flight Course Model

- Banner drag modeling based on experimental data for towing in vertical orientation.
- Uncertainty propagation to account for model limitations and environmental variability.

Table 4.1 Mission Model Segment Breakdown

Mission Phase	Modeled Parameters	Key Outputs
Takeoff	Ground roll distance, rotation speed, thrust margin	Take-off distance, energy used
Climb	Rate of climb, climb angle, power setting	Altitude gain, energy consumption
Straight Segment	Cruise speed, lift coefficient, drag polar	Lap time, power draw
Coordinated Turn	Turn radius, bank angle, load factor	Turn time, structural load, energy penalty
Banner Tow (M3)	Additional drag, deployment/release timing	Speed reduction, lap count impact
Landing	Approach speed	Mission completion, battery reserve

4.3 Weight Estimation

Aircraft weight was estimated using historical MIST DBF data and preliminary component specifications. The estimated battery weight is approximately 1.1 lb. For M2, which represents the highest energy-demand mission, the propulsion system was sized to comply with the 100 W-h battery energy limit. The payload of 24 passengers and 8 cargo pucks targeted in section 3.4.4 weighs approximately 4.05 lb, totaling 5.15 lb of combined payload and battery weight. Based on prior team experience and historical empty-weight fractions of successful DBF aircrafts, an empty weight fraction of 0.60 was assumed for the preliminary design, resulting in a maximum takeoff weight of 12.87 lb for M2. Based on these results and constraint-based sizing, the target design parameters for the aircraft are summarized in Table 4.2. These preliminary weight values will be refined during detailed design but provide a feasible baseline for aerodynamic, structural and propulsion sizing in subsequent sections.

Table 4.2 Target Design Parameters

Parameters	Target Value	Notes
MTOW (M2)	12.87 lb	Includes 24 pax + 8 cargo + battery
Empty Flyable Weight	7.72 lb	Airframe without battery or payload
Payload Capacity	4.05 lb	24 passengers + 8 cargo pucks
Battery Weight	1.1 lb	Tiger 4500 mAh 6S LiPo
Propulsion System Power	1800 W	Peak power at full throttle

4.4 Wing Sizing

A comprehensive trade study was conducted to optimize the wing geometry within the competition's 5-ft wingspan constraint. Aspect ratios (AR) from 3.0 to 7.0 and wing areas (S) from 3.5 to 9.0 ft² were evaluated using a weighted scoring model that balanced aerodynamic efficiency, structural mass, wing loading and Reynolds-number effects. As depicted in Figure 4.3, the analysis identified a theoretical performance optimum at AR = 5.3 and S = 5.2 ft², achieving a normalized score of 100.0.

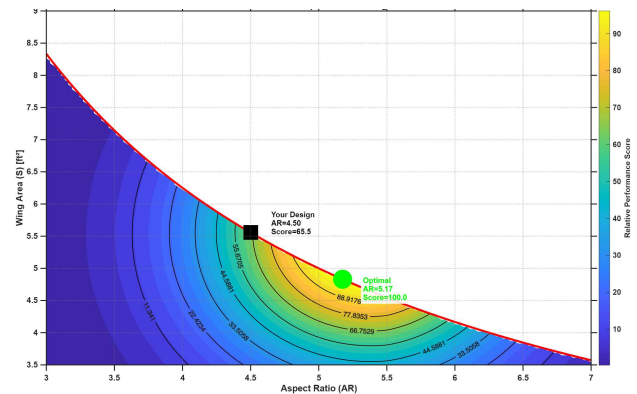


Figure 4.3 Wing Geometry Design Space

The selected configuration, AR = 4.5 and S = 5.53 ft², scores 93.5, representing a deliberate engineering compromise that prioritizes structural simplicity, manufacturing ease and mission reliability over peak theoretical performance. While higher AR (5.0-6.0) offer superior aerodynamic efficiency, they introduce significant structural complexity. The selected lower AR (4.5) reduces wing-root bending moment, simplifying structural design and improving durability. The slightly larger wing area (5.53 ft²) increases low-speed lift margin, enhancing controllability during banner-tow (M3) and high-payload takeoffs (M2). The configuration remains aerodynamically efficient, with an estimated impact of $\leq 6.5\%$ on theoretical performance score, while significantly improving constructability and flight-test robustness. The moderate aspect ratio simplifies construction and alignment, reduces manufacturing complexity, and provides more forgiving flight characteristics.

4.5 Aerodynamic Analysis

4.5.1 Airfoil Selection

For the wing, four candidate airfoils (NACA 2412, Clark Y, FX63-137, SD7062) were analyzed using XFLR5 [3] at Reynolds numbers corresponding to M2 ($Re \approx 300,000$) and M3 ($Re \approx 245,000$). Lift-to-drag polars revealed distinct performance profiles. The FX63-137 and SD7062 exhibited the highest $C_{L_{max}}$ and wider low-drag buckets, offering the greatest peak aerodynamic efficiency and tolerance for angle-of-attack variation. Clark Y demonstrated a more moderate $C_{L_{max}}$ and a narrower but still favorable low-drag region, while the NACA 2412 provided stable but conservative performance across all metrics [4].

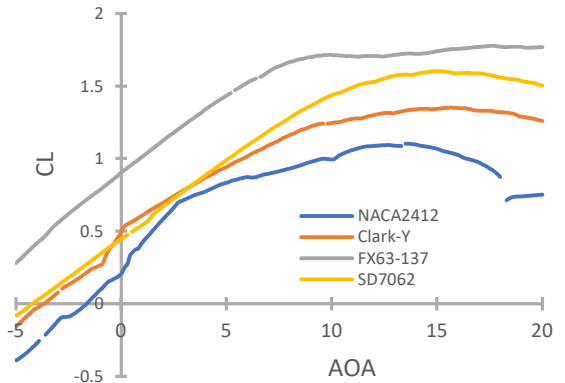


Figure 4.4 CL vs AOA graphs

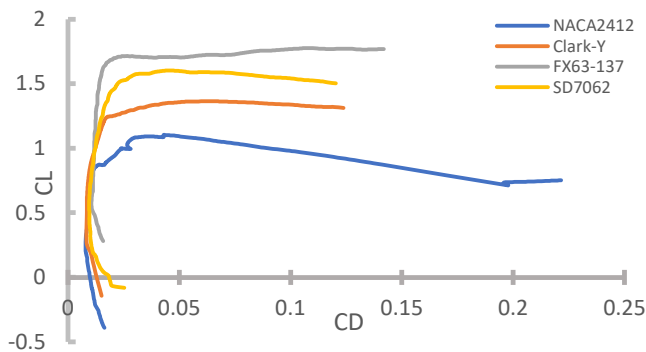


Figure 4.6 Drag Polars of airfoils (M2)

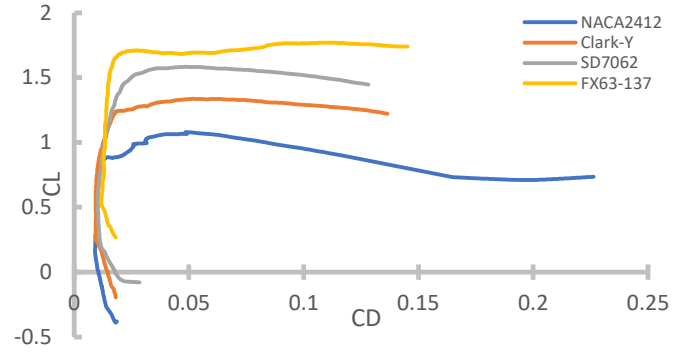


Figure 4.5 Drag Polars for airfoils (M3)

While the FX63-137 and SD7062 offered superior theoretical performance, their fully under-cambered geometries are difficult to manufacture as they are highly sensitive to construction inaccuracies; minor deviations in rib shape or covering tension can degrade their designed laminar flow and stall characteristics. The Clark Y, with its flat-bottom lower surface, provides a stable construction datum, ensuring consistent wing incidence and twist are reliably built into every rib. Furthermore, its 11.7% thickness provides ample volume for a long, lightweight spar, providing a simpler, more predictable load path than thinner airfoils. The Clark Y's gentle, trailing-edge-initiated stall also offers a critical safety margin during the low-speed, high-load-factor turns. Consequently, the Clark Y was selected for its optimal balance of adequate aerodynamic efficiency, structural integration and manufacturable reliability.

The NACA 0009 airfoil was selected for the horizontal and vertical tails due to its proven performance and predictable behavior in prior implementations within our team. It being symmetric provides neutral pitching characteristics and linear control response, which are desirable for stability and trim. It is also a widely adopted because of its reliable low-Re performance and well-documented aerodynamic data for validation [5].

4.5.2 Lifting Surface Analysis

The aircraft lifting surfaces were analyzed using Athena Vortex Lattice (AVL) [6], which models lifting surfaces as bound vortex sheets to compute lift coefficient, induced drag, span efficiency, and trim deflections under steady flight conditions. AVL is appropriate for preliminary design trade studies where relative aerodynamic trends are of primary interest.

Manufacturability strongly influenced planform selection. While purely rectangular wings are easiest to fabricate, moderate taper can significantly improve span efficiency, particularly important under the 60-in wingspan constraint, which results in

relatively low aspect ratio and increased sensitivity to spanwise lift distribution. High lift capability is required during takeoff, turning flight, and banner tow operations.

The wing planform was parameterized using taper ratio $\lambda = c_t/c_r$ and taper start location y_1 , defined as the spanwise location beyond which the wing begins to taper. Wingspan was fixed at 60 in, and wing area was held constant at 800 in² based on preliminary sizing. For each configuration, the wing was modeled with a rectangular inboard section extending to y_1 , followed by a linearly tapered outboard section. All other geometry (empennage, fuselage, airfoils, reference dimensions) was kept constant to isolate planform effects.

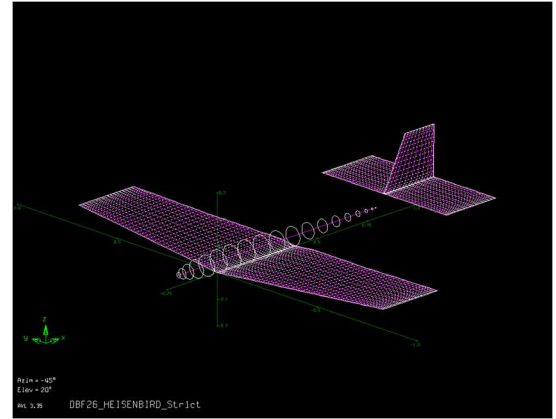


Figure 4.7 Model of 'HeisenBird' in AVL

Nine discrete configurations were evaluated, corresponding to combinations of three taper ratios and three taper start locations, at a fixed near-stall angle of attack of $\alpha = 12^\circ$, and the resulting total lift coefficient was extracted from the AVL solution. Since AVL is an inviscid solver, these results are interpreted as relative lift capability trends rather than absolute stall predictions.

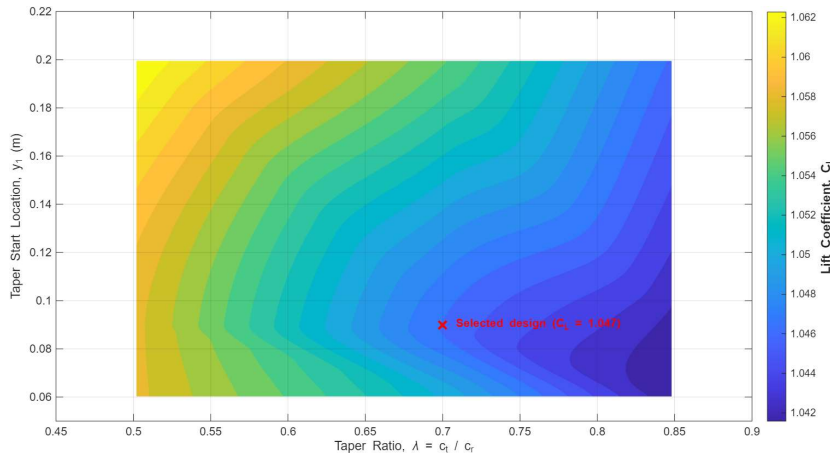


Figure 4.8 AVL-based parametric contour of wing lift coefficient as a function of taper ratio and taper start location. The selected design point is indicated

coefficient within approximately 2% of the maximum observed value, while maintaining simple geometry and favorable structural characteristics. The selected design point is highlighted in and corresponds to an actual AVL evaluation.

Following planform selection, horizontal and vertical tails were sized in AVL to satisfy static stability and trim requirements. The final empennage configuration was verified near stall conditions with the aircraft trimmed using elevator deflection. Overall, the lifting surface analysis confirms that the selected planform provides a balanced combination of high lift capability, favorable spanwise load distribution, manufacturability, and compatibility with the aircraft's stability and control requirements.

4.5.3 Drag Analysis

A component-based drag buildup method was used to estimate aircraft drag during cruise for each mission. Aerodynamic coefficients from XFLR5 with three-dimensional corrections were combined with induced drag from AVL and experimentally measured banner drag data to quantify the contribution of individual aircraft components (wing profile, wing induced, fuselage, empennage, landing gear, payload, and interference drag). The resulting drag breakdown is shown in Figure 4.9.

For M1 and M2, the dominant drag component was wing profile drag, reflecting the relatively clean configuration and moderate lift coefficients flown during cruise. Fuselage and empennage drag contributions remained secondary due to the

compact fuselage design and conservative tail sizing.

M3 exhibited a markedly different drag distribution, dominated by banner drag and wing induced drag. Induced drag increased significantly relative to M2 because the aircraft operated at a higher lift coefficient to maintain level flight at a lower cruise speed (approximately one-third of the M2 cruise speed), resulting in a corresponding rise in induced drag. Therefore, banner drag reduction was prioritized as a higher-leverage design variable. Detailed banner drag testing and optimization are discussed in 7.4.

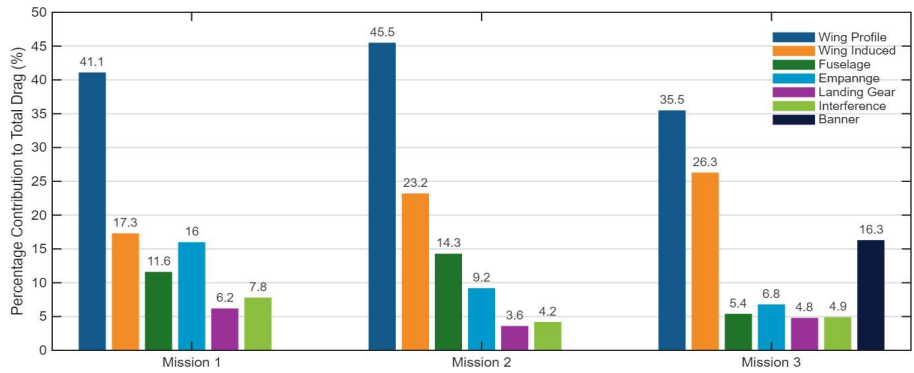


Figure 4.9 Aircraft drag breakdown during cruise for each mission

These drag contributions of individual components from Figure 4.9 were integrated to construct the aircraft's overall drag polar, which was used to generate lift-to-drag ratio (L/D) curves for all missions (Figure 4.10), clearly illustrating the aerodynamic efficiency penalty of banner towing.

4.5.4 Lift-to-Drag Performance

The derived drag polars were used to analyze cruise efficiency across missions. During M2 cruise, the aircraft operated at approximately 87% of its maximum lift-to-drag ratio, indicating near-optimal cruise efficiency. In contrast, M3 exhibited a significantly lower (L/D) due to increased induced drag and banner drag. Banner drag was explicitly included in the (L/D) curve of M3, while the M1 and M2 curves represent clean configurations.

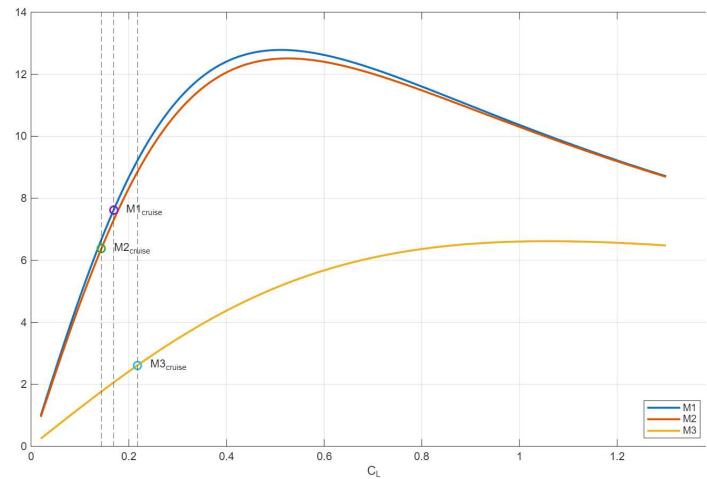


Figure 4.10 L/D as a function of lift coefficient for all missions

The significant separation between the curves highlights the aerodynamic penalty associated with banner towing and reinforces the importance of minimizing banner drag to improve M3 endurance.

4.5.5 Trim and Induced Drag Analysis

AVL was used to perform trim and Trefftz-plane analyses for steady, level cruise in each mission. Mission-specific mass and CG configurations were modeled. A lift-constrained trim approach was used. The required lift coefficient was calculated analytically and imposed as a constraint in AVL, while angle of attack and elevator deflection were allowed to vary freely with zero pitching moment about the CG.

From the trimmed solutions, AVL was used to compute the trim angle of attack (α), elevator deflection (δ_e), induced drag coefficient, and span efficiency factor (e). As AVL is an inviscid solver, parasite drag was estimated separately using the component-based buildup method. The total drag coefficient for each mission was obtained by summing the induced drag from AVL with the

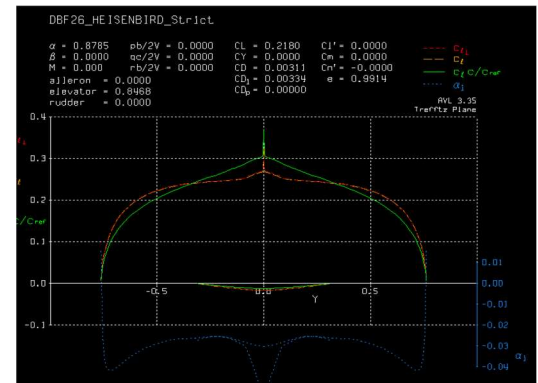


Figure 4.11 Trefftz-plane analysis during M3

analytically estimated parasite drag, including banner drag for M3.

The resulting trim parameters, lift coefficients, and total drag coefficients are summarized in Table 4.3. A representative Trefftz-plane analysis for M3 cruise is shown in Figure 4.11, confirming efficient spanwise lift distribution despite increased induced drag during banner towing.

Table 4.3 Trimmed cruise parameters and aerodynamic coefficients for all missions

Parameters	M1	M2	M3
e	0.9246	0.9513	0.9914
α [°]	0.4164	0.3031	0.8785
δ_e [°]	-0.4172	0.1756	0.8468
C_L	0.1690	0.1664	0.2180
C_D	0.0400	0.0398	0.149 (w/o banner)

4.5.6 Pressure Contour Analysis

A CFD analysis was performed on the HeisenBird using SolidWorks Flow Simulation [7] with an external flow setup, while internal cavities were excluded. Ambient conditions were set to 20°C (68°F) and 101325 Pa to represent typical mid-April weather in Wichita. The freestream velocity was 98.43 ft/s at zero angle of attack. The pressure contour shows a clear low-pressure region on the upper surface of the wing. This is mainly due to the cambered Clark Y airfoil combined with the initial wing incidence. As a result, a pressure difference develops between the upper and lower wing surfaces, which leads to lift generation. The nacelle and fuselage geometry guide the flow smoothly along the body. The flow remains largely attached, which reduces the likelihood of separation. Based on these observations, the team decided to retain the current configuration for further design development.

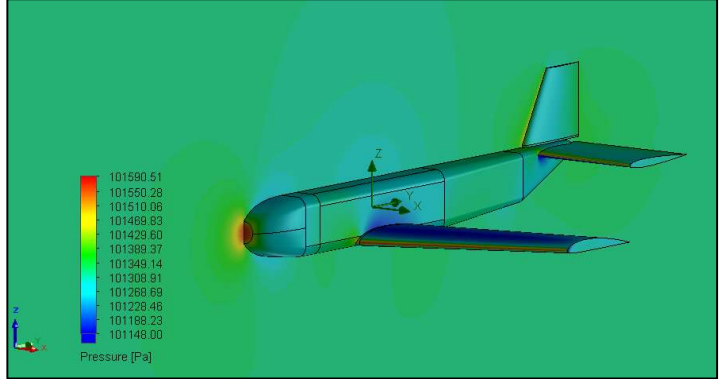


Figure 4.12 Surface pressure distribution over the aircraft

4.6 Stability and Control

4.6.1 Static Stability Analysis

The static stability of the aircraft was analyzed to ensure positive, predictable handling characteristics across all mission configurations. Stability derivatives were computed using AVL for trimmed cruise conditions in Missions 1, 2 and 3.

The aircraft was initially sized for adequate longitudinal and lateral-directional stability using standard tail volume coefficients ($V_H = 0.60, V_V = 0.05$). The CG was placed near the wing's quarter-chord point. This configuration was then iteratively refined using AVL analysis and flight test observations, converging on final horizontal and vertical tail volume coefficients of 0.70 and 0.04, respectively. This refinement resulted in a static margin of approximately 18% for M1 and M2, providing a robust stability baseline.

Table 4.4 Static stability derivatives of the aircraft during cruise for M1–3 obtained using AVL

Stability Derivative	M1	M2	M3
$C_{L_{\alpha}}$	4.4324	4.4328	4.4292
$C_{m_{\alpha}}$	-0.7963	-0.4621	-0.03177
$C_{Y_{\beta}}$	-0.1191	-0.1192	-0.1200
C_{Y_r}	0.1300	0.1259	0.1225
$C_{L_{\delta_e}}$	0.01052	0.010516	0.010513
$C_{m_{\delta_e}}$	-0.02420	-0.02340	-0.02242
C_{L_r}	0.04990	0.04825	0.05511
C_{L_q}	8.3702	7.6705	6.8867
C_{m_q}	-9.7951	-9.0854	-8.3631
$C_{l_{\beta}}$	-0.04350	-0.04359	-0.04943
$C_{n_{\beta}}$	0.05104	0.04905	0.04704
C_{Y_p}	-0.01521	-0.01855	-0.001443
C_{l_p}	-0.3576	-0.3584	-0.3572
C_{n_p}	-0.01593	-0.01536	-0.02057
C_{n_r}	-0.07411	-0.07020	-0.06628

A detailed 3-D model including the wing, horizontal tail, vertical tail and fuselage was used. For each mission, the aircraft mass properties and CG location were adjusted to reflect the appropriate payload and banner configuration. Prior to derivative calculation, the aircraft was trimmed in steady, level cruise by enforcing the required lift coefficient and zero pitching moment about the CG. Once trimmed, AVL's linearized aerodynamic model was used to compute static stability derivatives with respect to small perturbations in angle of attack, sideslip angle, control surface deflections and angular rates. The resulting nondimensional stability derivatives are summarized in Table 4.4. The computed stability derivatives confirm the aircraft is statically stable and controllable across all missions.

The pitching moment derivative C_{m_α} is negative for all missions, confirming positive longitudinal static stability. Although the magnitude of C_{m_α} decreases from M1 to M3 due to higher trimmed angle of attack and increased drag from the banner. Lift curve slope C_{L_α} remains consistent across missions, indicating stable aerodynamic behavior despite changes in loading configuration.

The aircraft exhibits stable lateral-directional characteristics. The negative value of C_{l_β} confirms a positive dihedral effect for spiral stability, while the positive C_{n_β} ensures positive weathercock stability. Increased coupling between axes was observed during M3 as a result of higher angle of attack and banner-induced aerodynamic effects; however, all coupling remained within acceptable limits based on pilot feedback.

Flight testing confirmed that mounting the banner near the aircraft's aerodynamic center minimized its effect on stability and control during M3. Overall, the AVL analysis and flight test observations demonstrate that the aircraft maintains adequate static stability and controllability across all mission configurations.

4.6.2 Dynamic Stability Analysis

The dynamic stability analysis assesses the aircraft's transient response to disturbances during cruise, determining whether oscillations damp out or diverge over time. Using AVL, the system eigenvalues were computed for the trimmed cruise conditions of M2 and M3. Based on the location of eigenvalues in the complex plane, a root locus plot was generated to analyze the five classical dynamic stability modes: roll subsidence, spiral, phugoid, Dutch roll and short-period oscillation. For a mode to be dynamically stable, its eigenvalue must lie in the left half of the complex plane. Key response parameters, including damping ratio (ζ), natural frequency (ω_n), and time constant (τ), were derived and benchmarked against MIL-F-8785C handling-quality standards for Class I (small, lightweight) aircraft [8].

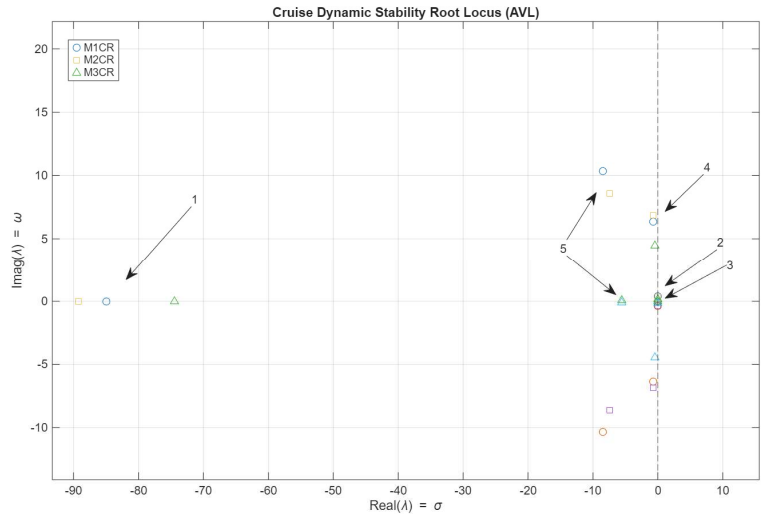


Figure 4.13 Root locus plot of the five dynamic stability modes during cruise for all missions; roll subsidence (1), spiral (2), phugoid (3), Dutch roll (4), and short-period (5)

For each mission, the aircraft was trimmed in steady, wings-level cruise by constraining the lift coefficient and enforcing zero pitching moment about the center of gravity. Linearized stability analysis was then performed using AVL to extract the system eigenvalues in stability axes. Figure 4.13 shows the root locus plot of the dynamic modes during cruise for all missions. All eigenvalues lie in the left half of the complex plane, indicating dynamically stable cruise behavior.

Table 4.5 Dynamic stability parameters of the aircraft during M2 cruise ($V = 108$ ft/s). Values in parentheses indicate the applicable MIL-F-8785C handling-quality requirements

Mode	ζ	ω_n (rad/s)	$\zeta\omega_n$ (rad/s)	τ (s)
Roll	—	—	—	0.013 ($\tau < 1.4$)
Spiral	—	—	0.0181	55.3
Phugoid	0.060 ($\zeta > 0.04$)	0.141	0.00847	—
Dutch Roll	0.097 ($\zeta > 0.08$)	4.47	0.435	—
Short Period	0.999 ($0.35 \leq \zeta \leq 2.00$)	5.61	5.61	—

Dynamic stability parameters, including damping ratio (ζ), natural frequency (ω_n), damped frequency ($\zeta\omega_n$), and time constant (τ), were computed from the AVL eigenvalues and are presented in Table 4.5 and Table 4.6 for M2 and M3 cruise, respectively. The corresponding MIL-F-8785C handling-quality requirements are shown in parentheses for comparison. The

dynamic stability characteristics of M1 were found to be comparable to those of M2 and are therefore not presented separately. Parameters not applicable to a given dynamic mode are intentionally omitted.

Table 4.6 Dynamic stability parameters of the aircraft during M3 cruise ($V = 88$ ft/s). Values in parentheses indicate the applicable MIL-F-8785C handling-quality

Mode	ζ	ω_n (rad/s)	$\zeta \omega_n$ (rad/s)	τ (s)
Roll	–	–	–	0.011 ($\tau < 1.4$)
Spiral	–	–	0.0149	67.2
Phugoid	0.052 ($\zeta > 0.04$)	0.331	0.0171	–
Dutch Roll	0.103 ($\zeta > 0.08$)	6.92	0.714	–
Short Period	0.655 ($0.35 \leq \zeta \leq 2.00$)	11.40	7.45	–

All oscillatory modes satisfy the MIL-F-8785C requirements for a Class I (small and lightweight) aircraft. Although the spiral mode exhibits a relatively long time constant, it remains stable and within acceptable limits for cruise flight. The short-period mode in M3 shows a reduced natural frequency due to the altered mass distribution and trim condition; however, it remains adequately damped and compliant with the specified handling-quality criteria. Overall, the aircraft demonstrates satisfactory dynamic

stability characteristics during cruise for all mission configurations. Overall, the aircraft satisfies the dynamic stability requirements for a Class I aircraft during cruise for all mission configurations.

4.7 Propulsion Package Selection

The propulsion system (motor, ESC, battery, propeller) was designed to meet the most demanding mission performance requirements, specifically, the high takeoff-weight of M2 and the energy-constrained endurance of M3. Sizing followed a performance-driven, iterative process: Establish static thrust and power needs based on takeoff and cruise analysis; Use propeller and motor models to evaluate candidate combinations; Assess the impact of each candidate on predicted lap count and mission score within the 100 W-hr energy constraint.

4.7.1 Static Thrust Requirement

The propulsion system was sized to meet takeoff performance requirements for the most demanding mission configuration, M2, with a takeoff gross weight of 12.87 lb. A static thrust target was established using a first-principles takeoff performance model. For this analysis, a conservative design takeoff field length (TOFL) of 30 ft was assumed. Although there is no takeoff distance limit, 30 ft represents a practical and repeatable operational target.

The reference takeoff velocity was calculated assuming liftoff occurs at 20% above stall speed to ensure reliable, repeatable takeoff performance. The maximum lift coefficient at takeoff $C_{L_{max}} = 1.35$, was obtained from the aerodynamic analysis in 4.5.1. Using the mission takeoff weights, the reference takeoff velocity $V_{T_{01}}$ was computed: 36.97 ft/s for M1, 44.65 ft/s for M2 and 37.83 ft/s for M3. To account for blown lift effects, the induced velocity through the propeller disk V_e was estimated using actuator disk theory: $V_e = \sqrt{\frac{2T}{\rho A V_{T_{01}}}}$. To account for the additional lift generated by the propeller slipstream and to reduce the takeoff velocity requirement, an effective takeoff velocity defined as: $V_{T_0} = V_{T_{01}} - V_e$.

The required static thrust was determined using the effective takeoff velocity and estimated takeoff field length. An initial freestream velocity, V_0 , of 16.4 ft/s was assumed to represent light headwind conditions. A thrust safety margin of 10% was applied to account for modeling uncertainty and operational variability. The required static thrust was calculated: $T_{static} =$

$$\frac{m((V_{T_0})^2 - (V_0)^2)}{2 \cdot \text{TOFL} \cdot F S_{T_0}}$$

Based on this analysis, the required static thrust was determined to be approximately 4.57 lbf for M1, 10.46 lbf for M2, and 5.04 lbf for M3.

4.7.2 Turning Radius

Several methods were investigated to improve the aircraft's turning capability, including the use of slotted flaps, leading-edge slats, vortex generators, and slotted airfoil sections. These devices were evaluated based on their potential to reduce turn radius and improve controllability at high bank angles. After considering the associated penalties, it was determined that the incorporation of slotted flaps would introduce additional weight and manufacturing complexity that would outweigh the achievable gains in turning performance. As a result, the aircraft was designed to rely solely on conventional control surfaces for maneuvering.

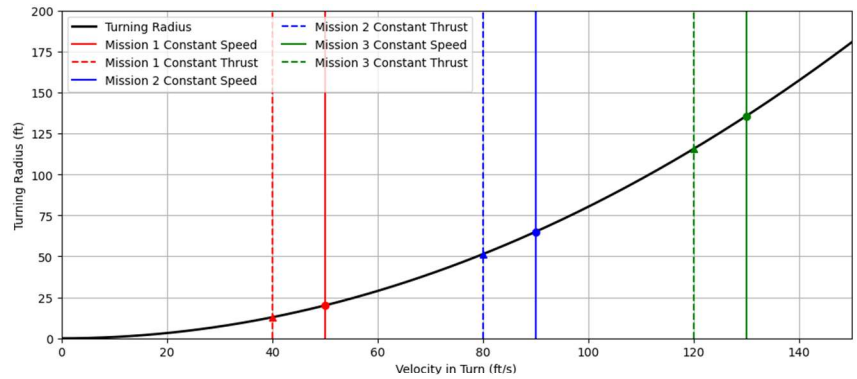


Figure 4.14 Turning Radius for Constant Speed and Thrust

The maximum allowable structural load factor for the aircraft was established as 4g, which corresponds to a maximum bank angle of approximately 75°. This load factor was used to evaluate the aircraft's turning performance under aggressive maneuvering conditions. Using the level coordinated turn relationship, the turning radius was computed as a function of velocity: $R = \frac{v^2}{g \tan \theta} = \frac{v^2}{g \sqrt{n^2 - 1}}$. A plot of turning radius versus velocity was generated for the three mission configurations, as shown in Figure 4.14. Two turning conditions were analyzed:

- Constant speed turns, where thrust is increased during the maneuver to compensate for the rise in induced drag.
- Constant thrust turns, where thrust is held constant and the aircraft experiences a reduction in velocity due to increased induced drag.

The solid curves in Figure 4.14 represent the turning radius assuming constant speed, while the dashed curves correspond to constant thrust operation. Mission-specific velocities for both cases are indicated on the plot.

For all missions, the resulting constant-thrust turning speeds remain above the aircraft stall speed at the selected bank angle. Therefore, there is no risk of stall during the turn under the evaluated conditions. This confirms that the aircraft is capable of safely executing high-load-factor turns using conventional control surfaces alone.

4.7.3 Power Curve

Two propellers, the Falcon 16×7E and Falcon 15×8E [9], were selected to provide sufficient thrust to overcome aerodynamic drag while achieving the desired operating speeds for each mission configuration. Total aircraft drag for all missions was obtained using aerodynamic analysis, after which propeller performance models were developed to evaluate thrust availability as a function of flight speed and motor RPM.

Propeller performance data were organized with respect to airspeed and rotational speed to enable direct comparison between available thrust and mission drag requirements. M2 operating at a higher gross weight due to the onboard electronics payload, requires increased thrust to maintain its target flight condition. As a result, a propeller with relatively higher effective pitch characteristics was required to satisfy this demand. Conversely, M3 experiences increased drag due to the banner, necessitating higher motor RPM to achieve the target airspeed.

The thrust–drag performance models presented in Figure 4.15 illustrate the behavior of the selected propellers across the operating envelope. The Falcon 15×8E propeller corresponds to the M2 configuration, where higher thrust is required, while the Falcon 16×7E propeller is utilized for M1 and M3, where efficient operation across a broader speed range is prioritized.

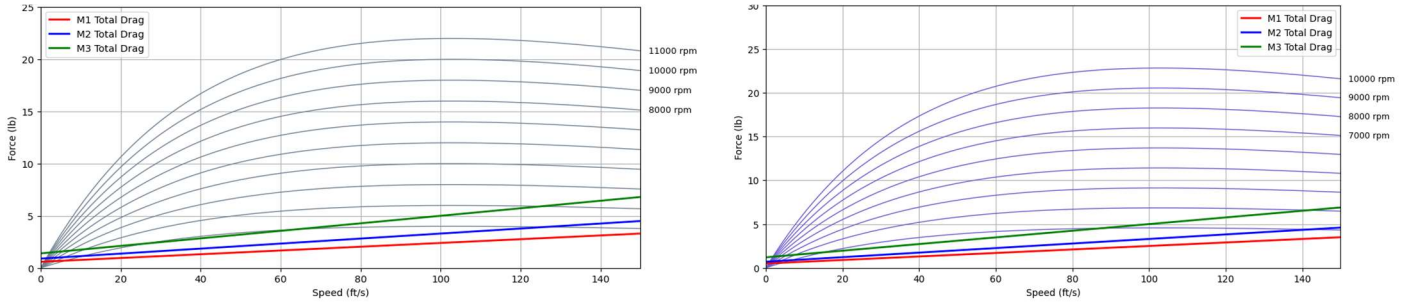


Figure 4.15 Thrust, Drag, and Speed Plots for (left) 16X7E Propeller and (right) 15X8E Propeller

4.7.4 Rate of Climb Performance

Figure 4.16 shows the rate of climb (ROC) for each mission. The ROC was calculated using Raymer [10] using airspeed, V , available thrust, T , drag, D and weight, W : $ROC = V \left(\frac{T-D}{W} \right)$.

M1 and M3 exhibit higher peak ROC compared to M2 due to significantly lower aircraft weight. M2, operating at maximum payload weight, shows reduced excess thrust and consequently lower climb performance. At high speeds, ROC approaches zero for all missions as $T \approx D$, corresponding to the maximum level-flight speed limit.

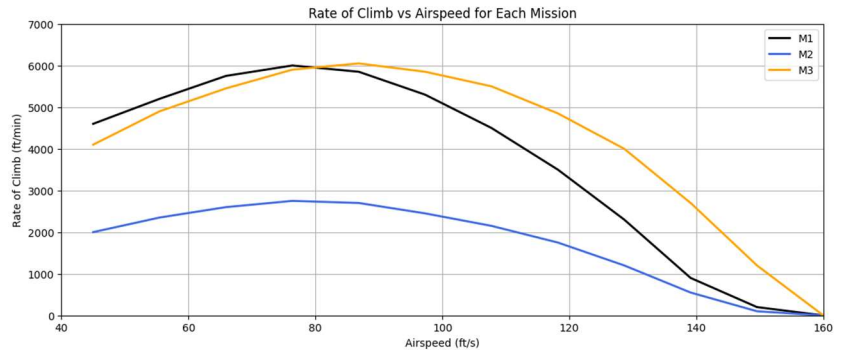


Figure 4.16 Rate of climb for each mission

4.7.5 Motor and Propeller Selection

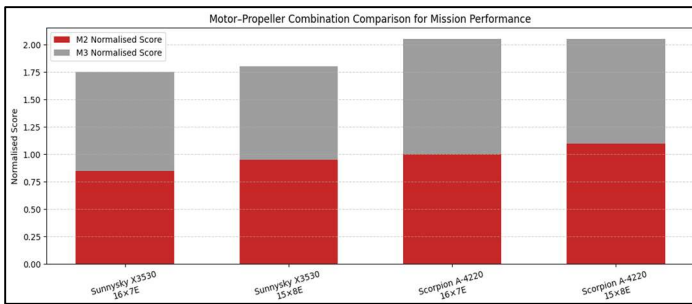


Figure 4.17 Propeller/Motor Combinations vs M2/M3 Scores

A propulsion trade study was conducted to identify a motor–propeller combination capable of meeting the aircraft's mission performance objectives. Several candidate configurations were evaluated using a mission-level performance model that accounted for thrust capability, propulsive efficiency, and their impact on overall mission scoring. M2 favored configurations that could deliver higher dynamic thrust at increased cruise speeds, while M3 emphasized efficient operation at reduced throttle settings to maximize usable flight time. To enable direct comparison, normalized mission performance scores were calculated for each configuration. The results, summarized in Figure 4.17, indicate that the Scorpion A-4220 motor [11] provides the strongest overall performance across both missions. Accordingly, the Falcon 15×8E propeller was selected for M2, while the Falcon 16×7E propeller was selected for M1 and M3.

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4.7.6 Battery Selection

With the propeller and motor configuration established, battery selection was performed to ensure compatibility with the propulsion system across all mission profiles. A lithium-polymer (LiPo) battery was selected due to its high-power density and ability to sustain the discharge rates required for both high-thrust and endurance-focused flight segments. The primary

consideration in battery selection was achieving an appropriate balance between voltage and capacity while remaining within the allowable total energy limit.

The total energy capacity of the battery was calculated using: $\text{Energy}_{\text{wh}} = 1000 \times \text{mAh} \times \text{Voltage}_{\text{nominal}}$, which relates the nominal voltage and rated capacity. A single 4500 mAh LiPo battery was selected for use across all missions, simplifying system integration and reducing overall system complexity. The chosen battery remains within the 100 Wh energy limit while providing sufficient power delivery and endurance margin for Missions 1, 2 and 3.

4.7.7 Electronic and Speed Controller (ESC) Selection

Following battery selection, an appropriate electronic speed controller (ESC) was chosen to ensure reliable power delivery and control of the propulsion system. The HOBBYWING FlyFun 120A 6S V5 ESC was selected due to its current handling capability, voltage compatibility, and proven reliability in high-power electric aircraft applications. The ESC provides sufficient current margin to accommodate peak power demands during takeoff and aggressive maneuvering while maintaining efficient operation during cruise. Its compatibility with the selected motor and battery configuration ensures stable throttle response and robust thermal performance across all mission profiles.

4.7.8 Finalized Propulsion Package

The finalized propulsion package was selected after iterative thrust analysis, current draw estimation, and mission energy requirement calculations. The configuration ensures sufficient static thrust for takeoff at maximum takeoff weight, reliable cruise performance, and safe electrical operation within component limits. Table 4.7 summarizes the selected motor, ESC, battery, propeller options, and fuse system used across all missions.

Table 4.7 Propulsion packages for all missions

Motor	ESC	Battery	Propeller	Fuse
Scorpion 4220 540 kV	HOBBYWING FlyFun 120A 6S V5 ESC [12]	6S 4500 mAh LiPo battery	Falcon 16×7E Propeller (M1, M3) Falcon 15×8E Propeller (M2)	100A MAXI Fuse

4.8 Operational Wind Limits and Gust Disturbance Analysis

Historical wind data for mid-April operating conditions at Wichita Dwight D. Eisenhower National Airport (ICT) were analyzed to evaluate gust disturbance margins and crosswind operability of the aircraft [13]. While the aircraft maintains an average cruise speed of approximately 30 m/s across all missions, low-altitude flight phases, such as, takeoff, landing and turning segments remain susceptible to

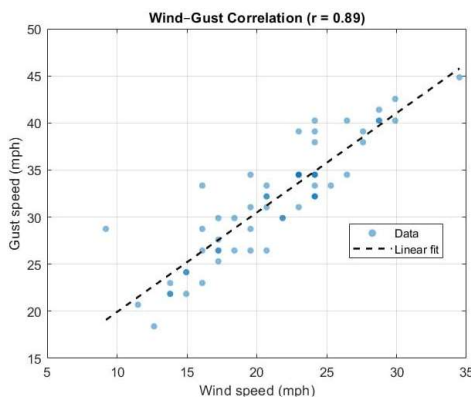


Figure 4.19 Correlation between steady wind speed and gust magnitude

atmospheric disturbances that affect controllability and pilot workload.

Figure 4.18 illustrates the temporal variation of steady wind speed and

associated gusts over representative operating days. The data indicate that although steady wind levels remain moderate for most of the operating window, intermittent gust events introduce transient disturbances that can affect flight-path tracking and handling quality. To ensure consistent controllability and safe operation during these critical phases, a conservative flight operating wind limit of 25 mph was adopted.

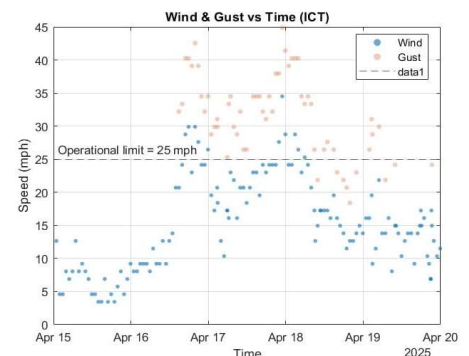


Figure 4.18 Wind and gust speed variation over time at ICT during representative operating conditions

Figure 4.19 shows the strong positive correlation between steady wind speed and gust magnitude (Pearson correlation coefficient, $r = 0.89$), which demonstrates that gust intensity scales predictably with mean wind conditions, supporting the use of a wind-dependent gust increment when defining disturbance margins. Accordingly, control authority and structural margins were evaluated with sufficient allowance to accommodate these predictable gust effects without compromising stability or mission execution.

Overall, the wind environment analysis confirms that the selected operating limits provide adequate robustness against gust-induced disturbances while maintaining safe and repeatable flight performance. This approach establishes clear go/no-go criteria for flight testing and competition operations under realistic environmental conditions.

4.9 Mission Uncertainties

As with all preliminary DBF performance predictions, the mission model is most reliable for relative trade studies but may deviate from field results due to idealized assumptions. Aerodynamic polars from XFLR5/AVL assume a clean, rigid configuration and may underpredict parasite and interference drag from surface finish, landing gear, payload interfaces and banner hardware. Propulsion maps derived from static testing and eCalc/MotoCalc [14, 15] may not fully capture in-flight thermal effects, voltage sag and wiring losses. Environmental variabilities, density altitude, gusts, runway condition and pilot technique in turns, throttle modulation and banner deployment introduce additional scatter in lap time and energy usage.

Despite these uncertainties, the model is considered sufficiently reliable at this stage because it is based on validated aerodynamic methods, calibrated static thrust measurements, conservative performance margins and sensitivity analyses that bound the effect of key variables. Agreement between independent tools (AVL, drag build-up, propulsion modeling) and preliminary ground tests provides directional confidence in predicted trends and mission feasibility. Accordingly, the preliminary results establish a robust baseline from which the subsequent chapters will further refine, validate and update performance predictions through hardware-level verification.

5 DETAILED DESIGN

Building on the selected configuration from the conceptual and preliminary phases, the detailed design stage focused on refining structures and subsystems to translate analytical predictions into a lightweight, robust, and manufacturable aircraft. Structural members, control mechanisms, and mission-specific modules were optimized to improve stiffness-to-weight ratio, aerodynamic efficiency, and operational reliability. This phase extended beyond analytical refinement into fabrication, ground validation, and flight testing, where test feedback directly informed iterative modifications to address manufacturing constraints and improve overall system performance.

5.1 Dimensional Parameters

The dimensional parameters presented in this section represent the finalized geometric configuration derived from the combined outcomes of the conceptual and preliminary design phases. Wing planform selection, tail sizing, stability targets, propulsion integration constraints, and mission-specific packaging requirements were consolidated to define the aircraft's definitive dimensions. The values summarized in the following table reflect the integrated configuration adopted for detail design, fabrication and validation, ensuring consistency with aerodynamic analyses, structural sizing, and competition constraints.

Table 5.1 Final Dimensions and Component Specifications

Wing		Horizontal Tail		Vertical Tail		
Airfoil	ClarkY	Airfoil	NACA0009	Airfoil	NACA0009	
Span (in)	60	Span (in)	29.06	Span (in)	7.88	
Aspect Ratio	4.5	Aspect Ratio	2.5	Aspect Ratio	1.02	
MAC (in)	13.49	MAC (in)	9.8	MAC (in)	7.92	
Planform (Shape)	Tapered	Planform Area (in²)	239.88	Planform Area (in²)	60.96	
Planform Area (in²)	800	Taper Ratio	1	Taper Ratio	0.58	
Taper Ratio	0.7	Incidence Angle (°)	0	Incidence Angle (°)	0	
Incidence Angle (°)	2	Tail Arm (in)	31.50	Tail Arm (in)	31.50	
Leading Edge Sweep Angle (°)	5.89	Control Surfaces & High-Lift Devices				
			Span		Chord	
		Ailerons	15.31		3.54	
		Rudder	7.91		2.99	
Fuselage		Elevator	29.06		2.99	
Total Length (in)	54.91	Propulsion				
Max. Width (in)	6.5	Motor		Propeller		
Max. Height (in)	6.5	Model	Scorpion A-4220 V2	Manufacturer	Falcon	
Tail Length (in)	10.77	Effective kV	540	M1, M2, M3	15X8, 16X7	
Nose Length (in)	9.02	Max. Power (W)	2553	Propulsion Battery		
		Weight (lbs.)	0.5577	Model	Tiger 6S	
				Capacity (mAh)	4500	
		No-Load Current (A)	1.54	Max. Discharge Rating (C)	30	
Landing Gear		Internal Resistance (mΩ)	18.2	Weight Per Pack (lbs.)	1.102	
Overall Height (in)	7.8	ESC				
Track Length (in)	18.75	Model	HOBBYWING FlyFun 120A	Continuous Current (A)	120	
Wheel Base (in)	16.78	Weight (lbs.)	0.205			
Nose Gear Angle (°)	0	Controls				
Transmitter Model	Radiolink AT10 II	Receiver Model	R12DS	Avionics Battery Model	Tiger 1000mAh	
Servo Models	Ailerons	MG996R		Pack Voltage (V)	8.4	
	Rudder			Cell Count	2	
	Elevator			Pack Weight (lbs.)	0.11	
	Nose steer					
	Banner Deploy/Release					

5.2 Structural Characteristics and Capabilities

5.2.1 Flight Envelope: V-n Diagram Analysis

The V-n diagram defines the structural flight envelope as shown in Figure 5.1. The aircraft is designed for a positive limit load factor of +4.0g and a negative limit of -1.6g, with corresponding ultimate loads of +6.0g and -2.4g using a 1.5 safety factor. Stall boundaries were computed using $C_{L_{max}} = 1.27$ and $C_{L_{min}} = -1.0$, yielding a 1g stall speed of 12.5 m/s (41 ft/s) and a maneuvering speed V_A of 25 m/s (82 ft/s). The normal operating range extends to the cruise speed of 34 m/s (112 ft/s), followed by the caution range up to the never-exceed speed V_{NE} of 52 m/s (170 ft/s). A 10 m/s (33 ft/s) vertical gust produces load factor variations of approximately $\pm 0.8g$ at cruise. All mission profiles operate between 27-33 m/s (89-108 ft/s) with expected load factors below 2.0g, remaining well

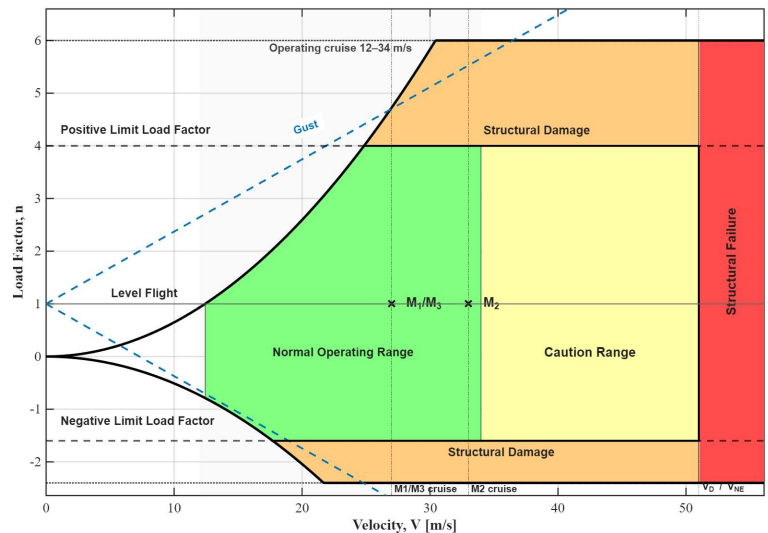


Figure 5.1 V-n Diagram of HeisenBird

at cruise. All mission profiles operate between 27-33 m/s (89-108 ft/s) with expected load factors below 2.0g, remaining well

within the normal envelope. Structural damage occurs between limit and ultimate loads, while failure occurs beyond ultimate values, providing adequate safety margins for competition operations.

5.2.2 Detailed Wing Design

Shear Force and Bending Moment Distribution

The shear force and bending moment distributions were obtained for the semi-span wing modeled as a cantilever beam with the root fixed. An elliptic lift distribution was applied using the aircraft MTOW of 12.87 lb. and a 2.5g limit load factor, with the load distributed over the 2.5 ft semi-span. The SFD shows a maximum shear force of approximately 16 lbf at the wing root, decreasing smoothly to zero at the tip, while the BMD indicates a maximum root bending moment of around -17 lb-ft that similarly reduces to zero at the wingtip. The smooth profiles confirm distributed aerodynamic loading.

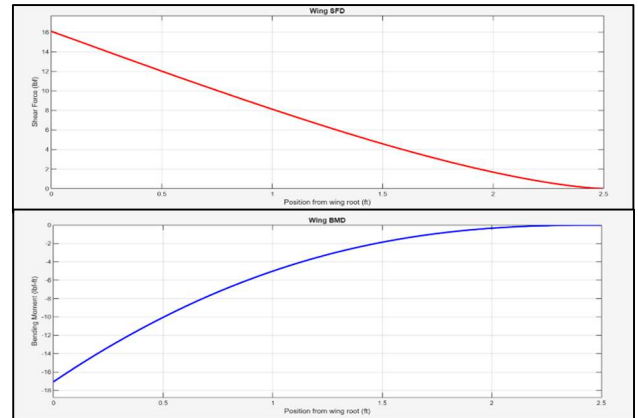


Figure 5.2 SFD & BMD of Wing Under 2.5g MTOW Load

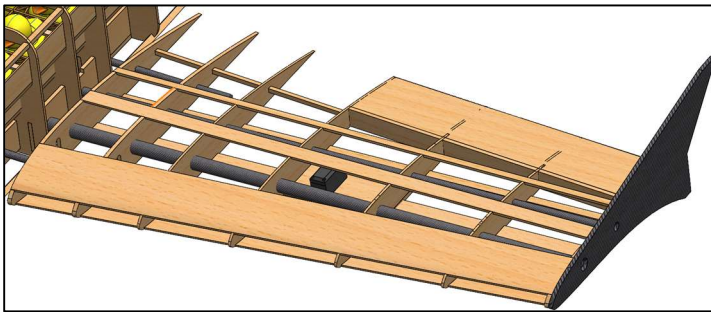


Figure 5.3 Wing Structure

The wing has a half-span of 26.75 in including the winglet, with a root chord of 15.37 in and a tip chord of 10.76 in, resulting in an aspect ratio of 4.5. The winglet thickness is 0.35 in. The primary structure consists of a 0.55 in outer diameter carbon fiber main spar and a 0.39 in outer diameter secondary spar running along the span. Two additional 0.31 in outer diameter carbon spars extend from the fuselage into the wing root for alignment and structural continuity. A built-in 3° dihedral is achieved using two 2.36 in long, 0.63 in outer diameter carbon sleeves installed at an angle inside the wing box, into which the main spar telescopes. The wing is secured from the underside of the fuselage using two M5 bolts engaging embedded M5 T-nuts inside the wing box.

The root rib is fabricated from 0.2 in plywood, while all other ribs are 0.12 in balsa. The leading edge is molded with 0.08 in balsa sheeting, and 0.2 in balsa stringers for longitudinal support. Two winglets are 3D printed with CF-PLA. A 0.2 in plywood servo plate is integrated within the rib structure. The wing surface is covered with Hanger 9 UltraCote heat-shrink film.

Wing Structural Analysis

A static structural study was performed on the wing structure in SolidWorks Simulation to verify its strength under critical maneuver and landing conditions. A 2.5g MTOW load was applied as an elliptically distributed downward load on the ribs, representing the lift distribution during a sharp turn or hard landing, while the wing was constrained at the root. The maximum

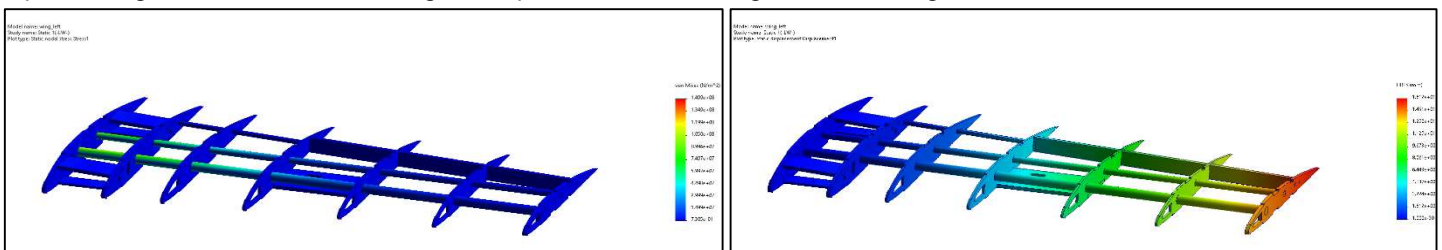


Figure 5.4 FEA of Wing Under 2.5g MTOW Load

von Mises stress in the wing structure was $1.5 \times 10^8 \text{ Pa}$ (21800 psi) at the spar root. Using a conservative allowable strength of 500 MPa (72500 psi) for a typical quasi-isotropic CFRP laminate, the estimated factor of safety is around 3.3 under the 2.5g limit load case. The maximum wingtip deformation was 16 mm (0.63 in), remaining within acceptable limits. These results confirm adequate strength and stiffness margins, ensuring safe maneuvering capability and predictable handling.

Winglet Aerodynamic Analysis

A forward wingtip fence was analyzed using ANSYS Fluent [16] with the $k-\omega$ SST model to assess its effect on wingtip flow and aerodynamic efficiency. The computational domain extended 5 chord lengths upstream, 10 chord lengths downstream, and 5 chord lengths laterally, and AOA was varied by resolving freestream velocity components while keeping the geometry fixed. Velocity magnitude and pressure contours show that the winglet restricts spanwise flow near the wingtip, producing a weaker and more diffused vortex compared to the baseline case, thereby reducing induced drag. The lift coefficient increases nearly linearly at low angles of attack, indicating stable behavior in the normal operating range, with a peak value of $C_L \approx 2.0 \times 10^{-3}$ near 80° , after which nonlinear

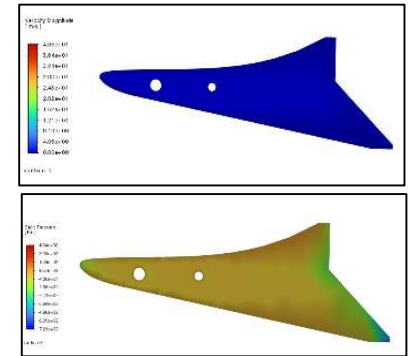


Figure 5.6 Forward Winglet Flow Field: Velocity(top) and Pressure(bottom)

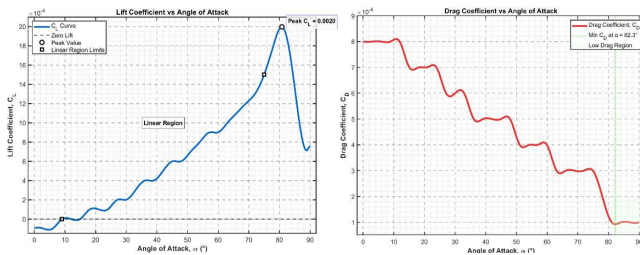


Figure 5.5 Aerodynamic Performance of Winglet Configuration

effects appear. The drag coefficient decreases with increasing angle of attack and reaches a minimum of $C_D \approx 1.0 \times 10^{-4}$ near 82.3° , demonstrating improved aerodynamic efficiency. In addition to reducing induced drag, the winglet provides a beneficial contribution to directional stability, justifying its inclusion in the final design without increasing wingspan or structural complexity [17].

Wingtip Vortex Analysis

Figure 5.7 compares the wingtip flow field without and with a winglet. In the baseline configuration, strong spanwise flow rolls around the wingtip, forming a coherent vortex that persists downstream and contributes to induced drag and reduced lift efficiency. With the winglet installed, the vortex core is weaker and more diffused, indicating reduced spanwise flow and wake rotation. This reduction in vortex strength corresponds to lower induced drag and improved lift efficiency at the wingtip. In addition, the winglet provides a small but beneficial contribution to directional stability. Based on these observed improvements, winglets were selected to enhance overall aerodynamic efficiency and stability without increasing wingspan.

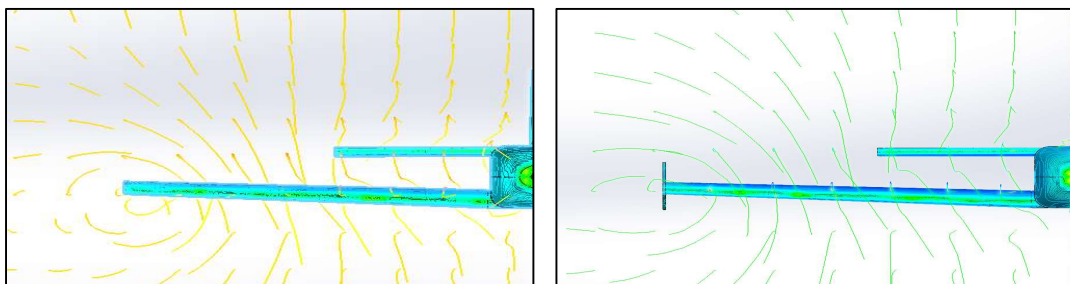


Figure 5.7 Wingtip Vortex Comparison: Without (left), With (right)

Wing Mount Structure

A static structural study was performed on the fuselage wing box in SolidWorks Simulation to verify that it can withstand the loads transferred from the wing spars during high-load cases such as a sharp turn and hard landing. The wing box is built

from reinforced bulkheads and internal support members and serves as the primary structural interface where the wing spars attach to the fuselage. To represent a critical design condition (limit load), a downward force equal to 2.5g MTOW was applied at the spar attachment region. This loading captures the wing-root reaction loads produced when the wing generates increased lift (maneuvering) and when the airframe experiences elevated vertical acceleration at touchdown. The surrounding fuselage attachment faces were constrained to simulate the wing box being supported by the fuselage structure. As shown in Figure 5.8, peak von Mises stresses occur near the spar attachment and adjacent bulkhead/support junctions, where the bending load enters the box and is redistributed through the frames. The maximum stress from the study was approximately $5 \times 10^4 \text{ Pa}$ (7.25 psi), which is far below the material yield strength of $2 \times 10^7 \text{ Pa}$ (2900 psi), confirming a large structural margin. Therefore, the wing box design is sufficiently strong for the 2.5g maneuver/landing load case, and the current bulkhead-supported wing–fuselage integration was retained without additional reinforcement.

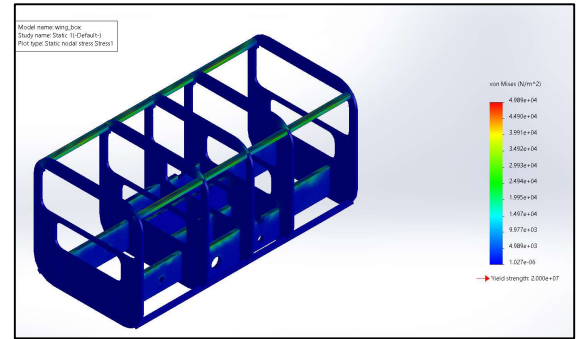


Figure 5.8 Wing Mount FEA Result

5.2.3 Detailed Motor Mount Design

A static structural study was performed on the motor mount using SolidWorks Simulation to evaluate the stress distribution caused by the maximum thrust of 7 N produced by the Scorpion A-4220-540KV V2 motor with a Falcon 16×7 wooden propeller. The motor mount was modeled as a set of 5 mm (0.2 in) balsa bulkheads, which serve as the primary load-bearing elements transferring thrust loads into the fuselage. The thrust force was applied at the motor mounting face in the axial direction, while the bulkhead mounting points were constrained to represent attachment to the fuselage structure. As shown

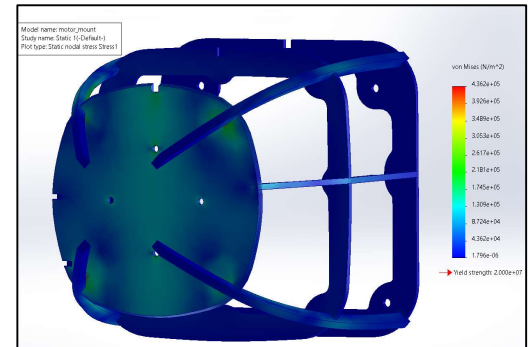


Figure 5.9 Motor Mount FEA Result

in Figure 5.9, the highest von Mises stress occurs around the motor mounting plate and along the bulkhead connections, where thrust loads are introduced and redistributed. The maximum stress was approximately $4.4 \times 10^5 \text{ Pa}$ (64 psi), which is well below the plywood material yield strength. This confirms that the 5 mm (0.2 in) plywood bulkhead motor mount provides sufficient strength and stiffness to safely withstand maximum propulsion loads with a large margin of safety.

5.2.4 Detailed Fuselage Structure Design

The fuselage has an overall length of 54.46 in (nose tip to tail end) and is based on a 6.5 in × 6.5 in square cross-section bulkhead geometry. The structure was constructed using laser-cut 5 mm (0.2 in) plywood bulkheads arranged to define the passenger and cargo compartments. These bulkheads were interconnected with 5 mm (0.2 in) plywood longerons to form a rigid internal frame. After assembly, 2 mm balsa sheeting was molded over the framework to produce a smooth external contour, and the fuselage was covered with Hanger 9 UltraCote heat-shrink film.

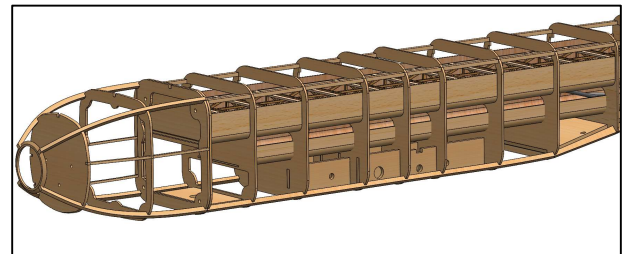


Figure 5.10 Fuselage Structure

The wing mount region was locally reinforced to sustain combined aerodynamic and landing loads. Five closely spaced 5 mm (0.2 in) plywood bulkheads were installed in this section along with multiple 5 mm (0.2 in) plywood support beams

containing spar pass-through holes. The MLG plate was integrated within this reinforced zone so that wing bending loads and landing impact loads are reacted through the same structural cluster. A separate 5 mm (0.2 in) plywood plate was installed in the forward fuselage to support the nose landing gear.

A side-sweep was incorporated aft of the wing for improved airflow, followed by an upsweep to provide rotation clearance during takeoff and to house the banner module. At the tail upsweep, a detachable banner stowing module was integrated for M3 operations. Based on CG calculations, the battery was positioned in the nose section, and an avionics access door was incorporated to provide access to the battery, receiver, and wiring.

5.2.5 Detailed Empennage Design

The tail structure was constructed using a built-up configuration with symmetric NACA 0009 sections. The horizontal stabilizer is reinforced with two 0.20 in outer diameter carbon fiber spars passing fully through the tail mount, while the vertical stabilizer uses a single 0.20 in outer diameter carbon fiber spar extending 1.50 in into the mount. The tail mount consists of three 0.20 in plywood bulkheads, two 0.20 in plywood servo plates, and four longerons forming a rigid box structure. The assembly is secured to the fuselage using four M5 bolts engaging pre-installed M5 T-nuts to ensure positive load transfer and prevent rotation.

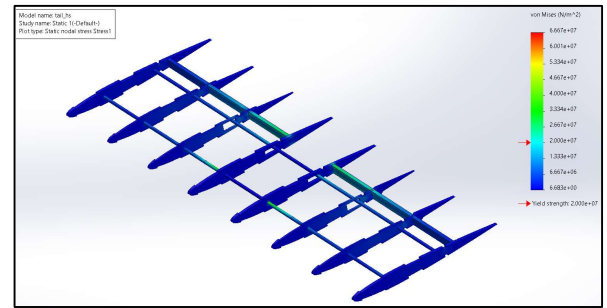


Figure 5.11 Horizontal Stabilizer FEA Result

Each horizontal stabilizer half contains four ribs, with a 0.20 in plywood root rib and three 0.12 in balsa ribs uniformly spaced along the span. The vertical stabilizer follows the same rib configuration. Rounded leading edges were molded using 0.08 in balsa sheeting to maintain airfoil fidelity. Elevator and rudder surfaces are attached using a custom lug hinge system, and both servos are mounted directly within the tail mount structure. The entire assembly is covered with Hanger 9 UltraCote heat-shrink film.

The horizontal stabilizer was analyzed using a static structural study in SolidWorks Simulation to evaluate its ability to withstand the negative lift required for longitudinal stability. A downward load of 11.2 lbf was applied to the stabilizer to represent the required tail download, while the root was fixed to model attachment to the tail structure. The stabilizer was designed with two carbon-fiber spars, which act as the primary load-carrying members under bending. As shown in Figure 5.11, the highest von Mises stress occurs near the root of the spars, where bending moments are maximum. The peak stress remained below the carbon-fiber yield strength with most of the structure operating at significantly lower stress levels. This confirms that the dual-spar configuration effectively distributes the applied load and that the horizontal stabilizer has sufficient strength to safely support the required tail download without structural modification.

5.2.6 Detailed Main Landing Gear Design

A static structural FEA analysis was conducted on the MLG to verify its ability to withstand harsh landing loads. The analysis considered a 2.5g load factor of MTOW, representing a severe landing condition. The upper attachment region was fixed to simulate the connection with the fuselage, while the landing load was applied at the wheel attachment locations. The MLG was modeled as a fully carbon-fiber structure, designed to absorb landing loads through elastic deformation. As shown in Figure 5.12, the maximum von Mises stress occurs near the upper bend region of the gear, where bending moments are highest, with a peak value of 15800 psi (109 MPa). This stress level remains well below a conservative carbon-fiber yield

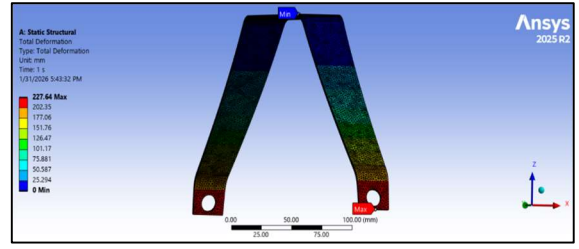
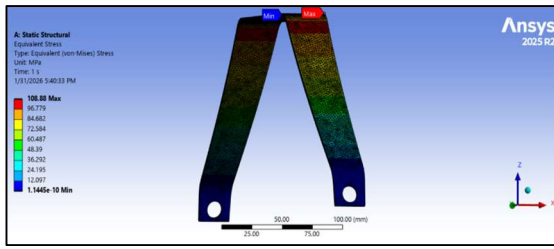


Figure 5.12 FEA of MLG: Stress(left), Deformation(right)

strength of approximately 72500 psi (500 MPa), resulting in an estimated factor of safety of 4.6. The maximum deformation under the 2.5g landing load was 8.96 in (227.64 mm), occurring at the lower ends of the gear legs, indicating significant energy absorption without material failure. Based on these results, the carbon-fiber MLG demonstrates sufficient strength and compliance to safely withstand the design landing load and was deemed structurally adequate for the final configuration.

5.3 Subsystem Integration

5.3.1 Passenger Restraint System

The passenger restraint system was developed in full compliance with the requirements mandating that passengers be individually boarded, fully restrained in an upright vertical orientation, and prevented from movement during all phases of flight. All rubber duck passengers are seated within precision-profiled recesses machined into a lightweight balsa baseplate integrated into the single passenger compartment, where controlled geometric tolerances provide lateral and longitudinal frictional constraint without the use of rigid partitions. Vertical and rotational restraint is achieved using a unified top compression cover fabricated

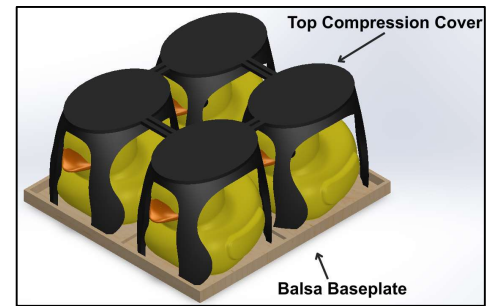


Figure 5.13 Passenger Restraint System

from PLA as shown in Figure 5.13. The cover incorporates four slender flexible extruded members per passenger that contact the duck body at four corner points and elastically deform to apply consistent compressive preload, accommodating dimensional variations in contest-provided ducks while increasing the normal force at the base interface to enhance sliding resistance under maneuver and landing loads. All individual compression elements are integrated into a single cover assembly, allowing simultaneous restraint of all passengers in one operation and reducing ground mission loading time. The complete passenger cover assembly is secured within the compartment using mechanically locking push latches, ensuring positive retention of the restraint structure to the airframe and preventing any relative motion during taxi, takeoff, flight, or landing. During the proposal phase, an adhesive-backed plate was initially considered as a secondary retention method; however, subsequent Q&A clarification [18, 19] prohibiting adhesive-based retention led to complete elimination of that concept and transition to a fully mechanical restraint solution.

5.3.2 Cargo Restraint System

The cargo restraint system was designed in accordance with the requirement that cargo be carried in a separate cargo bay and restrained to prevent movement during all phases of flight. A dedicated cargo compartment, structurally isolated from the passenger bay, houses a removable 3D-printed PLA cargo box whose internal dimensions closely match the nominal geometry of the standard hockey puck payload. The box arranges the cargo in a linear, line-by-line orientation with minimal clearance, thereby providing geometric constraint in the longitudinal and lateral directions without reliance on adhesives or secondary inserts. The cargo box incorporates an integrated sliding cargo door for rapid loading and unloading while

maintaining structural rigidity and positive retention during flight. As shown in Figure 5.14, the compartment is dimensioned such that the inserted box is constrained laterally and vertically by close-tolerance sidewalls and floor contact, preventing relative motion under maneuver and landing loads. Access to the compartment is provided by a rear-facing external door located beneath the tail upsweep, ensuring clear separation from the passenger compartment as required. For secure handling and retention, a handle is integrated at one end of the cargo box; when inserted, this handle interfaces with a dome-shaped groove incorporated into the main cargo door. Upon closure of the tail door, this interface acts as a mechanical locking feature that prevents forward, and backward displacement of the cargo box, effectively transferring inertial loads into the airframe structure.

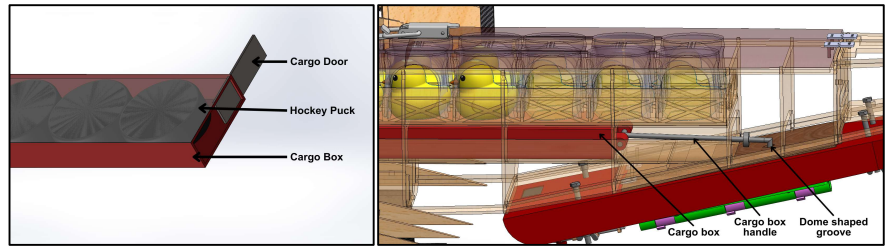


Figure 5.14 Cargo box with sliding door and integrated handle-groove locking interface installed in the tail cargo compartment

5.3.3 Banner Fabric

Multiple banner fabrics were evaluated to identify the lowest-drag configuration that satisfied structural, deployment, and tow stability requirements. The candidate materials included Taffeta 240T, Nylon 228T Taslan, and 240T Pongee, as shown in Figure 5.15. Material selection and experimental procedures are described in section 7.4 and 7.8.



Figure 5.15 Candidate banner fabrics evaluated during material selection: Nylon 228T Taslan (left), Taffeta 240T (mid), 240T Pongee (right)

Drag performance trends and comparative results are summarized in section 8.1. Across the tested Reynolds number range, Taffeta 240T consistently demonstrated the lowest drag coefficients relative to Nylon Taslan and Pongee. Nylon Taslan exhibited higher drag attributed to increased surface texture, while Pongee showed intermediate behavior. Based on these measured results, Taffeta 240T was selected as the final banner material.

The final banner configuration consists of a solid $AR_b = 5$, 5 ft × 1 ft banner without trailing-edge slits, as additional configuration comparisons discussed in section 8.1 indicated no measurable drag reduction from slit configurations under competition operating conditions. A 5 mm (0.2 in) carbon fiber rod is sewn into the leading edge to maintain dimensional stability during towing. Ring grommets installed at the upper and lower leading-edge corners connect to a fishing-line tow system that transitions from a Y-shaped bridle to a single tow line. To ensure stable vertical alignment in flight, a bolt integrated at the lower end of the carbon rod allows adjustable nut weights for trim tuning.

The university name “MIST” will be sewn onto both sides of the banner using contrasting fabric to ensure clear visibility while maintaining durability and structural integrity during towing operations.

5.3.4 Banner Module

The external banner stowage, deployment, and detachment module was designed as a removable M3-specific unit to simplify aircraft configuration and minimize drag during M1 and M2. The module, shown in Figure 5.16, is 3D printed in CFPLA and mounted on the aft fuselage beneath the tail upsweep, positioned close to the wing to reduce structural moment

arm effects. A half-cylindrical recessed cavity is integrated into the housing to compactly stow the rolled banner in compliance with external stowage requirements. A mechanically fastened claw mechanism, secured with machine screws, retains the banner prior to deployment. The module incorporates a servo-driven actuation system connected to a locking rod that controls all banner phases through three discrete positions: (i) full extension, where the rod engages the claw and positively locks the banner for takeoff; (ii) mid-position, where the rod partially retracts, allowing the claw to open and the banner to drop under gravity, with aerodynamic drag assisting full unfurling; and (iii) full retraction, where the rod releases the tow-line attachment, enabling complete banner detachment. The tow system uses a fishing line connected to the banner via a Y-shaped bridle, while the aircraft-side connection terminates in a lobster-claw clip for rapid ground attachment to a servo-actuated rod housed within a U-shaped block integrated into the module. This interface allows efficient ground mission handling while ensuring positive mechanical retention during flight. The tow point is located as close to the aircraft center of gravity as practicable to minimize trim change and pitching moment variation during deployment and release. The resulting module provides a lightweight, structurally robust, and fully mechanical solution that ensures secure stowage, reliable gravity-assisted deployment, controlled towing, and clean in-flight detachment with minimal aerodynamic and stability penalties.

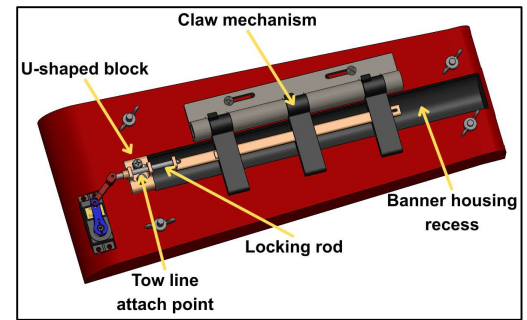


Figure 5.16 External banner stowage, deployment, and detachment module

5.3.5 Banner Claw Mechanism

The banner module consists of a housing enclosure and an integrated claw mechanism designed to retain the banner during stowage and release it upon command. Actuation is achieved through a control rod, which both operates the claw and serves as the tow interface when the tow rope end is attached. When deployed, the claw is exposed to freestream airflow up to 98.4 ft/s, resulting in corresponding dynamic pressure loads. The entire module is manufactured using CFPLA through additive manufacturing to provide adequate strength and durability while maintaining low structural weight.

A static structural FEA was conducted on the banner claw mechanism to evaluate its strength when deployed in the freestream. The claw was subjected to an equivalent aerodynamic pressure corresponding to an aircraft speed of 98.4 ft/s (dynamic pressure ≈ 550 Pa), applied to the exposed surfaces in the direction of airflow. The claw was fixed at the mounting plate to represent attachment to the fuselage. As shown in Figure 5.17, the maximum von Mises stress occurs at the leading edge of the claw, where aerodynamic loading first impinges and load transfer into the mount is highest. The peak stress was 1.41 psi, which is several orders of magnitude lower than the typical tensile strength of CFPLA 5,800 psi, resulting in an extremely large factor of safety. The maximum deformation was only 4.31×10^{-6} in, indicating negligible deflection under aerodynamic loading. Based on these results, the claw mechanism experiences minimal stress and deformation during banner deployment and was deemed structurally adequate for operation in the freestream.

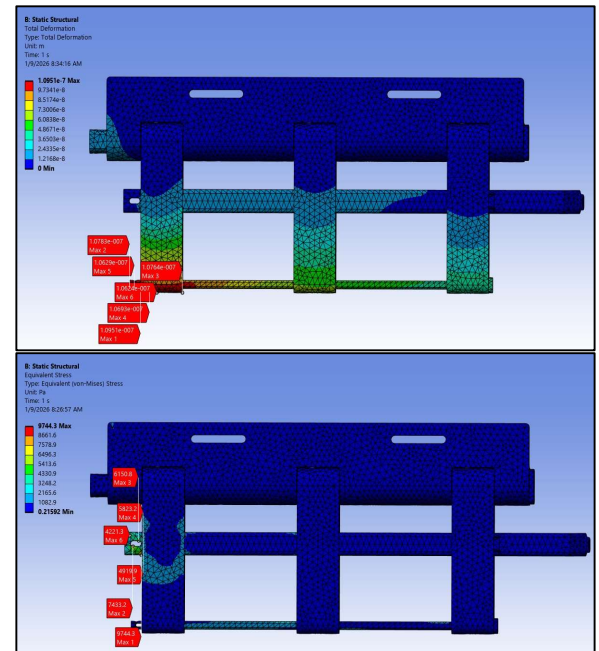


Figure 5.17 FEA of Banner Claw Mechanism: Stress(top), Deformation (bottom)

5.3.6 Propulsion System

The propulsion system was selected to deliver high static thrust for safe takeoff performance while maintaining electrical reliability and mass efficiency. A 6S 4500 mAh 30C LiPo battery powers the system, providing sufficient discharge capability to support peak current demands during takeoff without

Table 5.2 Final Propulsion Package Breakdown

Propulsion Items	Models	Mass (lbs.)
Battery	Tiger 4500 mAh 6S 30C LiPo	1.1
Motor	Scorpion A-4220 520 KV V2	0.74
ESC	HOBBYWING FlyFun 120A	0.26
Propeller	Falcon 16×7E	0.1
	Falcon 15×8E	0.11
Fuse	100A MAXI Fuse	0.02
Total Mass (lbs.)		2.33

excessive voltage sag. The Scorpion A-4220 520 KV V2 motor was chosen for its high torque characteristics, which allow efficient operation with large-diameter propellers. A HOBBYWING FlyFun 120A ESC ensures adequate current margin and thermal stability during sustained high-power operation. Two propeller configurations, Falcon 16×7E and Falcon 15×8E, were tested to balance thrust and pitch speed depending on mission requirements. A 100A MAXI fuse is integrated as a protective element to prevent overcurrent damage to the battery and ESC. Table 5.2 summarizes the finalized propulsion components and their individual masses. The total propulsion mass was carefully monitored, as it directly affects wing loading, takeoff distance, and center of gravity placement.

5.3.7 Avionics and Wiring

Figure 5.18 shows the wiring layout for the onboard avionics and propulsion system of HeisenBird. The MIST DBF team intentionally simplified the wiring architecture so that cable runs occupy minimal space and fit entirely beneath the passenger compartment. This layout ensures compliance with DBF rules, where the access door is restricted only to the passenger compartment and does not expose the cargo bay or other internal sections. The simplified routing also improves repairability and reduces troubleshooting time during competition operations. The aircraft uses a single tractor propulsion configuration. The Scorpion A-4220-V2 motor is connected to a HOBBYWING FlyFun 120A 6S V5 ESC through a 100A blade fuse and is powered by a Tiger 4500 mAh 6S 30C propulsion battery. Avionics power is supplied by a separate Tiger 2S 1500 mAh 30C battery, which feeds six servos, the ESC control circuit, and a Radiolink R12DS receiver. A dedicated switch is installed between the avionics battery and the receiver to allow safe connection and disconnection during ground handling and maintenance.

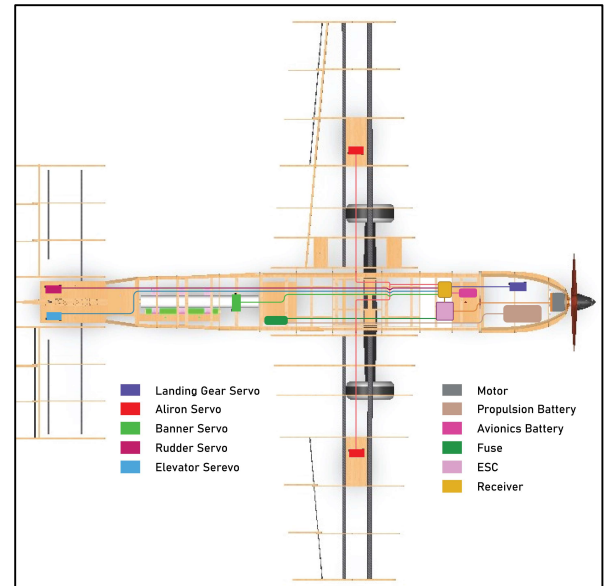


Figure 5.18 HeisenBird Wiring Diagram

A dedicated switch is installed between the avionics battery and the receiver to allow safe connection and disconnection during ground handling and maintenance.

5.4 Weight and Mass Balance Analysis

A proper center of gravity (CG) location is critical for flight stability and control. In the design process, CG analysis was conducted using SolidWorks models, assigning appropriate weights and densities to each component, and was then validated against the physical aircraft. The coordinate system used to model the CG is shown in Figure 5.19.

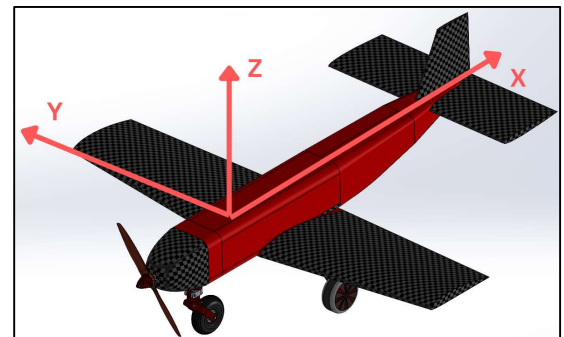


Figure 5.19 HeisenBird Coordinate System

Table 5.3 shows each component's mass and CG location with respect to this coordinate system. For an aircraft configured to carry a significant payload, such as passengers and luggage, it is essential to position that payload's CG close to the

aircraft's empty CG location to maintain longitudinal stability.

Design features like multiple luggage compartments and adjustable battery placement allow for necessary CG adjustments.

In practice, batteries can also be used as ballast to ensure the CG falls within the required static margin. Mission requirements can necessitate significant adjustments; for instance, when transitioning from a payload-carrying mission to one without passengers, battery packs may be shifted aft to compensate for the changed weight distribution.

In practice, batteries can also be used as ballast to ensure the CG falls within the required static margin. Mission requirements can necessitate significant adjustments; for instance, when transitioning from a payload-carrying mission to one without passengers, battery packs may be shifted aft to compensate for the changed weight distribution.

Table 5.3 Weight and Mass Balance Table of HeisenBird for all Missions

Aircraft Component	Mass (Kg)	X (in)	Y (in)	Z(in)
Fuselage	1.6	14	0	-2.91
Wing	.66	6.4	0	-4.576
Motor	.4	-15	0	-3
V.T.	.065	37.3	0	3.33
H.T.	.2	37.6	0	-5
Nose Landing Gear	.16	-3.68	0	-11.12
Main Landing Gear	.265	5	0	-11.6
Prop	.05	-16.3	0	-3
Empty	3.4	8.95	0	-4
Battery	.6	(-15.54)	0	(-3)
M1	4	4.32	0	-3.81
Battery	.6	(-15.54)	0	(-3)
Pax	.44	8.73	0	-1.1
Cargo	1.36	8.5	0	-3.1
M2	5.84	5.84	0	-3.375
Battery	.6	(-15.54)	0	(-3)
Banner	.18	58.4	0	-5.5
M3	4.18	7.21	0	-3.91
Quarter Chord, X (in) = 4.32				
CG Location and Stability Margin Per Mission				
	M1	M2	M3	
CG Location (in)	4.32	5.84	7.21	

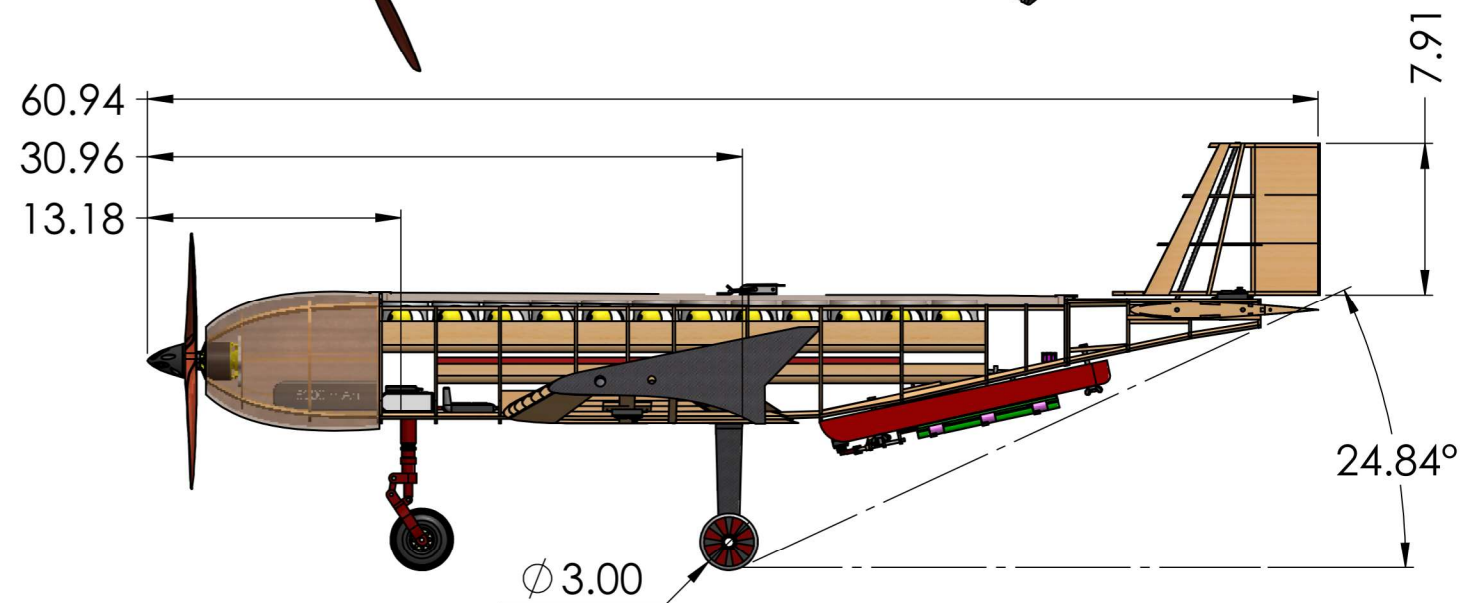
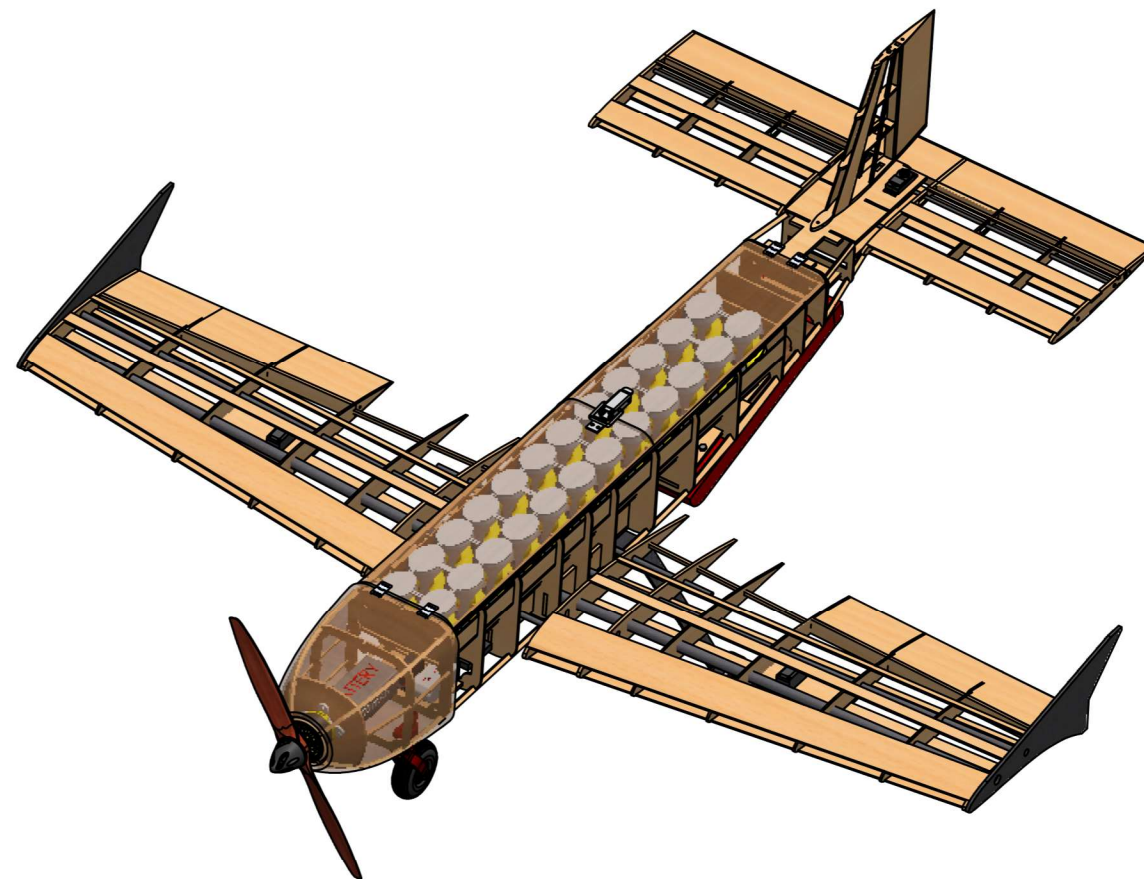
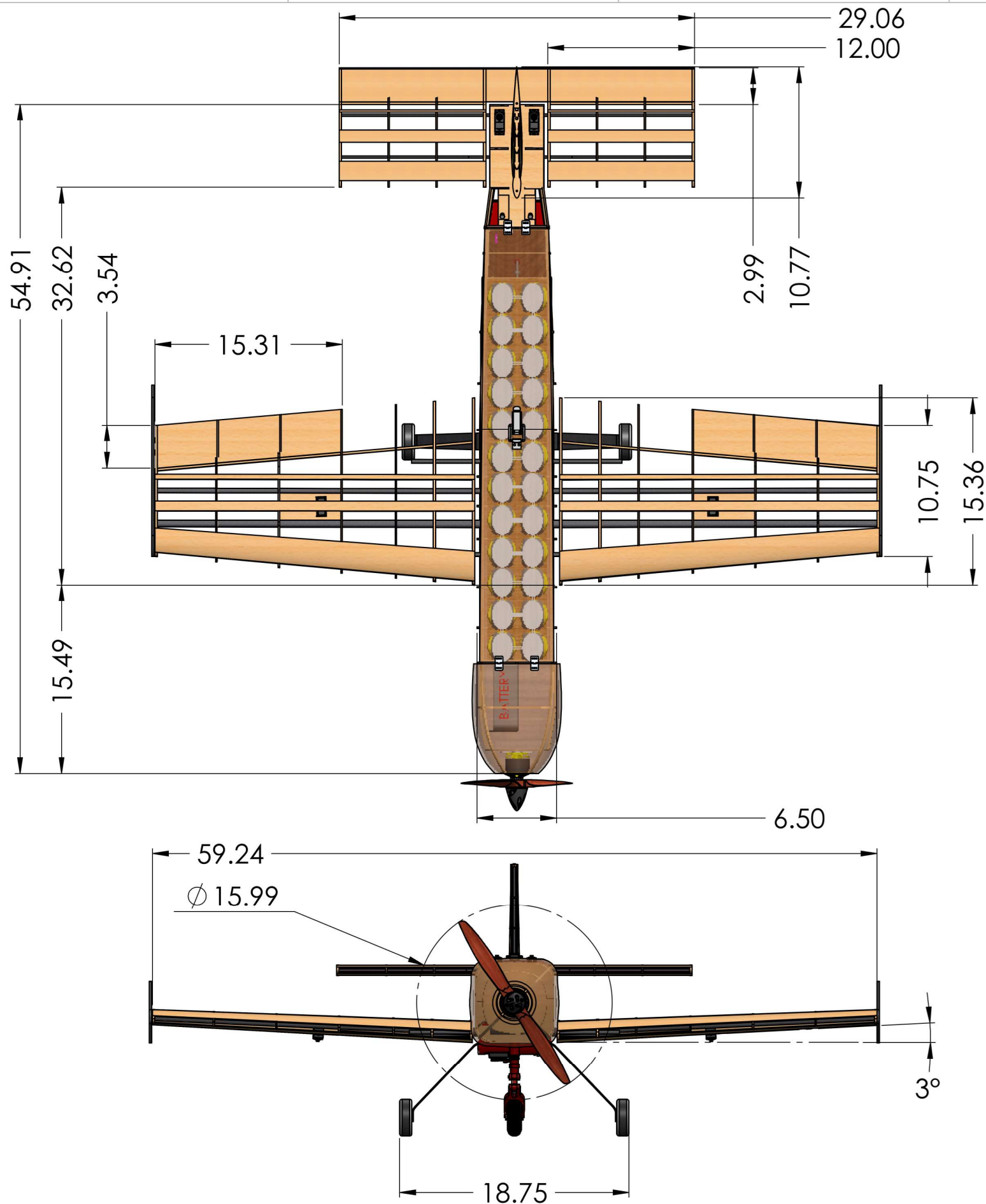
5.5 Flight and Mission Specific Performance

Final flight and mission performance were predicted using the final aircraft configuration parameters as shown in Table 5.4 in conjunction with the mission model, aerodynamic analyses (XFLR5/AVL), and propulsion sizing tools and weight, cruise speed, lap count, energy consumption, lap time and takeoff performance were computed for all missions. The resulting performance metrics and projected mission scores were benchmarked against conceptual estimates and further validated through flight testing to ensure consistency between analytical predictions and actual aircraft behavior.

Table 5.4 Predicted Final Aircraft Performance and Mission Score

Parameters	M1	M2	M3	GM
Gross Weight (lb)	8.82	12.87	9.2	N/A
Wing Loading (lb/ft ²)	1.5876	2.3166	1.656	
Take-off Distance (ft.)	26	399	28	
Static Thrust (lbf.)	14.3	13.05	14.3	
Avg. Cruise Speed (ft./s)	89	108	89	
Max. Speed (ft./s)	102	120	102	
Stall Speed (ft./s)	28.9	35	29.6	
C _{Lmax}	1.36	1.36	1.36	
C _{Lcruise}	0.186	0.167	0.194	
(L/D) _{max}	11.79	11.79	11.79	
(L/D) _{cruise}	5.82	5.28	6.03	
Turn Radius (ft.)	246.2	363	246.2	
Time taken for a 360° Turn (s)	17.38	21.09	17.38	
Rate of Turn (°/s)	20.71	17.07	20.71	
Rate of Climb (ft./s)	65	45	60	
Banked Turn Speed (ft./s)	34.4	41.6	35.1	
Max. Current (A)	95	101.2	95	
Number of Laps	3	10	8	
Lap time (s)	35	30	37	
Mission Duration (s)	105	300	300	
Mission Scores	1	1.643	2.144	0.25
Total Mission Score	5.03			

5.6 Drawing Package



		MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY			
DRAWN BY	SAFWAN, SHAHIK	TITLE 3-VIEW DRAWING			
CHECKED BY	MUSFIQUR RAHAMAN				
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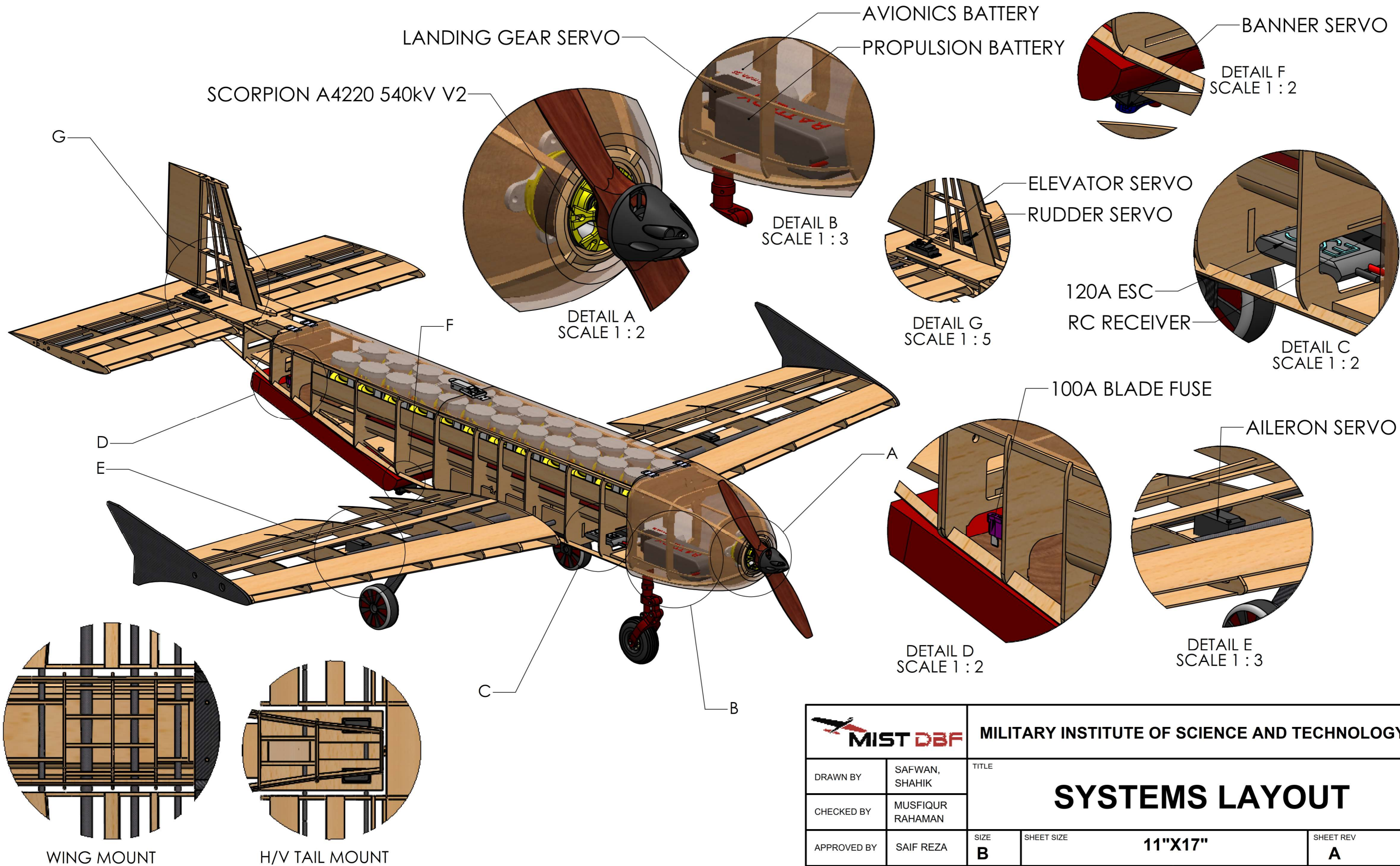
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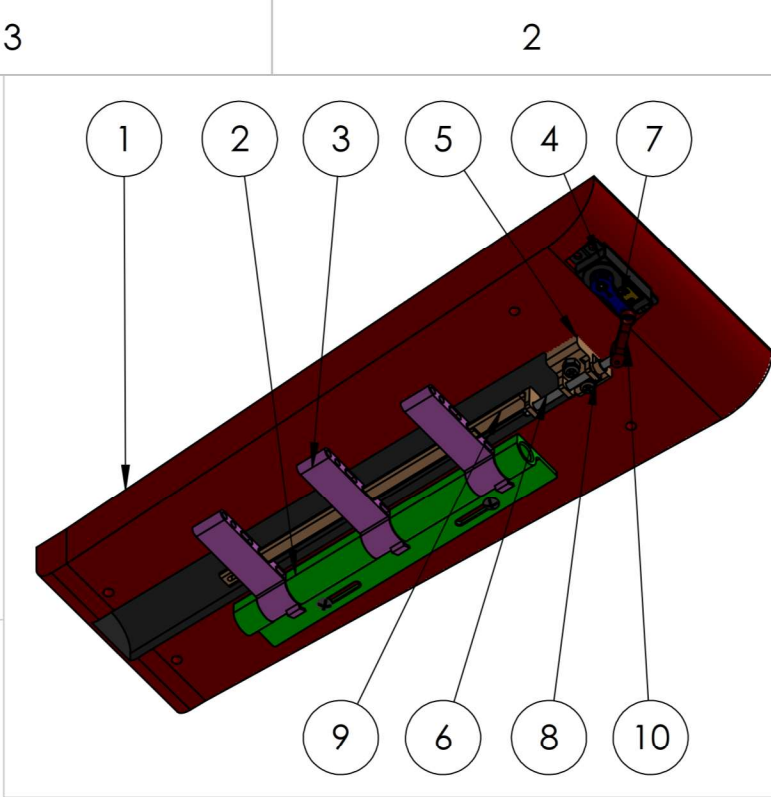
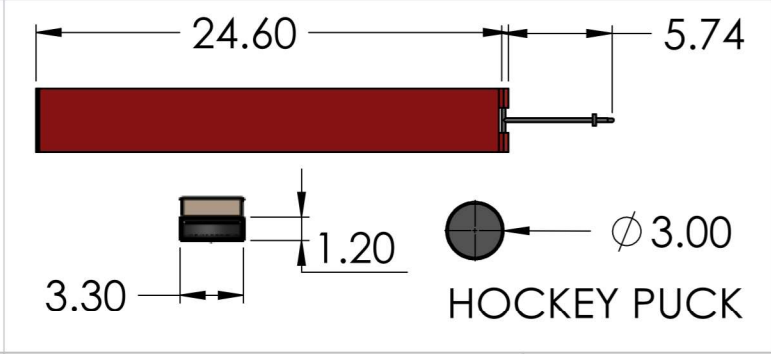
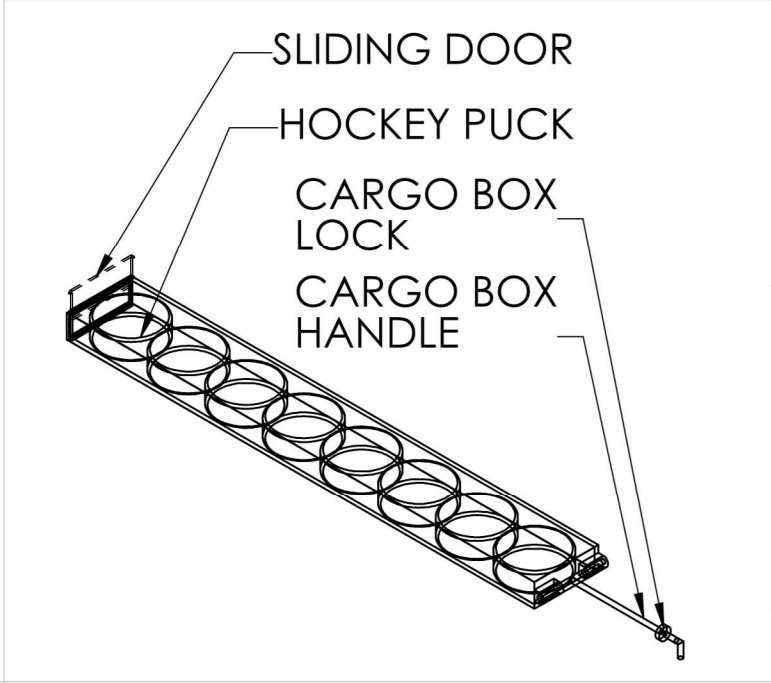
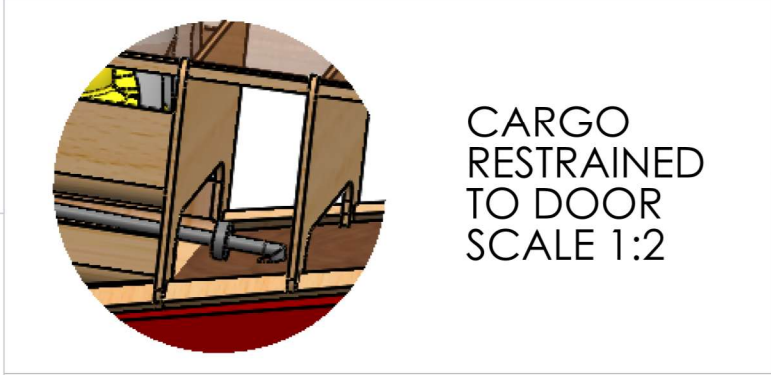
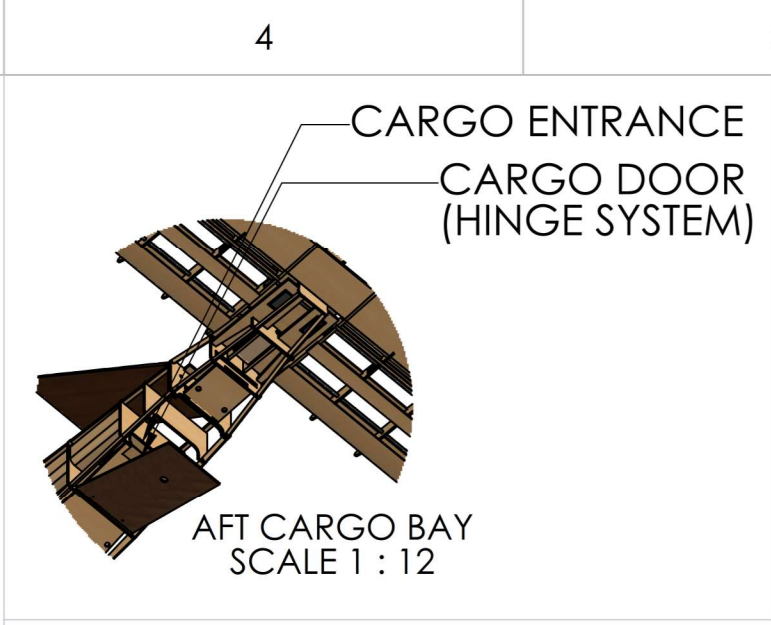
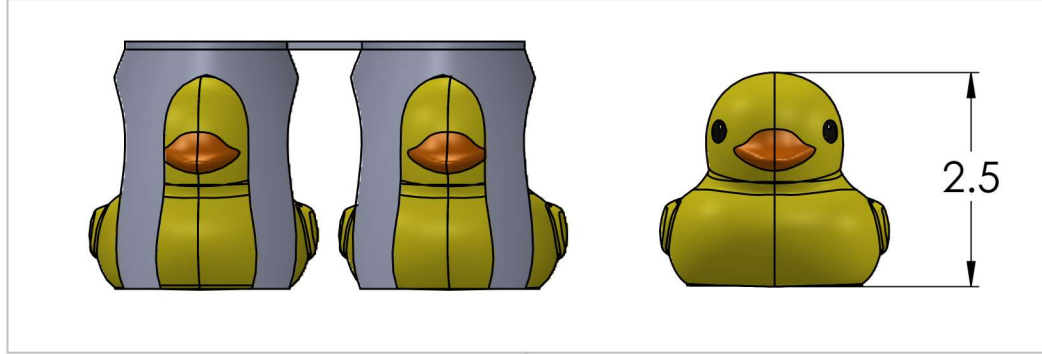
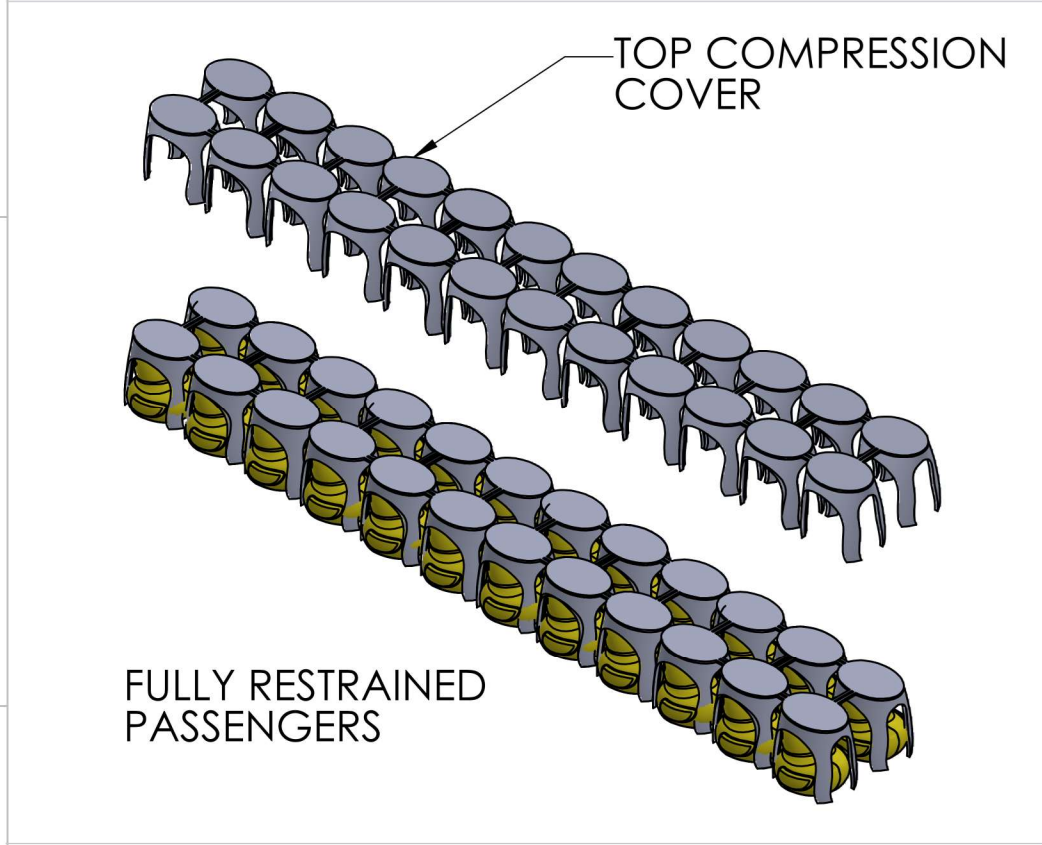
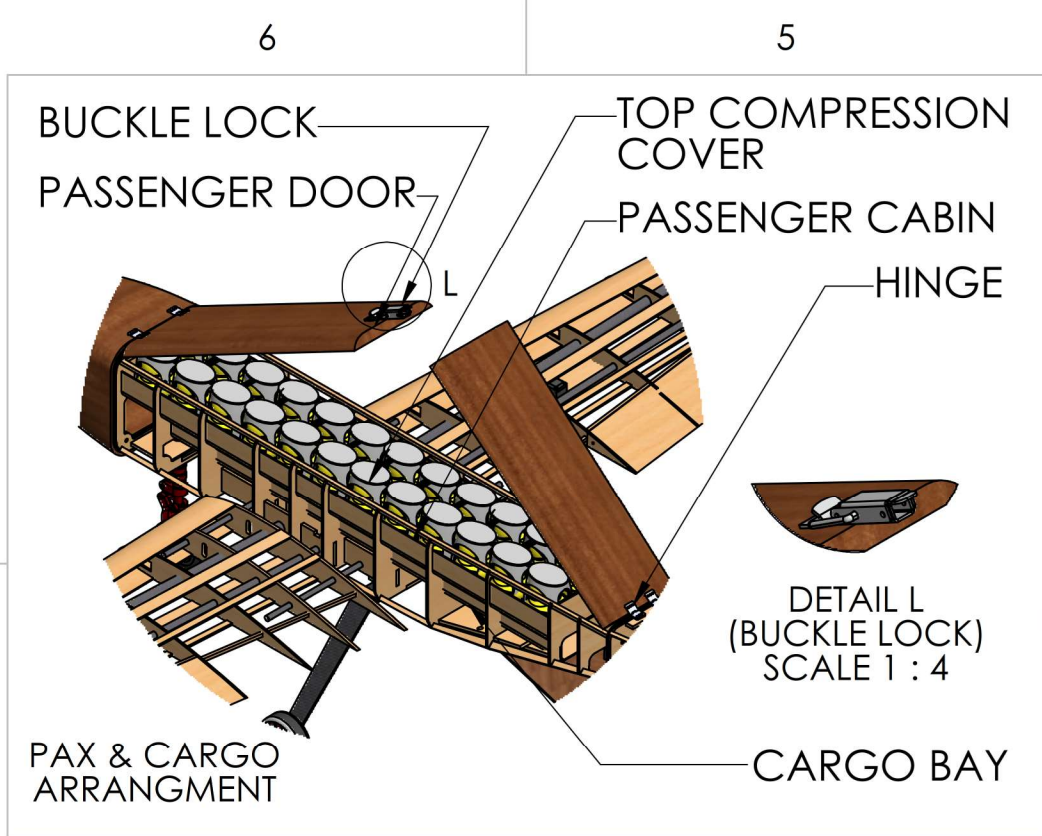
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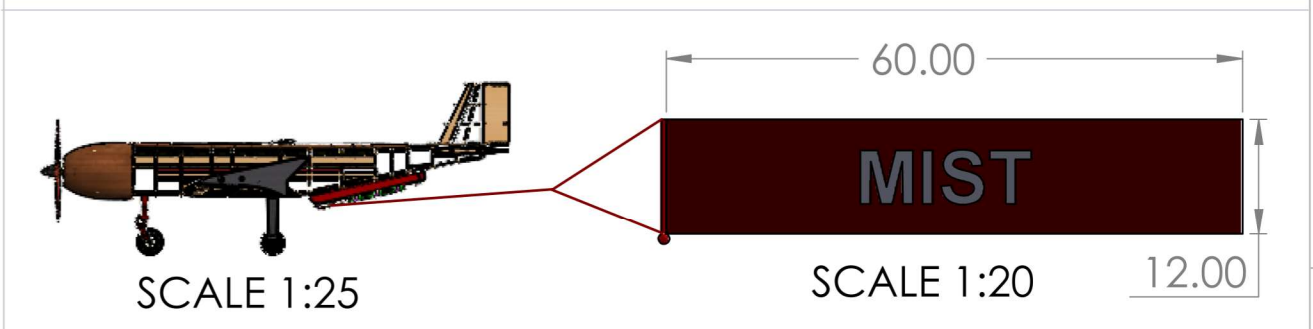
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		MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY			
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CHECKED BY	MUSFIQUR RAHAMAN				
APPROVED BY	SAIF REZA	SIZE B	SHEET SIZE 11"X17"	SHEET REV A	
DATE	19/01/2026	SCALE 1:6		SHEET 3 OF 4	



NO.	PART NAME	QTY.
1	MODULE	1
2	HINGE MECHANISM	1
3	CLAW	3
4	MG996 SERVO	1
5	STRING ATTACHMENT	1
6	ACTUATING ROD	1
7	SERVO ARM	1
8	M3 NUTS	2
9	CLAW LOCK	1
10	CONNECTING ROD	1



		MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY		
DRAWN BY	SAFWAN, SHAIK	TITLE SUBSYSTEM LAYOUT		
CHECKED BY	MUSFIQUR RAHAMAN			
APPROVED BY	SAIF REZA	SIZE B	SHEET SIZE 11"X17"	SHEET REV A
DATE	19/01/2026	DIMENSIONS ARE IN INCHES		SHEET 4 OF 4

6 MANUFACTURING PLAN

The manufacturing plan was developed to balance rapid prototyping, structural reliability, repairability, and competition readiness under limited time and budget constraints. An additional critical constraint was international transportation, as the aircraft had to be shipped from Bangladesh to the USA for the DBF competition. As a result, the aircraft was intentionally designed with detachable major components to allow safe transport, compact packing, and rapid reassembly at the competition site. Manufacturing activities were initiated prior to final design freeze to allow early validation of aerodynamic configuration, mission-specific ground operations, and assembly logic.

6.1 Manufacturing Processes Investigated

Several manufacturing processes were evaluated during the aircraft's development to balance cost, weight, manufacturability, and performance. An initial Prototype 0 using Depron and foam core materials enabled rapid, low-cost aerodynamic assessment and early flight testing, but insufficient rigidity and higher structural mass made it unsuitable for the competition aircraft. A built-up balsa and plywood monocoque construction was then investigated due to the team's prior experience, favorable strength-to-weight characteristics, low cost, ease of repair, and compatibility with laser-cut components for improved precision. A molded fiberglass composite fuselage was also explored; however, despite its high structural strength, it imposed significant weight, time, and budget penalties and was therefore discarded. Additive manufacturing was evaluated and adopted for non-structural components where complex geometries were required. Laser cutting and CNC foam cutting were investigated to improve dimensional accuracy, with laser cutting outsourced for plywood and balsa parts and an in-house CNC foam cutter used during early prototyping. Based on these evaluations, a hybrid manufacturing approach was selected, combining traditional built-up construction with selective use of advanced manufacturing methods.

6.2 Selected Manufacturing Processes

Material and manufacturing decisions were made using a structured weighting approach to ensure consistency across all major components. Each candidate material and fabrication method were evaluated against the performance and practicality metrics listed in Table 6.1 and Table 6.2 to determine the most suitable combination for the aircraft.

Table 6.1 Manufacturing Process Selection

Criteria	Weight (%)	Built-up Balsa & Plywood	Foam Core (Depron/XPS)	Molded Composites (GFRP)	3D Printing
Weight	30	8	6	9	5
Durability	20	8	5	10	6
Time	15	7	9	3	6
Simplicity	10	8	9	4	6
Experience	10	9	8	4	6
Cost	15	9	8	2	6
Total Weighted Score	100	8.10	7.05	6.25	5.70

Priority was given to minimizing structural mass, followed by achieving adequate durability. Manufacturing turnaround time was considered next to support rapid iteration, with additional consideration given to process simplicity, the team's prior experience, and overall cost.

Table 6.2 Manufacturing Techniques for Different Parts

Component	Material	Manufacturing Technique
Wing	Balsa and Plywood	Laser Cutting, Manual Assembly
Fuselage		
Empennage		
Nose Section		
Landing Gear	Carbon Fiber	Purchased COTS
Spars		
Banner Module	CF-PLA	3D Printing
Skin/Covering	Balsa Sheet, UltraCote Covering Film	Balsa Molding, Hot Iron

6.2.1 Wing

The wing was manufactured using laser-cut balsa ribs and stringers arranged in an I-beam internal configuration, with a plywood root rib and servo mounting plates to accommodate higher local loads, while a commercial carbon fiber spar rod served as the primary load-carrying element. The structure was covered with UltraCote heat-shrink film to provide a smooth aerodynamic surface with minimal weight addition, and the wing was designed as a fully detachable assembly to allow easy removal and reinstallation without disturbing the fuselage, supporting international transportation requirements and enabling rapid repair or replacement during testing.

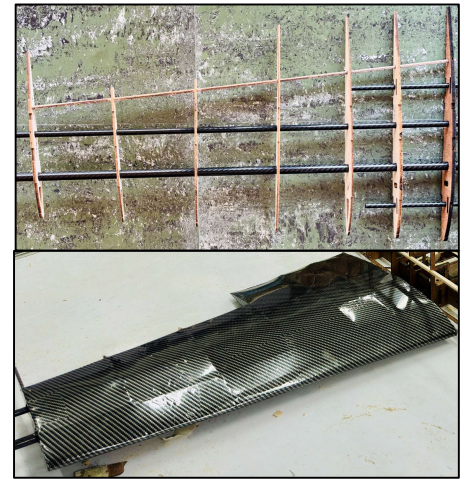


Figure 6.1 Manufactured Wing

6.2.2 Fuselage

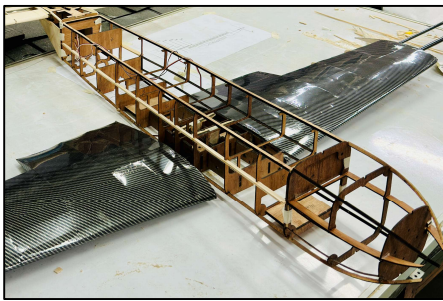


Figure 6.2 Manufactured Fuselage

The fuselage was constructed using a built-up monocoque structure with laser-cut plywood bulkheads, longerons, and internal support members, over which thin balsa sheets were wrapped to achieve a uniform external geometry before being covered with UltraCote film. The nose section was designed to be detachable to allow compact packing for international transport and improved access to internal components, reducing the risk of transport damage and enhancing maintainability during competition operations.

6.2.3 Empennage (Horizontal and Vertical Tail)

The horizontal and vertical tail surfaces were manufactured using the same built-up balsa construction philosophy as the wing. Both tail surfaces were designed to be fully removable, allowing safe transportation and precise alignment during final assembly. This approach also enabled rapid replacement in the event of damage during testing or competition operations.



Figure 6.3 Manufactured Empennage

6.2.4 Landing Gear

The landing gear was outsourced and manufactured from full carbon fiber to ensure high stiffness and strength with minimal mass. The landing gear assembly was designed to be detachable, allowing removal during transport and straightforward reattachment at the competition site without structural modification to the fuselage.

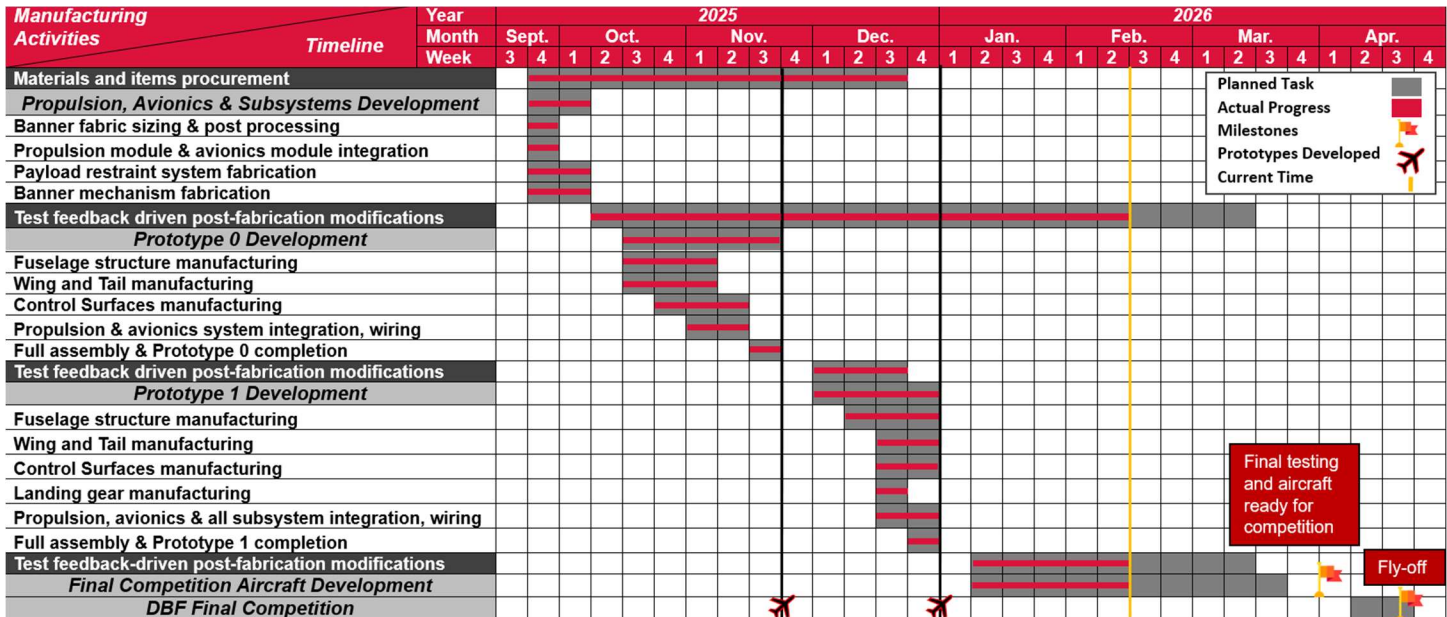


Figure 6.4 Main Landing Gear (left) and Nose Landing Gear (right)

6.3 Manufacturing Timeline and Milestones

In addition to structural fabrication, time was allocated for repeated assembly and disassembly cycles to validate fit, alignment, and repeatability of detachable components prior to international shipment, as shown in Table 6.3. This ensured that the aircraft could be reliably transported, reassembled, and made flight-ready within a limited timeframe at the competition site.

Table 6.3 Manufacturing Schedule



7 TESTING PLAN

The team conducted extensive ground and flight tests on the aircraft and its subsystems to validate the functionality of all systems, verify the predicted performance and inform design decisions through analysis of acquired data. Testing started during the preliminary design phase and continued throughout the subsequent design phases, guiding design improvements and producing successive prototypes that eventually evolved into the final aircraft.

7.1 Test Objectives

Table 7.1 Test Objectives

Test Type		Test Goals
Propulsion & Electronics	Static thrust test	Thrust and current draw are measured across multiple motor-propeller-ESC-battery combinations to identify the most efficient propulsion system
	Battery discharge test	Battery endurance and voltage stability are verified by discharging the selected battery under representative propulsion to confirm mission-duration energy compliance
	Servo and avionics test	Control-surface responses are observed while moving transmitter away to verify reliable TX–RX communication and control response
Aero-dynamic Tests	Banner- wind tunnel test	Banner drag forces are recorded using load cell for banner samples mounted in wind tunnel with varying wind speed to evaluate optimal banner material and design
	Banner- ground vehicle test	Banner drag forces are recorded using load cell for banner samples mounted on car roof with varying car speed to reconfirm banner drag and durability under dynamic airflow
	Banner mechanism test	Banner mechanism mockup is mounted and operated in wind tunnel with take-off and cruise velocities to validate reliability of banner stowage, deploy and release sequence
Ground Tests	Restraint system test	Payloads are loaded into aircraft; in-flight vibrations are simulated by shaking and observed for any displacement or loosening to ensure reliable attachment of payloads
	CG test	Aircraft is lifted by wingtips, observed for stability and adjustments are made until aircraft is stable and CG is found to verify aircraft stability and CG
	Structural load tests	Aircraft is loaded with max. payload, mounted on ground mission test stand and extra weight is applied at CG to verify structural integrity under large centered loading
	Landing impact test	Aircraft is dropped from scaled height and inspected for any structural damage to ensure landing gears can withstand the impact loads safely
	Assembly test	The whole aircraft is assembled, in-flight vibrations are simulated and aircraft is checked for damage or loose joints to ensure efficient and robust assembly
	Taxi test	Straight taxi runs are performed on ground, steerable landing gear response is observed to verify controlled taxing
Flight tests	Cruise speed test	Prototype is flown, its cruise speed is recorded using telemetry system to verify cruise speed with and without payload
	Steep turn test	Steep turns are performed with minimum bank angle of 20° and response is recorded data using telemetry system to verify minimum safe bank angle and turn radius
	Banner test	Banner drag forces are measured using strain gauge mounted on prototype in flight to finalize banner configuration
	Stability test	Prototype is flown, various maneuvers are performed and pilot's feedback and visual cues are recorded to confirm stable flight in all flight conditions
Full system test		Full mission attempts are performed with test prototype to verify all systems and mechanisms function as intended

the loading process. During testing, the wing exhibited smooth and controlled elastic deflection without any visible cracking, delamination, or structural damage.

Upon complete removal of the applied load, the wing returned to its original undeformed configuration, indicating the absence of permanent deformation. Based on the observed behavior under the applied loads and the full elastic recovery after unloading, the wing structure was determined to be structurally sound and adequate for safe flight operations.

7.3.2 Fuselage Structural Testing

Fuselage structural testing was conducted to verify the ability of the motor mount and surrounding structure to withstand propulsion-induced thrust and torque loads during worst-case operating conditions. The propulsion system was installed in its final flight configuration, and static ground tests were performed to represent takeoff and aggressive throttle transitions.

Thrust loading was simulated by applying a horizontal pulling force along the motor thrust axis using a calibrated fish scale. A load corresponding to the expected maximum static thrust, including a 1.5 safety factor, was applied incrementally. The motor mount and fuselage structure were continuously inspected for cracking, loosening, or permanent deformation.



Figure 7.2 Fuselage torque testing under sudden full-throttle application

To evaluate torque resistance, the aircraft was rigidly restrained on the ground and subjected to a sudden full-throttle application from zero throttle. This test was done to simulate the maximum transient torque experienced by the motor mount and fuselage during abrupt throttle inputs. The motor mount, fuselage skin, bulkheads, and fasteners were closely observed during and after the test for any signs of cracking, loosening, or permanent deformation, as shown in Figure 7.2.

No visible structural damage, cracking, or permanent deformation was observed during or after the torque test. The motor mount remained firmly attached, and no degradation of the surrounding fuselage structure was detected. Based on these observations, the fuselage and motor mounting system were determined to be structurally adequate to withstand propulsion-induced torque loads encountered during flight.

7.3.3 Landing Gear Testing

The landing gear was tested to verify its ability to withstand taxiing and landing loads at maximum takeoff weight (MTOW). With the aircraft configured at the design center of gravity, taxi tests were conducted by accelerating the aircraft up to the estimated stall speed, V_{stall} , to confirm straight-line tracking and to ensure no tendency to nose over. To evaluate landing impact loads, a controlled drop test was performed from a height of 5 inches at MTOW, corresponding to an equivalent 2.5g landing condition. The landing gear, attachment points, and surrounding fuselage structure were

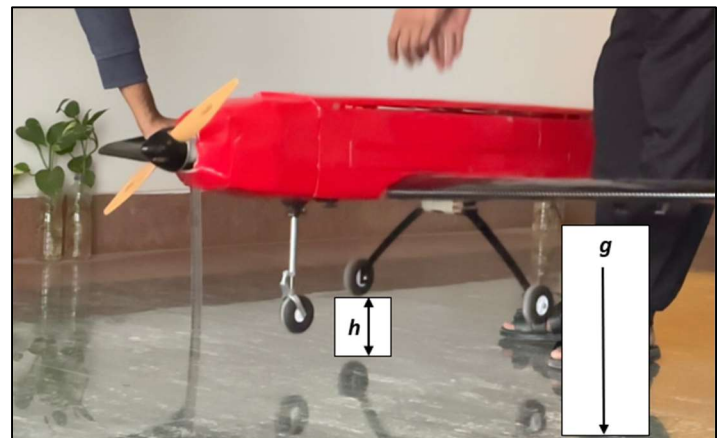


Figure 7.3 Landing drop test setup

inspected during and after testing, with no visible cracking, permanent deformation, or structural damage observed. These

results confirm that the landing gear system is structurally adequate for expected ground and landing loads, as shown in Figure 7.3.

7.3.4 Restraint System Testing

To ensure compliance with DBF 2026 requirements mandating that passengers remain upright and immobile and cargo remain restrained throughout all phases of flight, a restraint validation test was conducted by loading the declared passenger and cargo configuration into their respective compartments and subjecting the aircraft to repeated manual vibration and handling loads representative of taxi, flight maneuvers and landing. The passenger compression cover and cargo box were inspected for displacement, latch loosening or relative motion while monitoring structural interfaces and locking features under induced oscillatory inputs. No passenger misalignment, cargo translation or latch disengagement was observed, confirming that the mechanical restraint systems provide adequate and reliable retention throughout all flight conditions.



Figure 7.4 Passenger restraint testing

7.4 Aerodynamics Tests

To ensure reliable banner performance, a series of aerodynamic and mechanism validation tests were conducted under controlled and near real-world conditions. Both scaled wind tunnel experiments and full-scale ground-based evaluations were performed to understand drag behavior and deployment reliability across expected operating speeds. The detailed test results and quantitative analysis are presented in section 8.

7.4.1 Banner Drag – Wind Tunnel Testing

Preliminary aerodynamic testing of the banner was conducted in the AF 1300 subsonic wind tunnel to investigate banner drag characteristics and material-dependent aerodynamic behavior. The wind tunnel has a rectangular test section measuring 305 mm × 305 mm with a length of 600 mm, operating in the low-speed subsonic regime. Due to the limited test-section size, all banner models were tested at a reduced scale. Three banner fabrics, Taffeta 240T, Nylon 228T Taslan, and 240T Pongee, were evaluated based on their flexibility and suitability for lightweight banner towing applications.

The banner was mounted using a flexible attachment aligned with the freestream, allowing it to deform naturally under aerodynamic loading, as shown in Figure 7.5. Testing was carried out over a freestream velocity range of 10-30 m/s, and aerodynamic forces were measured using a three-component lift and drag balance integrated with the wind tunnel. The balance was calibrated prior to testing and zeroed before each run to remove fixture-induced bias. Force measurements were recorded once steady flow conditions were achieved at each test speed.

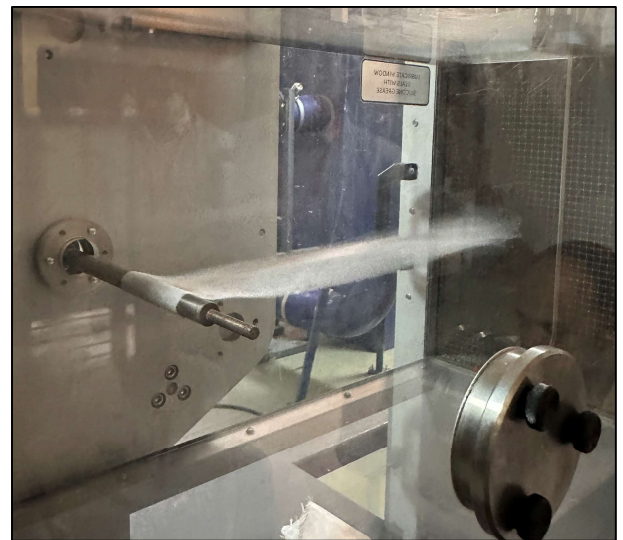


Figure 7.5 Banner drag testing in the AF 1300 subsonic wind tunnel using a three-component force balance

Because of the reduced scale, the achievable Reynolds numbers were significantly lower than those expected in flight, and wall interference effects were present. Consequently, the wind tunnel results were used primarily for comparative and trend-based analysis, including evaluation of relative drag behavior among different banner fabrics and observation of deformation

and stability characteristics, rather than for absolute drag prediction. Despite these limitations, the wind tunnel testing provided valuable insight into banner material selection and aerodynamic behavior and served as a baseline for subsequent full-scale ground vehicle testing.

7.4.2 Banner Drag – Ground Vehicle Testing

To obtain banner drag data under near full-scale and representative Reynolds number conditions, ground vehicle aerodynamic testing was conducted using two different platforms: a passenger car and a motorcycle. Testing on both vehicles allowed evaluation of banner behavior under differing wake characteristics and ensured robustness of the measured drag trends. Images of the car and bike test configurations are shown side by side in Figure 7.8.

A custom banner drag measurement rig was designed and fabricated for this study. The rig provided a rigid towing structure and a fixed measurement point for the banner tether. Drag force was measured using a calibrated load cell integrated into the rig, as shown in Figure 7.6. The drag forces generated by the banner were transmitted directly to the load cell through the rigid frame, ensuring accurate force measurement without structural compliance. Prior to each test run, the load cell was zeroed under no load conditions to eliminate measurement bias.



Figure 7.6 Banner drag measurement rig with integrated load cell

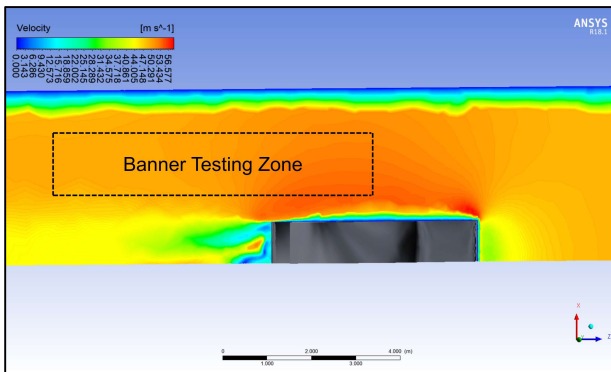


Figure 7.7 CFD analysis of vehicle wake showing banner testing region

During testing, the vehicle was accelerated to the target speed and held at a constant velocity for several seconds while drag readings were recorded. Multiple test runs were conducted to reduce the influence of transient effects such as acceleration, banner oscillation, and ambient wind disturbances. For the car-based tests, a computational fluid dynamics (CFD) analysis was performed in advance to identify the wake region downstream of the vehicle. The banner was positioned within the identified wake zone to ensure consistent and conservative drag measurements, as illustrated in Figure 7.7.

Although testing within the vehicle wake results in non-uniform flow conditions, this approach yields conservative drag estimates appropriate for design validation and mission performance modeling. The drag data obtained from both car and bike testing were therefore used as the primary basis for full-scale banner drag estimation, while the reduced-scale wind tunnel experiments provided complementary trend-based insights. The predicted banner performance derived from these ground-



Figure 7.8 Ground vehicle banner drag testing using a car (left) and bike (right)

based tests was subsequently validated through dedicated banner flight testing, which is discussed in a later section.

7.4.3 Banner Mechanism – Wind Tunnel Testing

A scaled banner sample was attached to a functionally representative mechanism mock-up mounted in the wind tunnel to evaluate stowage security, deployment and release performance under representative takeoff and cruise airspeed conditions. The servo-actuated system was triggered remotely at incremental airspeeds while high-speed video and visual observation were used to assess mechanism reliability, with minor flow disturbances introduced to simulate maneuvering effects through small angular offsets such as slightly tilting the mechanism mount. The scaled configuration demonstrated stable vertical orientation, repeatable deployment and detachment performance, minimal fabric fraying and mechanism structural degradation. The positive test outcomes confirmed the viability of the deployment concept and directly informed the design and fabrication of the final full-scale banner mechanism.

7.5 Propulsion and Electronics Tests

7.5.1 Static Thrust Testing

Static thrust testing was conducted to validate the performance of the selected propulsion system and to compare candidate propellers. The test setup consisted of a Scorpion A-4220-540 kV V2 motor, HOBBYWING FlyFun 120A 6S V5 ESC, and a Tiger 6S 4500 mAh 30C Li-Po battery. Two propellers were tested: Falcon 15×8E and Falcon 16×7E.

A custom-built static thrust test stand was fabricated to rigidly mount the motor-propeller assembly and isolate thrust forces along the measurement axis. Static testing was chosen to represent worst-case operating conditions due to the absence of freestream velocity. Each propeller was tested individually under static conditions, and steady thrust behavior was observed while operating within the safe limits of the propulsion system. The propulsion thrust testing setup is shown in Figure 7.9.

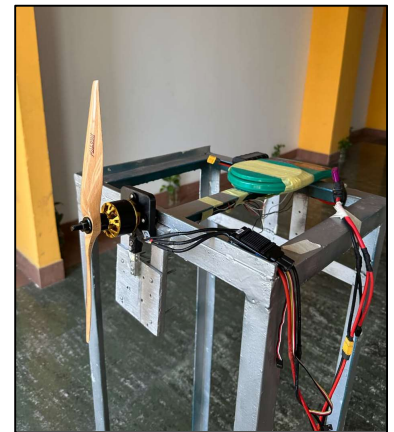


Figure 7.9 Custom-built static thrust test stand used for propulsion testing

The results of these tests were used to compare relative propeller performance and to confirm that the selected propulsion system provides sufficient thrust margin for takeoff and mission operations.

7.5.2 Battery Discharge Testing

Battery discharge testing was conducted to characterize the performance of the selected battery pack and to verify that the propulsion system is capable of completing each mission within the prescribed 5-minute mission time window. A 6S 4500 mAh Li-Po battery [20] was tested to evaluate its voltage–time behavior under discharge.

The discharge curve exhibits typical Li-Po characteristics, including an initial voltage drop after load application, a prolonged region of stable voltage, and a sharp voltage knee near the end of discharge. A conservative cutoff voltage of 22.8 V (3.8 V per cell) was selected to ensure safe operation of the propulsion and avionics systems. As shown in Figure 7.10, the battery reaches the cutoff voltage at approximately 6.5 minutes, providing sufficient voltage margin beyond the required mission duration.

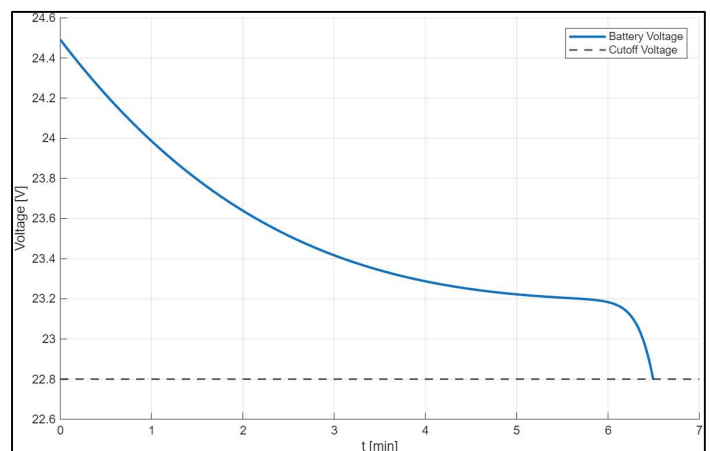


Figure 7.10 Battery voltage drop-off during discharge testing

These results confirm that the selected battery maintains adequate voltage stability throughout the entire mission time window, ensuring reliable propulsion performance under expected operating conditions.

7.5.3 Servo & Avionics Testing

The servo & avionics system was first validated at the subsystem level following fabrication of control surfaces and propulsion & electronics module, prior to aircraft assembly and flight testing, to reduce integration risk and ensure reliable control response and uninterrupted transmitter-receiver communication. Control surfaces were actuated through their entire deflection range while monitoring response smoothness and accuracy. A range check was performed in low-power mode while progressively increasing transmitter distance (up to 4000 ft.) to confirm stable link integrity and fail-safe behavior was verified by interrupting signal transmission and confirming immediate throttle cutoff and preset control positions. Following successful subsystem validation, fully assembled aircraft underwent a final integration check to ensure that installed wiring, routing, and structural mounting introduced no interference or degradation in performance. No signal dropouts, latency, or abnormal current draw were observed, demonstrating that the system was reliable and compliant for mission operations.

7.6 Mock Tests

Mock tests are structured in three progressive phases across all four missions to systematically validate performance, refine operational procedures and optimize mission scoring strategy. The first phase has been successfully implemented on Prototype 1 and upon completion of all phases, the final aircraft configuration will be finalized through iterative, feedback-driven design refinements. The test performance data are presented in section 8.

7.6.1 Ground Mission

GM testing was conducted to verify the team's ability to efficiently perform passenger loading and unloading as well as banner installation, deployment, and release in accordance with DBF Ground Mission requirements. The aircraft was positioned upright within the mission box, and all operations were performed without the propulsion battery installed. Multiple timed trials were conducted to evaluate crew coordination, payload handling procedures, and banner mechanisms, and to identify areas for improvement in task sequencing and execution. These tests confirmed reliable and repeatable GM operation while informing procedural and technical refinements aimed at reducing overall mission completion time.

7.6.2 M1 Test Flight

M1 mock flights were conducted by completing three laps within the five-minute window in the clean configuration to validate baseline flight performance and handling characteristics. The aircraft maintained consistent lap times, stable trim behavior and smooth landing performance, confirming satisfactory unloaded flight capability with no structural or control irregularities observed.

7.6.3 M2 Charter Flight

M2 mock flights were performed using the declared passenger and cargo configuration for a full 5-minute duration while recording current draw, lap count, and lap times to evaluate loaded performance and endurance. Measured electrical consumption and lap consistency aligned with predicted values, and no performance degradation or handling instability was observed, confirming adequate propulsion margin and structural integrity under payload conditions.

7.6.4 M3 Banner Flight

M3 mock tests executed the complete stow-to-deploy-to-release sequence within five minutes while monitoring aircraft stability, banner alignment, lap timing, lap count and propulsion response under tow conditions. The banner remained

vertically aligned with stable flight behavior throughout deployment and release and recorded performance data confirmed reliable mechanism operation and acceptable drag impact consistent with design expectations.

7.7 Flight Checklists

The team maintained a flight checklist, as shown in Table 7.3, in order to ensure safe, repeatable, and controlled aircraft operation by systematically verifying aircraft condition, subsystem functionality, and environmental readiness before, during and after each flight. Used consistently by the Testing Sub-team Lead, Pilot and Chief Engineer, the checklist enables reliable data acquisition, structured inspections, discrepancy tracking and reduced failure risk, facilitating safe transition from testing to competition operations. The checklist will be used during the fly-off contest as well.

Table 7.3 Flight Test Checklist

Flight Description					
Date:			Prototype:		
Flight / Mock Test Type:			Total Aircraft Weight (lb.):		
Wind Condition:			Aircraft CG From Nose (ft.):		
Preflight					
Propulsion and Electronics		Status (✓/X)	Assembly		Status (✓/X)
Throttle inhibit switch	On	☐	Wing, aileron, flaps	Secured	☐
Avionics and sensors	Installed	☐	Empennage and landing gear	Secured	☐
All internal wiring	Secured	☐	Propeller	Installed, tightened	☐
Avionics battery	Charge verified, secured	☐	Horizontal stabilizer and elevator	Secured	☐
Propulsion battery	Charge verified, secured	☐	Vertical stabilizer and rudder	Secured	☐
Receiver and receiver plugs	Secured	☐	Servos and control surface linkages	Secured	☐
ESC and controllers	Connected, secured	☐	Airframe	Damage free	☐
Motor	Connected, secured	☐	Banner (M3)	Stowed and secured	☐
Transmitter and avionics	On	☐	Banner mechanism (M3)	Installed and locked	☐
Avionics hatch	Closed	☐	Aircraft Interior		Status (✓/X)
Controls	Trimmed, centered	☐	Payload restraint system (M2)	Installed and locked	☐
Nominal and failsafe control surface	Checked	☐	Payload (M2)	Loaded, secured	☐
Control and telemetry range	Checked	☐	Payload access hatch (M2)	Closed	☐
Before Takeoff			After Landing		
Weather condition	Flyable	☐	Throttle inhibit switch	On	☐
Wind direction	Verified	☐	Flight timer	Stopped	☐
Runaway	Clear	☐	Arming plug	Disarmed	☐
Propeller area	Clear	☐	Propulsion and avionics battery	Unplugged	☐
Throttle inhibit switch	Off	☐	Transmitter data logging	Off	☐
Avionics	On	☐	RX and TX power	Off	☐
Transmitter data logging	On	☐	Payloads, banner and banner mechanism	Removed	☐
Arming plug	Armed	☐	Postflight		

TX and RX power	On	☐	Aircraft structure	Inspected	☐
Flight data	Recording	☐	Avionics and propulsion battery	Charge verified	☐
Control surfaces	Freely moving and correct	☐	Pilot and Chief Engineer debrief	Complete	☐
Flaps	As required	☐	Discrepancies (If Any)		
Takeoff intentions	Announced	☐			
Flight timer	Started	☐			
Before Landing					
Landing intentions	Announced	☐			
Flaps	Down	☐			
Runway	Clear	☐			

7.8 Flight Tests

The flight test campaign followed a progressive, risk-mitigation approach to validate performance and mission readiness. Initial cruise, steep turn, stability, and full-system tests verified aerodynamic behavior, controllability, and subsystem reliability under representative conditions. Two iterative prototypes supported this process. Prototype 0 established baseline stability and control characteristics through basic flight testing. Based on the collected test data, refinements were implemented in Prototype 1 development, which underwent short phased verification flight tests before extensive mock testing. This structured approach ensured systematic validation of stability, performance optimization and competition readiness.

Cruise testing was conducted in clean (Prototype 0) and mission-loaded (Prototype 1) configurations at steady throttle settings (70–80%) during level flight, with telemetry-recorded cruise speed and pilot-observed trim stability confirming that measured cruise speeds closely matched predicted values without noticeable performance degradation under payload conditions. Steep turn evaluations were performed beginning at a 20° bank angle and increasing incrementally, and the recorded altitude deviation, throttle response, and pilot feedback confirmed satisfactory, coordinated turns with minimal altitude loss and stable control response throughout sustained maneuvers. Stability tests were conducted in clean (Prototype 0) and full payload configurations (Prototype 1) where visual observation and recorded data confirmed predictable, stable handling with no oscillations or control saturation. A full mission configuration test with payload and banner integrated (Prototype 1) confirmed that all subsystems operated cohesively and that the aircraft maintained expected flight performance during complete mission execution.

7.8.1 Prototype 0 Testing

An initial prototype was constructed to validate preliminary aerodynamic sizing and to obtain early flight-test feedback on aircraft stability, control response, and subsystem integration. The prototype was intentionally designed as a low-risk, easily repairable platform to support rapid iteration during early testing.

Prototype 0 employed Depron foam construction for the fuselage and lightweight foam board for the wings, while maintaining geometric consistency with the subsequent competition prototype. A high-wing configuration was selected for this to simplify early stability evaluation and ground



Figure 7.11 Prototype 0

handling during initial flight tests. An avionics system equivalent to that of the final aircraft was installed to verify flight-control functionality and enable data logging.

Results from prototype zero testing informed several configuration refinements applied to later prototypes. In particular, a transition to a low-wing configuration was adopted to improve payload accessibility and integration, enabling more efficient passenger placement and handling in subsequent designs.

7.8.2 Prototype 1 Testing

Following the initial validation activities conducted with Prototype 0, a refined aircraft configuration was developed as the final competition prototype. Prototype 1 incorporated design updates informed by the results of Prototype 0 testing, with the objective of improving mission operability, payload handling, and overall system integration.

Prototype 1 adopted a low-wing configuration, replacing the high-wing layout used in Prototype 0. This change was driven by mission requirements, enabling improved payload accessibility, more efficient passenger accommodation, and simplified loading and unloading during mission operations. The fuselage and wing primary structure were constructed from plywood and balsa, and the aircraft was covered with Monokote, providing a lightweight yet robust airframe suitable for repeated ground and flight testing.



Figure 7.12 Prototype 1

All major onboard systems, including propulsion, avionics, banner deployment and release mechanism, and payload restraint system were integrated in their final competition configuration on Prototype 1. The aircraft was used as the primary test article for subsequent ground mission and flight testing, as described in section 7.3 and 7.8.3 respectively. Data collected using this prototype formed the basis for the performance results presented in section 8.

7.8.3 Banner Drag - Flight Testing

Flight tests were conducted with the banner attached to the aircraft to evaluate in-flight banner drag characteristics and to verify consistency with ground vehicle test trends under representative M3 operating conditions. Banner drag was measured using an inline load measurement device while varying banner configuration parameters, including material and the presence or absence of trailing-edge slits. The resulting drag coefficients, evaluated as a function of Reynolds number, showed good agreement with ground-based measurements and analytical estimates. No adverse effects on aircraft stability, controllability, or handling qualities were observed during banner towing, confirming the suitability of the selected banner configuration for flight. Detailed results and comparisons are presented in section 8.1.1.

8 PERFORMANCE RESULTS

8.1 Demonstrated Performance of Key Subsystems

8.1.1 Banner Drag Performance

The banner drag results presented in this section are based on the experimental test setups previously described in sections 7.4.1, 7.4.2, and 7.8.3. Full-scale ground vehicle testing was conducted to compare the aerodynamic performance of three

candidate banner fabrics, Nylon 228T Taslan, 240T Pongee, and Taffeta 240T, all tested at a fixed banner aspect ratio of $AR_b = 5$ and banner dimensions of 5 ft x 1 ft. The resulting variation of drag coefficient with Reynolds number is shown in Figure 8.1.

Across the tested Re range, corresponding to M3 operating speeds, all materials exhibit a decreasing drag coefficient with increasing Reynolds number. Taffeta 240T demonstrated the lowest drag over the entire Reynolds number range. It has smoother surface characteristics and reduced effective thickness, which lead to lower flow separation and reduced wake losses. Thus, Taffeta 240T was selected as the baseline banner material for subsequent configuration testing and flight evaluation.

Following material selection, a configuration study was performed using the Taffeta 240T banner to evaluate the effect of trailing-edge slits on drag performance. Both solid and slitted banner configurations were examined, with ground-based results used as a reference and flight-representative measurements obtained as described in section 7.8.3. The resulting drag trends are presented in Figure 8.2.

The results indicate that the introduction of trailing-edge slits reduced large-amplitude banner oscillations and improved tow stability, but it did not reduce the mean drag coefficient. In fact, the slitted configuration exhibited a slightly higher average drag compared to the solid banner. This increase is attributed to additional edge-induced losses and increased viscous dissipation associated with the slit geometry. The solid banner configuration maintained relatively low drag with acceptable dynamic behavior at higher Re, making it more suitable for maximizing M3 performance.

Based on the combined results of wind tunnel testing, ground vehicle testing and flight testing, a solid Taffeta 240T banner of 5ft length and 1ft width and no trailing-edge slits was selected for the final aircraft configuration. This design provides the lowest measured average drag within the operational Reynolds number range while maintaining stable tow characteristics, directly contributing to improved M3 scoring potential.

8.1.2 Propulsion Performance

The performance of the propulsion system was evaluated through static thrust testing and battery discharge testing to confirm that the selected configuration meets mission requirements. Static thrust test results, summarized in Table 8.1, show good agreement between experimental measurements and theoretical predictions for the tested propeller–motor combinations. Among the configurations evaluated, the Falcon 16×7E propeller demonstrated thrust and current characteristics most closely matching predicted values while operating within the safe limits of the propulsion system.

Measured static thrust levels provided sufficient margin for takeoff and climb across all mission configurations. Slightly higher experimental thrust values observed for certain combinations are attributed to conservative assumptions used during

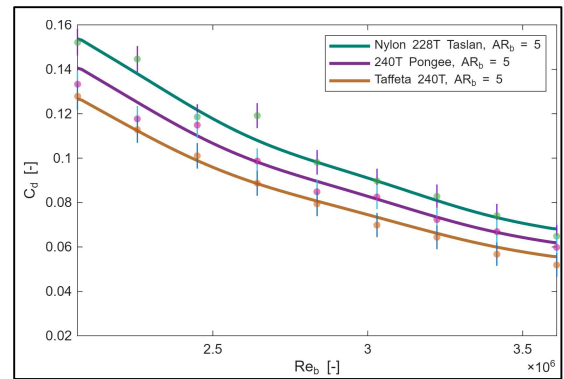


Figure 8.1 Ground vehicle test results showing variation of banner drag coefficient with Reynolds number for three candidate banner fabrics

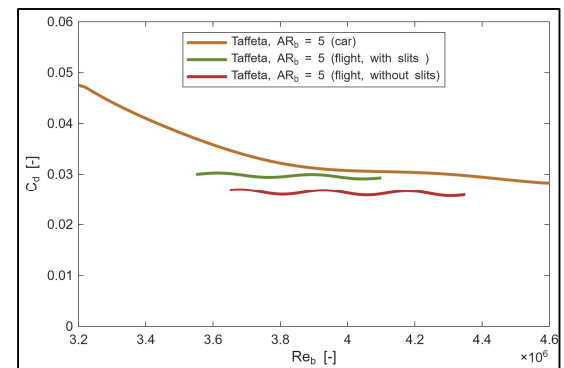


Figure 8.2 Comparison of drag coefficient trends for solid and slitted Taffeta 240T banner based on ground and flight-representative testing

propulsion sizing, including battery internal resistance. Overall, the results validate the propulsion system selection and confirm its suitability for mission operations without exceeding current or thermal constraints.

In addition to thrust performance, electrical endurance requirements were satisfied, as confirmed through battery discharge testing discussed in section 7.5.2. The measured voltage behavior under load demonstrated adequate margin to support the full mission duration while ensuring reliable operation of the propulsion system and onboard electronics.

Table 8.1 Experimental and Theoretical Static Thrust Performance of Evaluated Propeller Configurations

Propeller	Experimental			Theoretical	
	Minimum Voltage per LiPo cell (V)	Maximum Thrust (lb)	Maximum Current (A)	Maximum Thrust (lb)	Maximum Current (A)
Falcon 15×7E	3.82	12.20	88.50	12.75	92.00
Falcon 15×8E	3.77	13.05	101.20	13.80	105.00
Falcon 16×7E	3.79	14.30	95.00	15.10	99.00

8.1.3 Ground Mission Performance

GM performance was demonstrated through timed mock mission runs conducted following final aircraft assembly, as described in section 7.6.1. The total ground mission completion time was approximately 135 sec, consisting of 65 sec for passenger loading, restraint, and unloading, 25 sec for cargo loading, restraint, and unloading, and 40 sec for banner module installation, banner wrapping, and attachment. Based on the demonstrated completion time, the predicted GM score of 0.25 was satisfied. These results confirm reliable and repeatable execution of all GM tasks within DBF requirements, while demonstrating effective crew coordination and efficient ground handling capability. Further reductions in ground mission time are expected with continued crew training and improved task sequencing.

8.2 Demonstrated Flight and Mission Performance

Flight testing was conducted to demonstrate both flight characteristics and mission-level performance of the completed aircraft and to validate the design predictions developed during the preliminary and detailed design phases. Mock mission flights were performed for M1, M2, and M3 configurations to evaluate handling qualities, cruise performance, payload capability, and overall mission execution.

The aircraft exhibited stable and predictable flight characteristics throughout testing. Pitch and roll response remained responsive, and adequate thrust margin was observed across all configurations, including the fully loaded M2 condition. These qualitative observations are consistent with the stability and handling predictions discussed earlier in section 4.6.

A comparison of predicted and demonstrated mission performance parameters is presented in Table 8.2. Demonstrated cruise airspeeds were generally within close agreement with predicted values, with minor deviations attributable to wind conditions and banner drag effects. Average lap times increased slightly for M2 and M3 relative to predictions, resulting in a modest reduction in the number of completed laps. These differences are primarily attributed to operational factors such as takeoff distance, turn execution, and non-optimized flight paths rather than propulsion or aerodynamic limitations.

The aircraft successfully carried the full target payload for M2 and reliably completed banner deployment and towing for M3. Overall, the demonstrated flight performance shows good agreement with predicted values and confirms that the aircraft meets all mission requirements. Further improvements in demonstrated performance are expected through continued pilot practice and flight path optimization rather than configuration changes.

Table 8.2 Comparison of Predicted and Demonstrated Mission Performance (Demonstrated values obtained from mock mission flights; values reflect averaged performance across multiple runs)

Parameter	M1 Predicted	M1 Demonstrated	M2 Predicted	M2 Demonstrated	M3 Predicted	M3 Demonstrated
MTOW (lbs)	8.82	8.90	12.87	12.95	9.20	9.25
Payload Weight (lbs)	N/A	N/A	4.06	4.10	0.41	0.40
Passengers Carried	0	0	24	24	0	0
Cargo Units Carried	0	0	8	8	0	0
Banner Length (in)	N/A	N/A	N/A	N/A	60	60
Cruise Airspeed (ft/s)	89	93	108	106	89	86
Avg. Lap Time (s)	35	30	30	33	37	43
No. of Laps	3	3	10	9	8	7
Mission Score	1	1	1.643	1.572	2.144	1.98
Total Score	Predicted		5.03			
	Demonstrated		4.55			

8.3 Design Changes and Lessons Learned

Throughout the design-build-test cycle, several key design changes were implemented in response to prototype testing outcomes, rule clarifications, and operational considerations. These changes improved rule compliance, structural integrity, and overall mission reliability, while reinforcing the importance of iterative validation.



Figure 8.3 HeisenBird during a representative M2 mock flight

Early design concepts incorporated an aft belly cavity to house the banner. However, subsequent clarifications obtained through DBF committee Q&A revealed constraints on banner stowage and deployment that motivated a transition to a dedicated external banner housing, deployment, and detachment module. This change improved rule compliance, simplified banner handling, and enabled more reliable deployment and release during M3 operations.

During the proposal phase, adhesive-based sticky pads were considered as a secondary restraint for passenger retention. Following further Q&A clarification [18, 19] confirming that adhesives were not permitted, the passenger restraint system was redesigned to rely solely on a mechanical primary restraint. This modification ensured full rules compliance while maintaining secure passenger retention during flight and ground handling.

Configuration refinement was also driven by prototype findings. Prototype 0 employed a high-wing configuration, which simplified early integration and flight validation; however, ground handling and payload integration feedback indicated that a low-wing configuration provided improved payload accessibility and more efficient passenger accommodation for competition operations. As a result, the final aircraft adopted a low-wing layout to improve mission practicality and reduce turnaround time during loading and unloading.

Initial concepts placed the cargo compartment door on the side of the fuselage. Structural evaluation and prototyping indicated that this configuration introduced undesirable local structural weakening. As a result, the cargo access was relocated to a rear-facing door positioned beneath the tail upsweep, providing improved load paths, enhanced structural integrity, and more robust cargo handling. Similarly, the passenger compartment was initially designed with a single access

door. Prototype testing revealed localized torsional effects during loading and unloading operations. To mitigate this, the final configuration adopted two separate passenger doors, reducing torsional loading, improving structural symmetry, and simplifying ground mission operations.

Finally, the airframe manufacturing approach evolved from early foam-based mockups to a more flight-representative structure. While foam prototypes enabled rapid iteration during early development, the final aircraft structure was implemented using plywood and balsa primary structure with Monokote covering to improve durability, stiffness, and repeatability during testing and mission execution.

Collectively, these design changes demonstrate the value of a rule-informed, prototype-driven design process. The final configuration reflects a balanced solution emphasizing compliance, structural robustness, and operational efficiency, with each modification directly informed by test observations or official rule clarifications. The lessons learned through this process were incorporated into the final design and contributed to a robust, competition-ready aircraft solution.

The 2025-26 Team MIST DBF, shown in Figure 8.4, is prepared to compete in the 2026 AIAA Design, Build, Fly Fly-Off in Wichita, Kansas, and proudly presents HeisenBird, the culmination of this design effort.



Figure 8.4 Team MIST DBF 2025-26

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